

D

C

B

A

- 1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
- 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
- 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

K22

LAST_MODIFIED=Thu Aug 20 17:40:45 2009

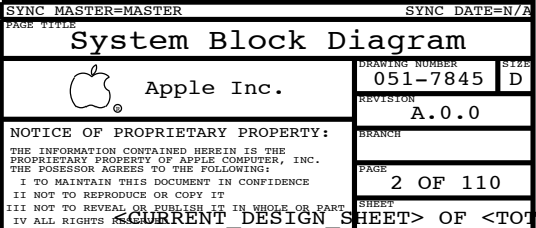
Page	(.csa)	Contents	Sync	Date
1	1	Table of Contents	MASTER	N/A
2	2	System Block Diagram	MASTER	N/A
3	3	Power Block Diagram	MASTER	N/A
4	4	BOM Configuration	MASTER	N/A
5	6	Power Conn / Alias	MASTER	N/A
6	7	HOLES & STANDOFFS	MASTER	N/A
7	8	UNUSED SIGNAL ALIAS	MASTER	N/A
8	9	SIGNAL ALIASES	MASTER	N/A
9	10	CPU FSB	MASTER	N/A
10	11	CPU TEST & MISC.	MASTER	N/A
11	12	CPU POWER, GND, DECAPS	MASTER	N/A
12	13	eXtended Debug Port (XDP)	MASTER	N/A
13	14	MCP CPU Interface	MASTER	N/A
14	15	MCP Memory Interface	MASTER	N/A
15	16	MCP MEMORY CNTRL & MISC	MASTER	N/A
16	17	MCP PCIe Interfaces	MASTER	N/A
17	18	MCP Ethernet & Graphics	MASTER	N/A
18	19	MCP PCI & LPC	MASTER	N/A
19	20	MCP SATA & USB	MASTER	N/A
20	21	MCP HDA & MISC	MASTER	N/A
21	22	MCP Power & Ground	MASTER	N/A
22	25	MCP Standard Decoupling	K51	12/08/2008
23	26	MCP Graphics Support	MASTER	N/A
24	28	SB Misc	MASTER	N/A
25	29	FSB/DDR3 Vref Margining	MASTER	MASTER
26	30	MEMORY CAPS	MASTER	N/A
27	31	DDR3 SO-DIMMs 0 & 2	MASTER	N/A
28	32	DDR3 SO-DIMM CONNECTOR B	MASTER	N/A
29	33	DDR3 SUPPORT AND BITSWAPS	K51	10/13/2008
30	34	PCI-E Wireless Connector	MASTER	N/A
31	37	Ethernet PHY (RTL8211CL)	K51	12/08/2008
32	38	Ethernet Support	MASTER	N/A
33	39	ETHERNET CONNECTOR	MASTER	N/A
34	41	FireWire LLC/PHY (XIO2213B)	MASTER	N/A
35	42	FW: 1394B MISC	MASTER	N/A
36	43	FIREWIRE CONNECTOR	MASTER	N/A
37	45	SATA Connectors	MASTER	N/A
38	46	EXTERNAL USB CONNECTORS	MASTER	N/A
39	47	Internal USB Connections	MASTER	MASTER
40	49	SMC	MASTER	N/A
41	50	SMC Support	MASTER	N/A
42	51	LPC+SPI Debug Connector	MASTER	N/A
43	52	SMBUS CONNECTIONS	MASTER	N/A
44	53	CPU/MXM CURRENT AND VOLTAGE SENSE	MASTER	N/A
45	54	MCP CURRENT AND VOLTAGE SENSE	K51	12/08/2008

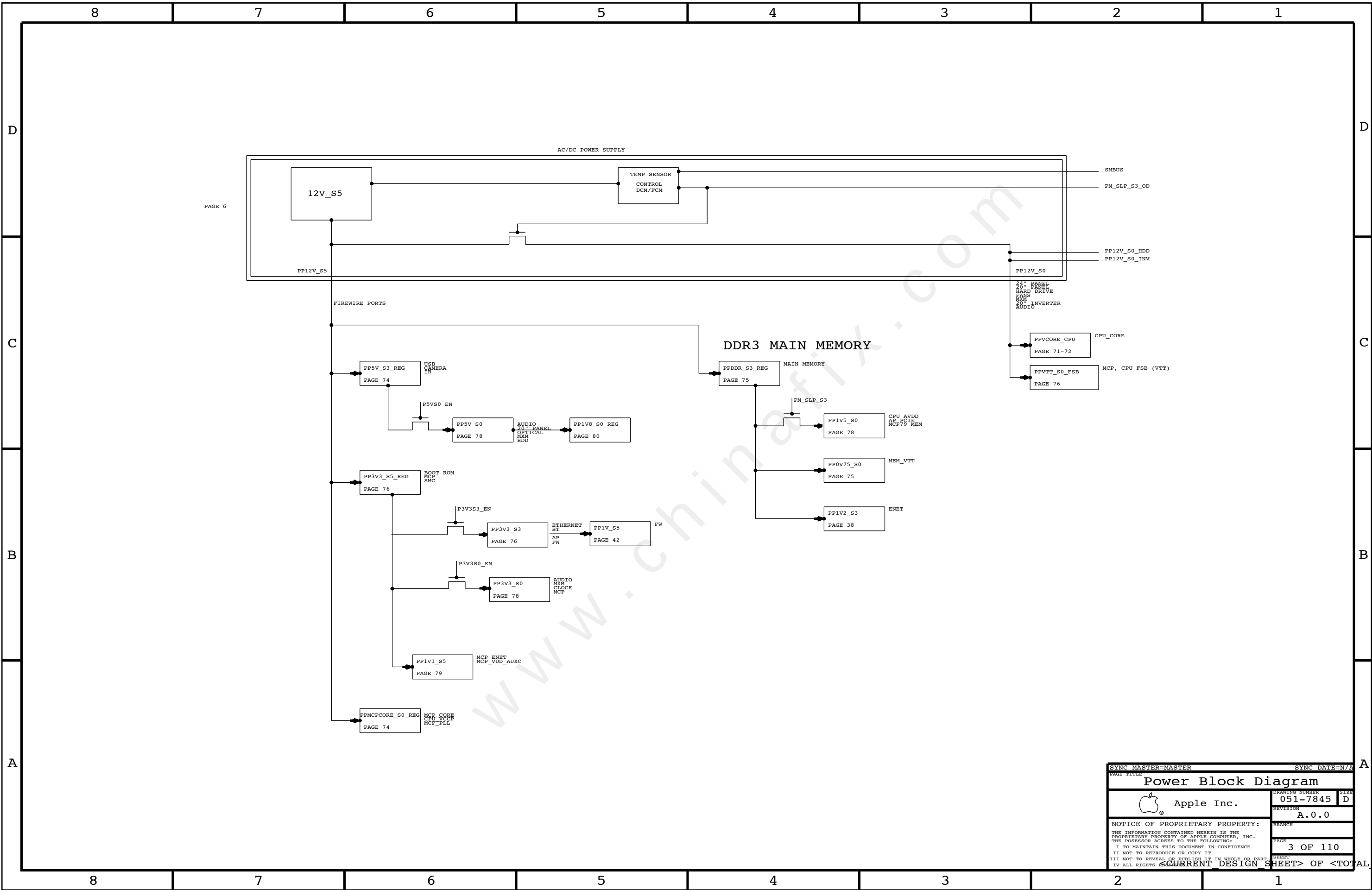
Page	(.csa)	Contents	Sync	Date
46	55	Thermal Sensors	MASTER	N/A
47	56	HD AND OD FAN	MASTER	N/A
48	57	CPU FAN	MASTER	N/A
49	61	SPI ROM	K51	12/08/2008
50	62	AUDIO: CODEC/REGULATOR	SKIPAUDIO	06/01/2009
51	63	AUDIO: FILTER/BUFFER	SKIPAUDIO	06/01/2009
52	64	AUDIO: SPEAKER AMP	SKIPAUDIO	06/01/2009
53	65	AUDIO: SPEAKER AMP	SKIPAUDIO	06/01/2009
54	66	Audio: MLB to I/O Conn.	SKIPAUDIO	06/01/2009
55	67	AUDIO: Detects/Grounding	SKIPAUDIO	06/01/2009
56	68	AUDIO: Mikey	SKIPAUDIO	06/01/2009
57	69	POWER SEQUENCING BLOCK DIAGRAM	K51	12/08/2008
58	70	PGOOD and Power Sequencing	MASTER	N/A
59	71	VREG: PPVCORE S0 CPU	MASTER	N/A
60	72	VREG: PPVCORE S0 CPU	MASTER	N/A
61	73	5V S3 REGULATOR	MASTER	N/A
62	74	MCP CORE REGULATOR	MASTER	N/A
63	75	1.5V DDR SUPPLY	MASTER	N/A
64	76	FSB VTT/3.3V S5 SUPPLIES	MASTER	N/A
65	78	S3 & S0 FETs	MASTER	N/A
66	79	1V1 S5 POWER SUPPLY	K51	10/31/2008
67	80	1V8 POWER SUPPLY	MASTER	N/A
68	84	MXM PCIe, DP & Power	K51	10/31/2008
69	85	MXM I/O	K51	10/31/2008
70	86	MXM PCIE CAPS	MASTER	N/A
71	87	MXM ALIASES	MASTER	N/A
72	89	LCD MUX & CHOKES	MASTER	MASTER
73	90	INTERNAL DISPLAY	MASTER	MASTER
74	91	DP MUX SUPPORT	MASTER	N/A
75	93	DISPLAYPORT SUPPORT	MASTER	N/A
76	94	DisplayPort Connector	MASTER	N/A
77	100	CPU/FSB Constraints	MASTER	N/A
78	101	Memory Constraints	MASTER	N/A
79	102	MCP Constraints 1	MASTER	N/A
80	103	MCP Constraints 2	MASTER	N/A
81	104	Ethernet Constraints	MASTER	N/A
82	105	FireWire Constraints	MASTER	N/A
83	106	SMC Constraints	MASTER	N/A
84	107	GRAPHICS CONSTRAINTS	MASTER	N/A
85	108	K22/K23 SPECIFIC CONSTRAINTS	MASTER	N/A
86	109	K22/K23 RULE DEFINITIONS	MASTER	N/A
87	110	K22/K23 ICT/FCT	MASTER	N/A

2			1		
REV	ECN	DESCRIPTION OF REVISION	CK APPD		
A	0000774858	PRODUCTION RELEASED	2009-08-21		



DRAWING TITLE			
SCH, K22, MLB			
Apple Inc.		DRAWING NUMBER	051-7845
		REVISION	A.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS <u>RESERVED</u> .		BRANCH	
		PAGE	1 OF 110
		SHEET	1 OF 110







D

C

B

A

D

C

B

A

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
630-9768	PCBA,MLB,GOOD,K22	K22,2P80GHZ_CPU,BASIC,IG
639-0036	PCBA,2.8 GHZ CPU,MXM,K22	K22,2P80GHZ_CPU,BASIC,MXM,K22_MXM
639-0184	PCBA,2.93 GHZ CPU,IG,K22	K22,2P93GHZ_CPU,BASIC,IG
639-0186	PCBA,2.93 GHZ CPU,MXM,K22	K22,2P93GHZ_CPU,BASIC,MXM,K22_MXM
639-0037	PCBA,3.0 GHZ CPU,IG,K22	K22,3P0GHZ_CPU,BASIC,IG
630-9878	PCBA,MLB,CTO,K22	K22,3P0GHZ_CPU,BASIC,MXM,K22_MXM
639-0183	PCBA,3.06 GHZ CPU,IG,K22(Investigation)	K22,3P06GHZ_CPU,BASIC,IG
639-0324	PCBA,MLB,3.16GHZ,MXM,K22	K22,3P16GHZ_CPU,BASIC,MXM,K22_MXM
639-0206	PCBA,MLB,3.33GHZ,IG,K22	K22,3P33GHZ_CPU,BASIC,IG
639-0207	PCBA,MLB,3.33GHZ,MXM,K22	K22,3P33GHZ_CPU,BASIC,MXM,K22_MXM
639-0392	PCBA,2.8 GHZ-2M CPU,IG,K22	K22,2P80GHZ_2M_CPU,BASIC,IG
639-0393	PCBA,2.8 GHZ-2M CPU,MXM,K22	K22,2P80GHZ_2M_CPU,BASIC,MXM,K22_MXM
607-4426	PCBA,MLB,DEV,K22	DEVELOPMENT,DEV_GROUP

BOM GROUPS

BOM GROUP	BOM OPTIONS
BASIC	COMMON,ALTERNATE,MCP7A,XDP,BETTER,MCP_ISL9563A,MLB_PNL_PWR,PRODUCTION
MCP7A	BOOT_MODE_USER,MEMRESET_HW,MEMRESET_MCP
DEV_GROUP	XDP_CONN,LPCPLUS,VREFMRGN,MCP_PWR_SENSE,MCP_CPU_TDIODE,PECI_SMB,MOJOMUX

BOARD STACK-UP

TOP	SIGNAL
2	GROUND
3	SIGNAL
4	POWER
5	POWER
6	SIGNAL
7	GROUND
BOTTOM	SIGNAL

COMMON

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0731	1	IC,GMCP,MCP7A-2A,B03,35X35MM,BGA1437,DT	U1400	CRITICAL	IG
338S0732	1	IC,MCP,MCP7A-DA,B03,35X35MM,BGA1437,DT	U1400	CRITICAL	MXM
341T0170	1	IC,EFI_BOOTROM,K22/K23	U6100	CRITICAL	
338S0765	1	IC,XIO22112AY,1394B,167BGA	U4100	CRITICAL	
338S0694	1	IC,RTL8251CA,GIGE TRANSCEIVER, 48P TQFP	U3700	CRITICAL	
825-7122	1	MLB_LABEL,48.0X4.8	X14	CRITICAL	

MCP -J SKU HAS INTEGRATED GPU
MCP -D SKU DOES NOT

CPUS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S3745	1	WLF,QXXX,Q8,2.80G,65W,1066,R0,3M,LGA	CPU	CRITICAL	2P80GHZ_CPU
337S3742	1	WLF,SLB9J,PRQ,2.83G,65W,1333,E0,6M,LGA	CPU	CRITICAL	2P83GHZ_CPU
337S3726	1	WLF,SLB9J,PRQ,3.0G,65W,1333,E0,6M,LGA	CPU	CRITICAL	3P0GHZ_CPU
337S3715	1	WLF,SLB9K,PRQ,3.16G,65W,1333,E0,6M,LGA	CPU	CRITICAL	3P16GHZ_CPU
337S3727	1	WLF,SLB9L,PRQ,3.33G,65W,1333,E0,6M,LGA	CPU	CRITICAL	3P33GHZ_CPU
337S3807	1	WLF,SLB9L,PRQ,2.93G,65W,1333,E0,6M,LGA	CPU	CRITICAL	2P93GHZ_CPU
337S3766	1	WLF,SLB9L,PRQ,3.06G,65W,1333,E0,6M,LGA	CPU	CRITICAL	3P06GHZ_CPU
337S3804	1	WLF,SL0U9,PRQ,2.80G,65W,1066,R0,2M,LGA	CPU	CRITICAL	2P80GHZ_2M_CPU

K22 PARTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-7845	1	SCH,K22,MLB	SCH1		K22
820-2494	1	PCBF,K22,MLB	MLB1		K22
341T0168	1	IC,SMC,K22	U4900	CRITICAL	K22

(338S0563 - BLNK)

ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0111	127S0060		C6211	AUDIO, NEED QUAL

SYNC MASTER=MASTER

SYNC DATE=N/A

BOM Configuration

Apple Inc.

DRAWING NUMBER

051-7845

SIZE

D

REVISION

A.0.0

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

BOMCH

PAGE

SHEET

4 OF 110

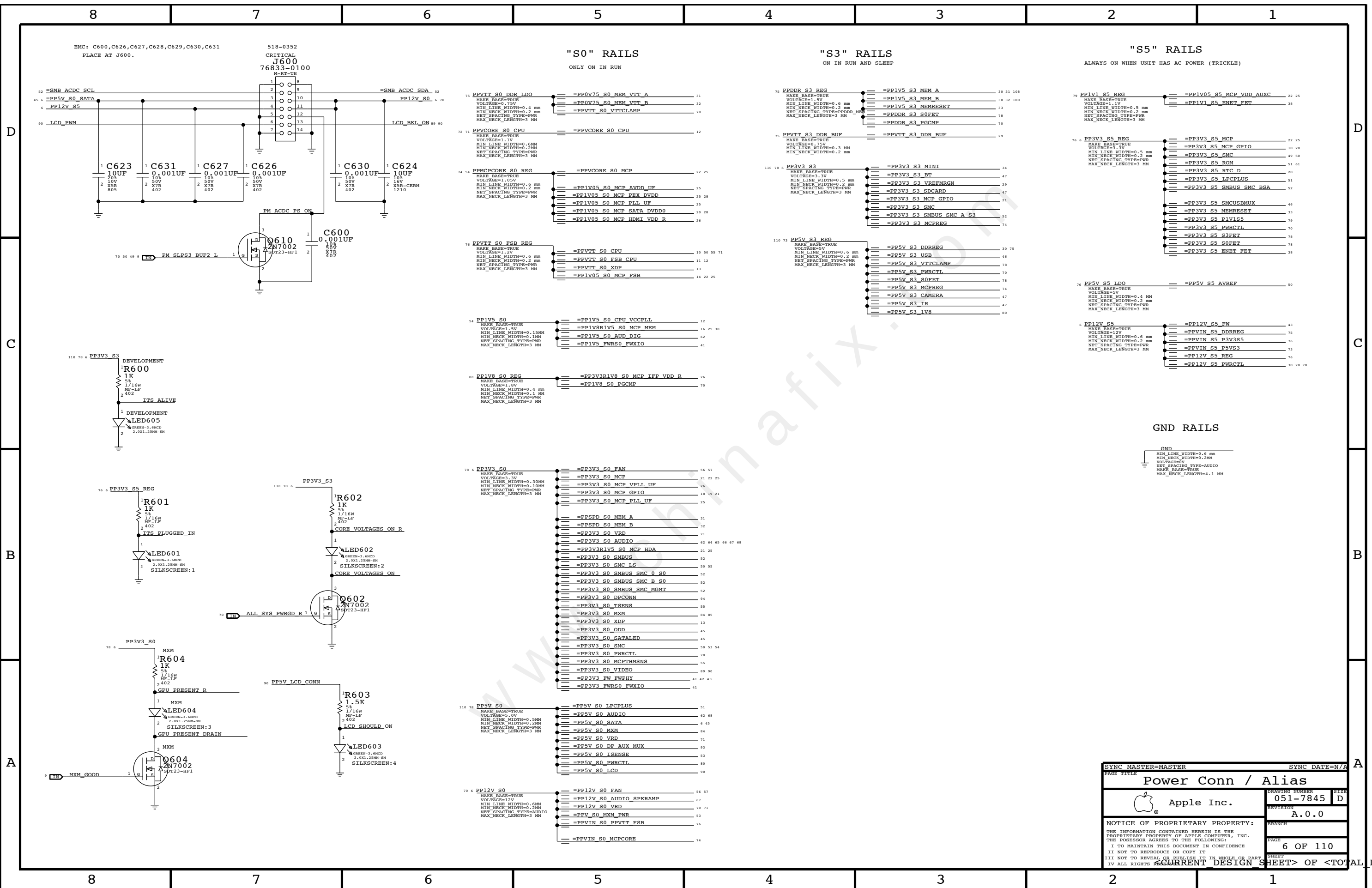
<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>



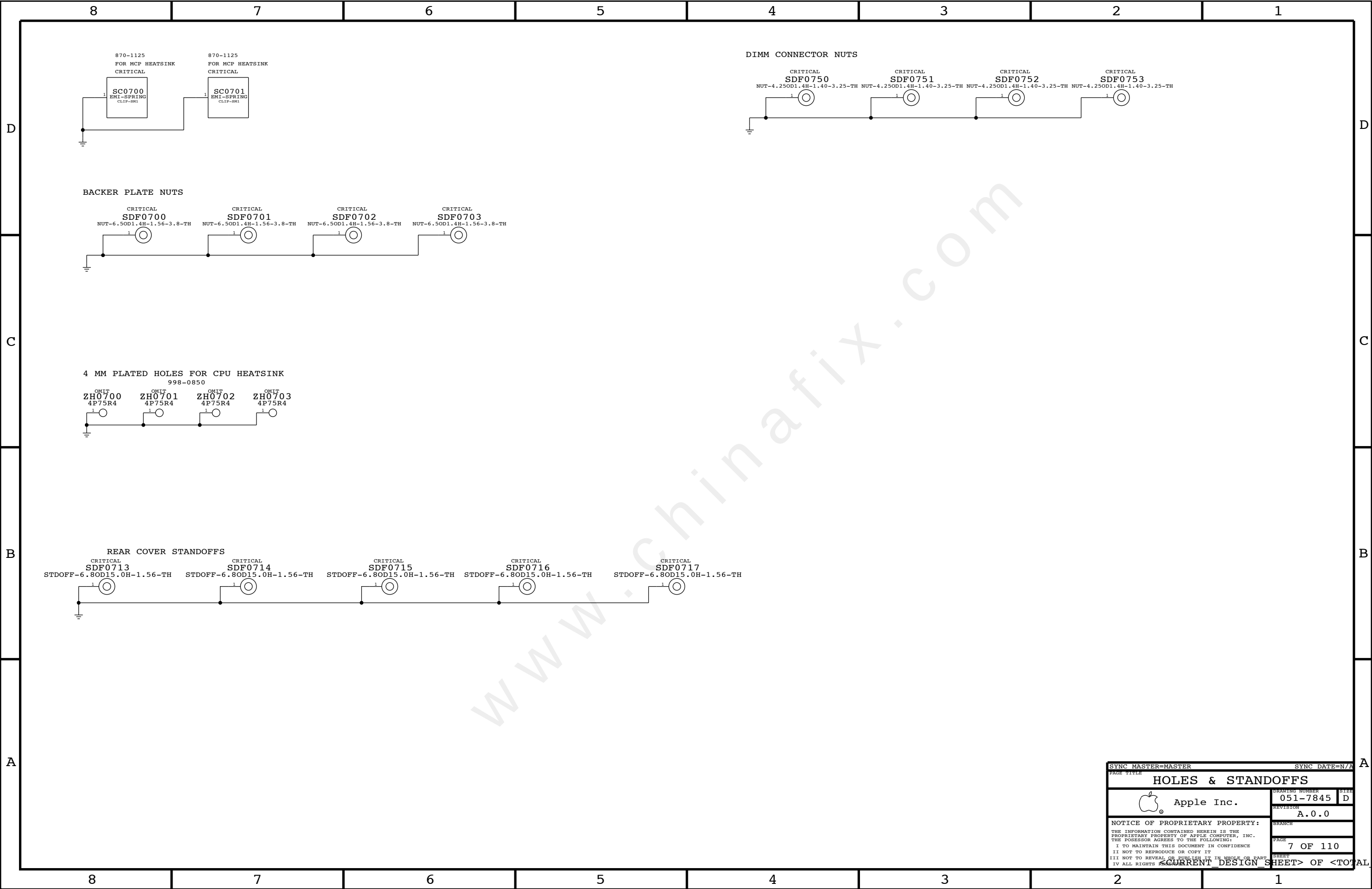
	8	7	6	5	4	3	2	1
D								
C								
B								
A								
	8	7	6	5	4	3	2	1

www.chinafix.com

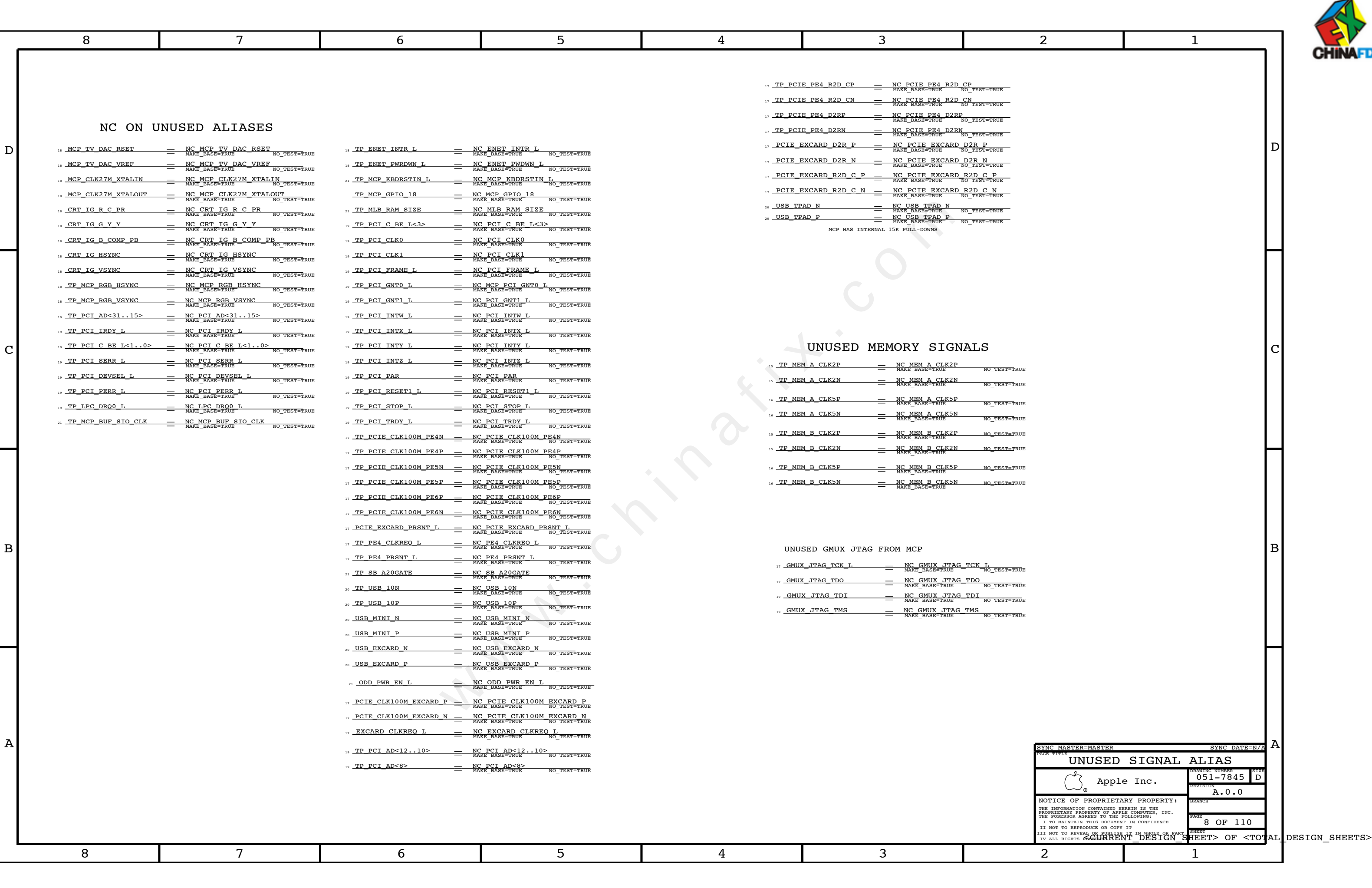
SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		5 OF 110	
		SHEET	
<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>			



SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
Power Conn / Alias			
DRAWING NUMBER		SHEET	
051-7845		D	
REVISION		A.0.0	
BRANCH		PAGE	
NOTICE OF PROPRIETARY PROPERTY:		6 OF 110	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		SHEET	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		IV ALL RIGHTS RESERVED	
II NOT TO REPRODUCE OR COPY IT		III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART	
CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>			



SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
HOLES & STANDOFFS			
	DRAWING NUMBER		SIZE
	051-7845		D
Apple Inc.	REVISION		
	A.0.0		
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
CURRENT DESIGN SHEET		PAGE	
		7 OF 110	
		SHEET	
		<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>	



D

CD

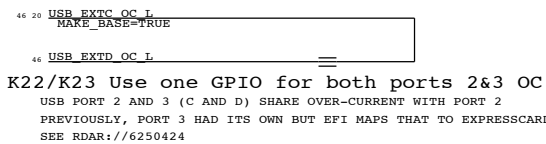
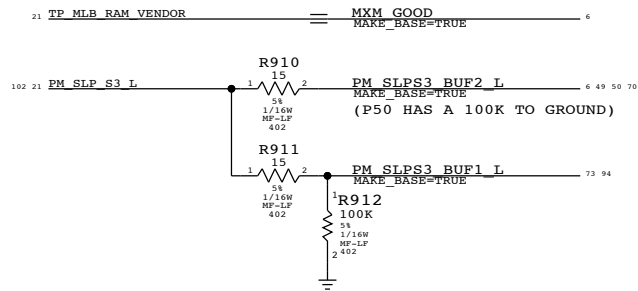
C

B

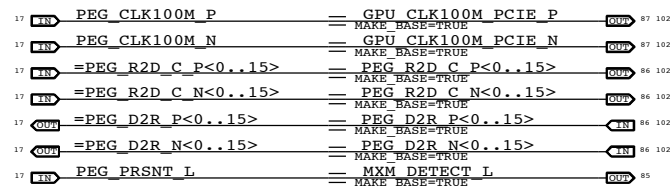
B

AA

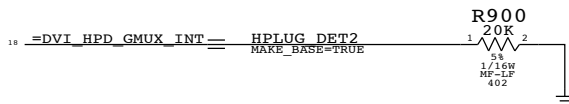
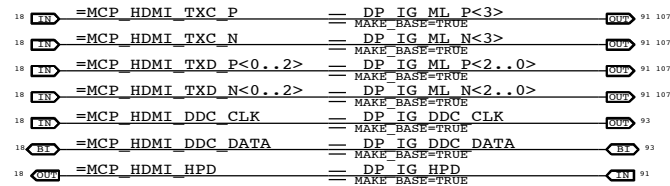
SIGNAL ALIAS



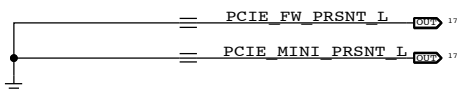
PEG Slot Support



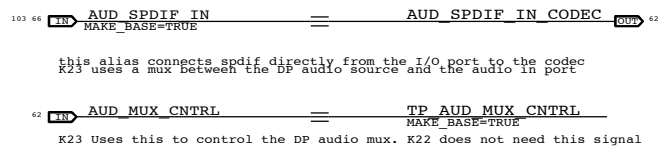
DisplayPort / TMDS Support



MCP79 PCIe PRSENT# Straps

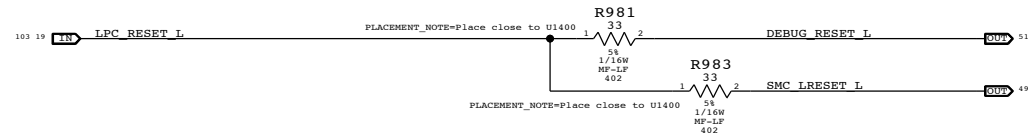


Audio Mux aliasing

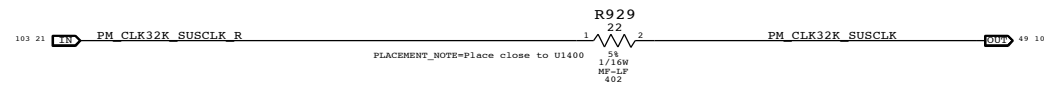
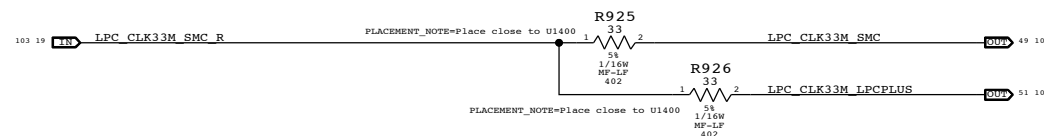
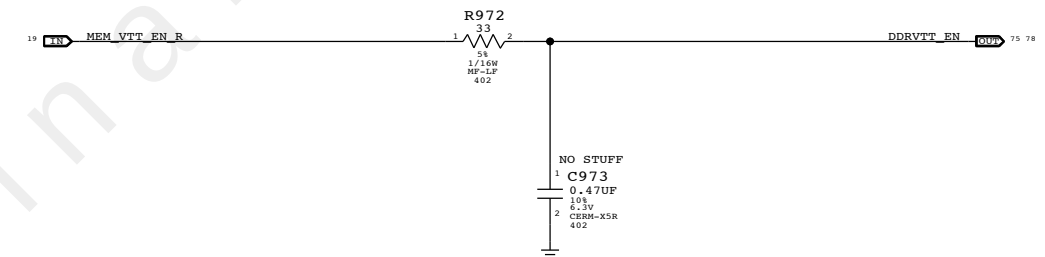
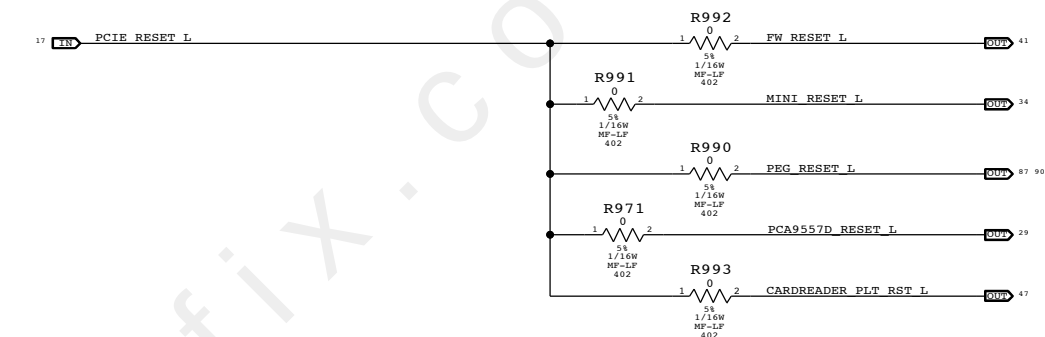


Platform Reset Connections

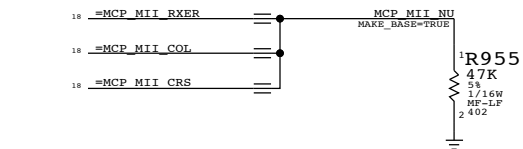
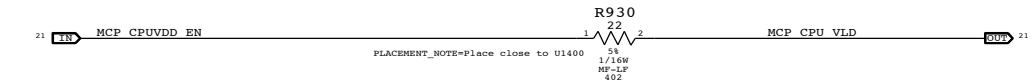
LPC Reset (Unbuffered)



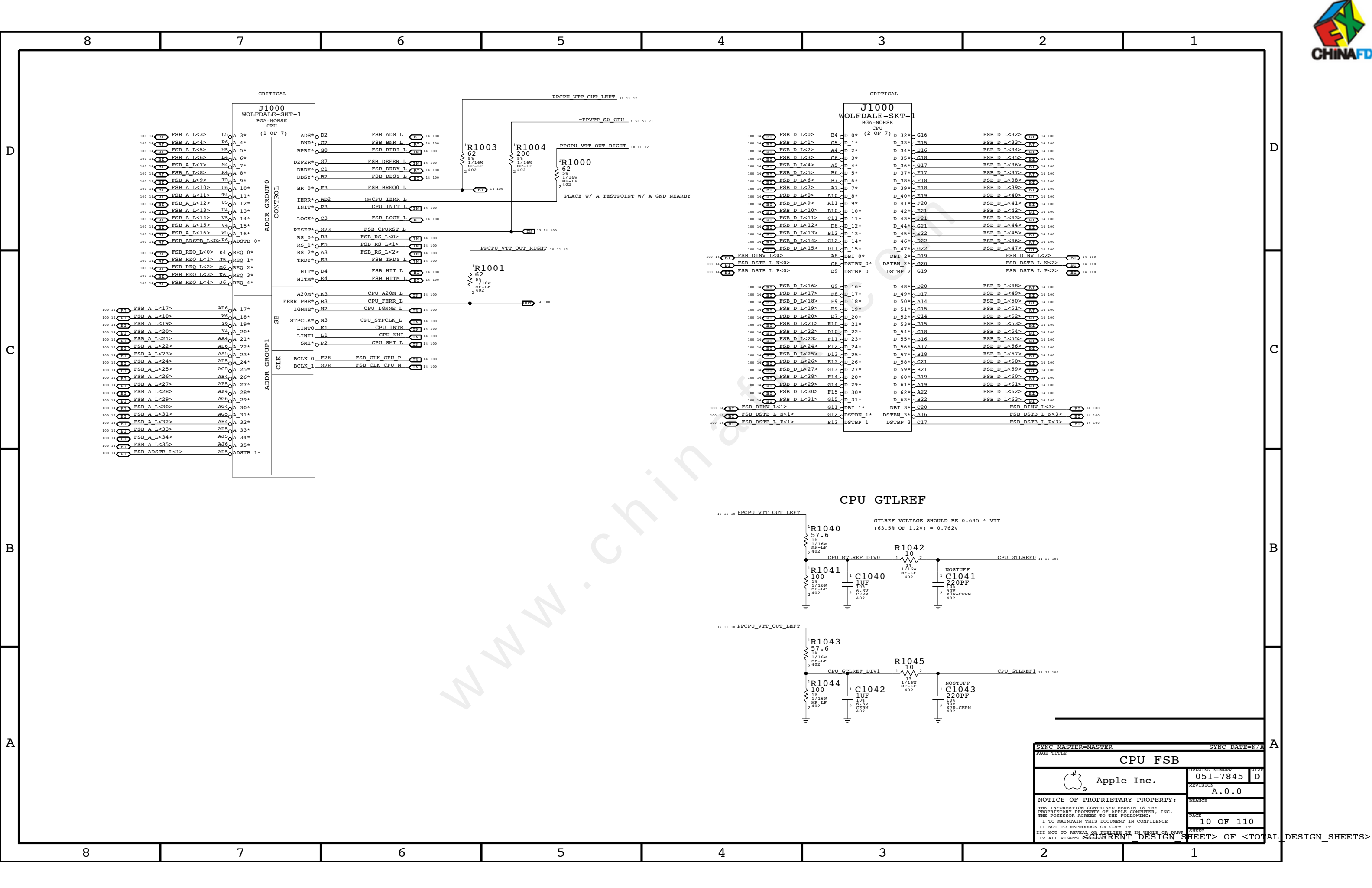
PCIE Reset (Unbuffered)

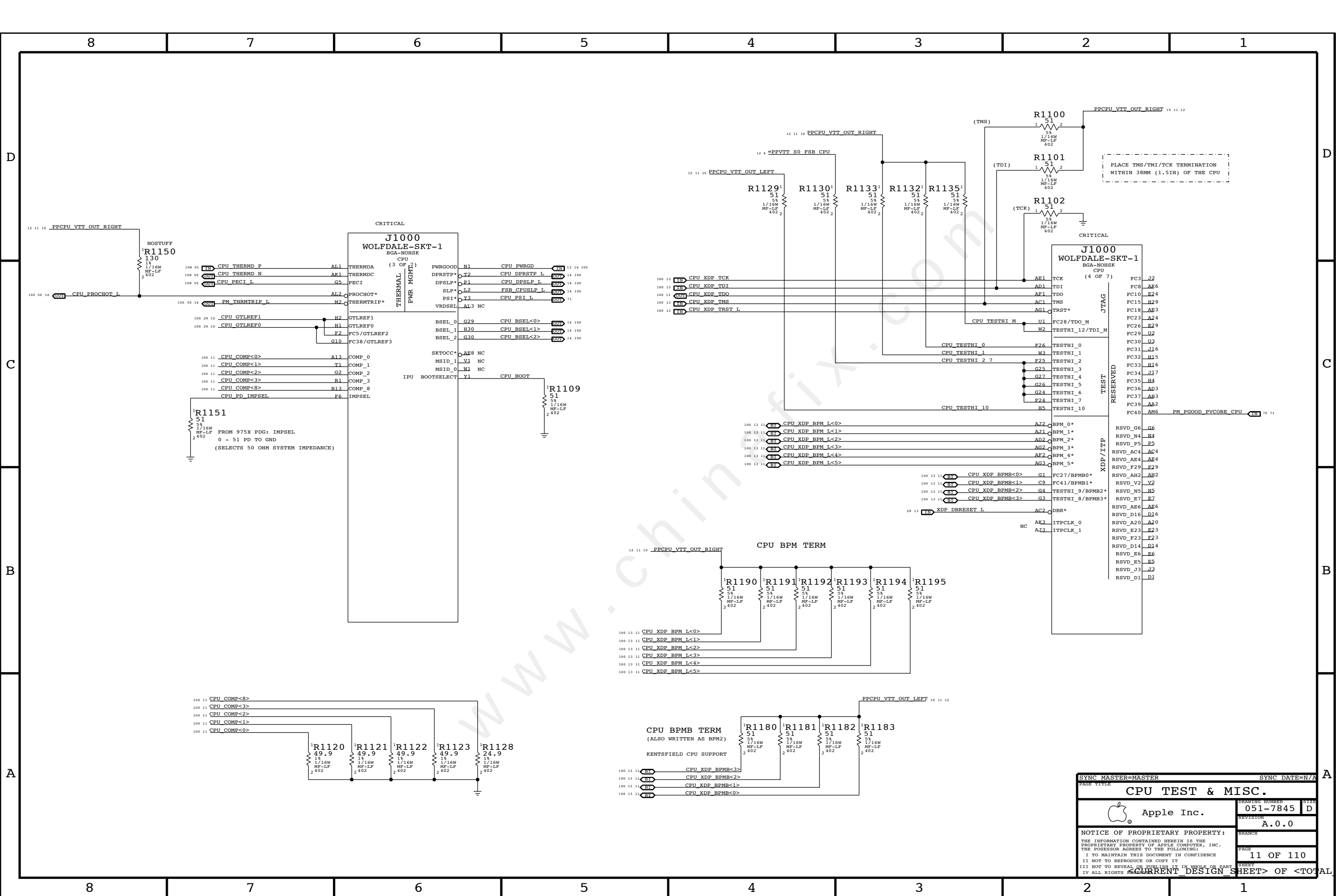


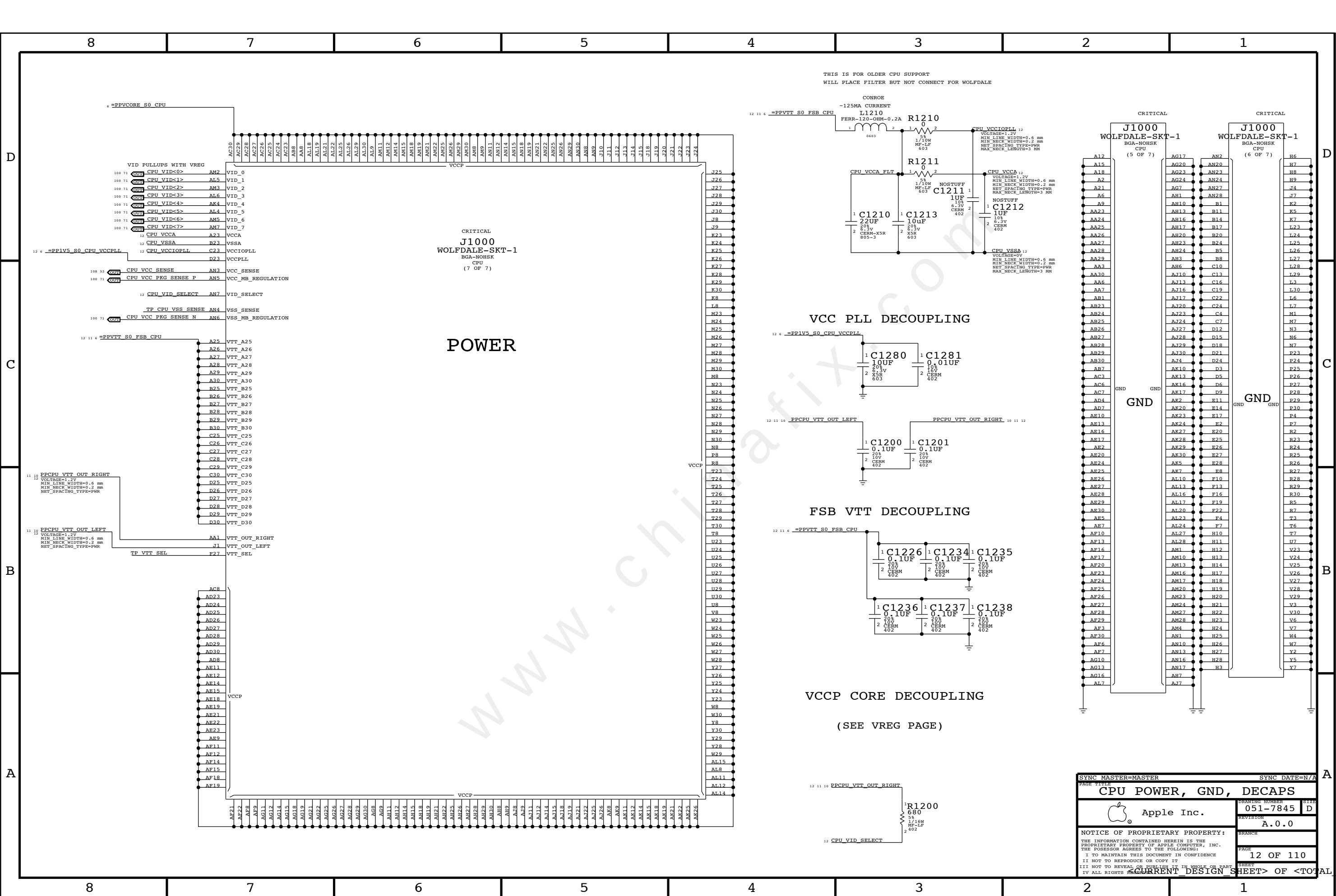
MCP_CPUVDD_EN WILL ASSERT AFTER MCP_PS_PWRGD IS UP

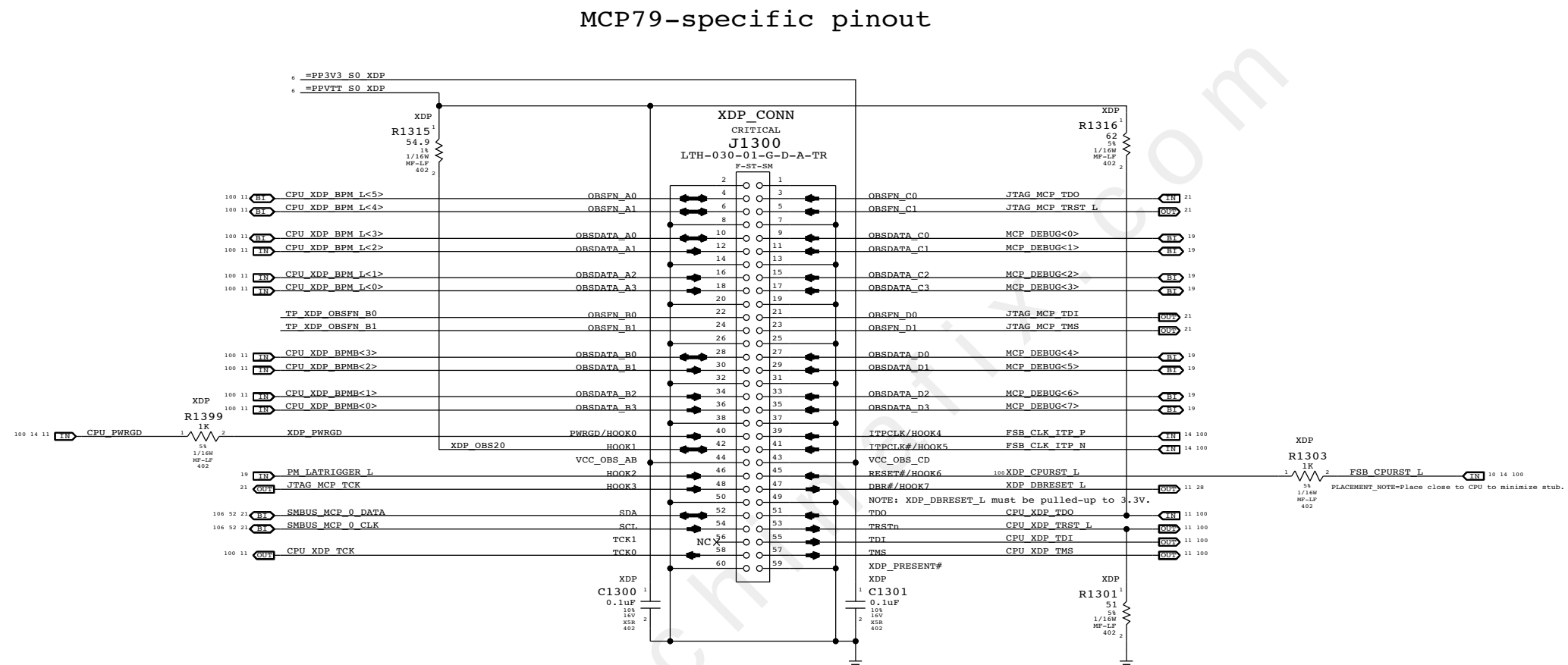


PAGE TITLE		PAGE NUMBER	
SYNCH MASTER=MASTER		SYNCH DATE=N/A	
SIGNAL ALIASES			
Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
		REVISION	
		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
PAGE		SHEET	
9 OF 110			
CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>			

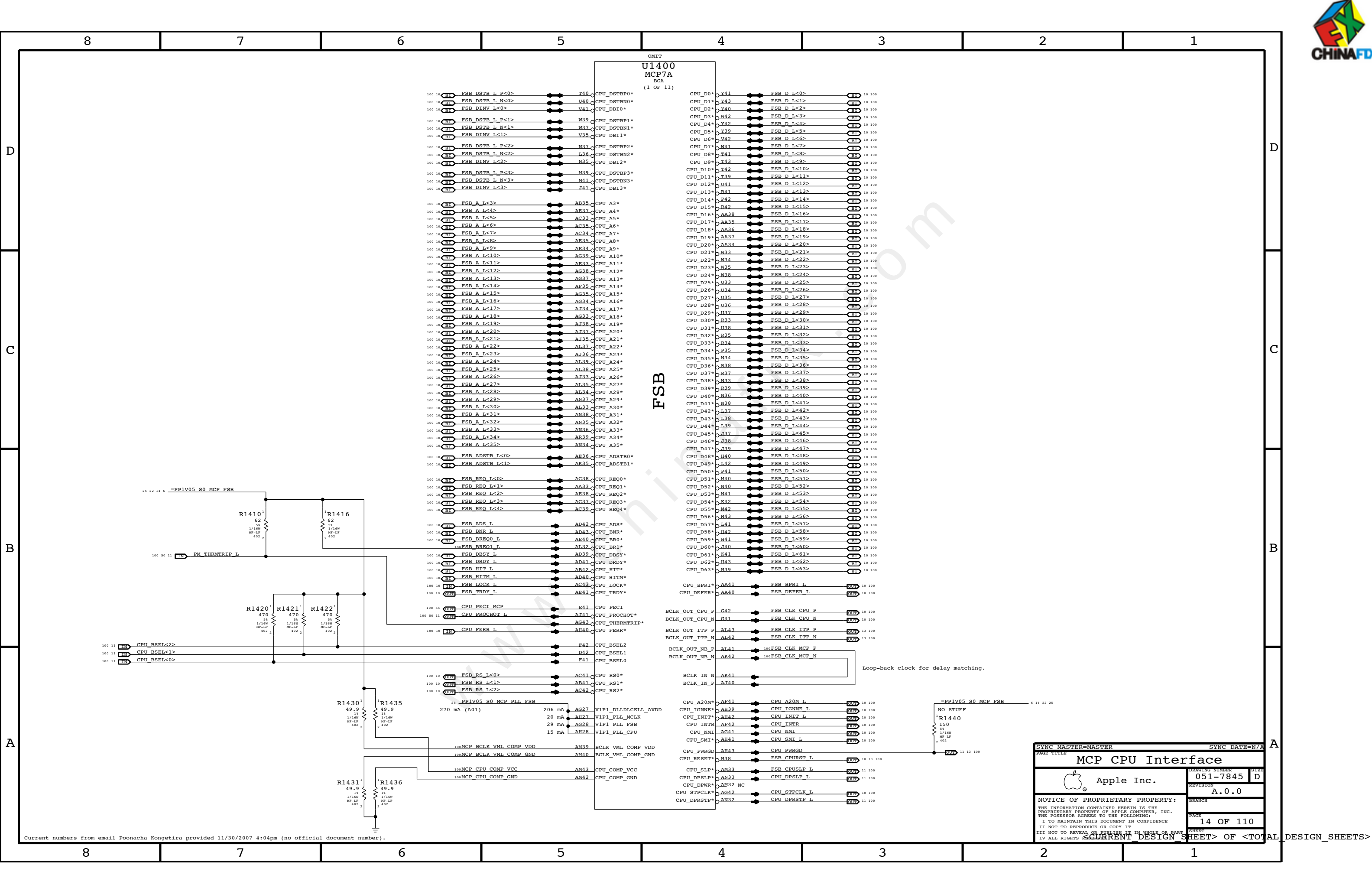


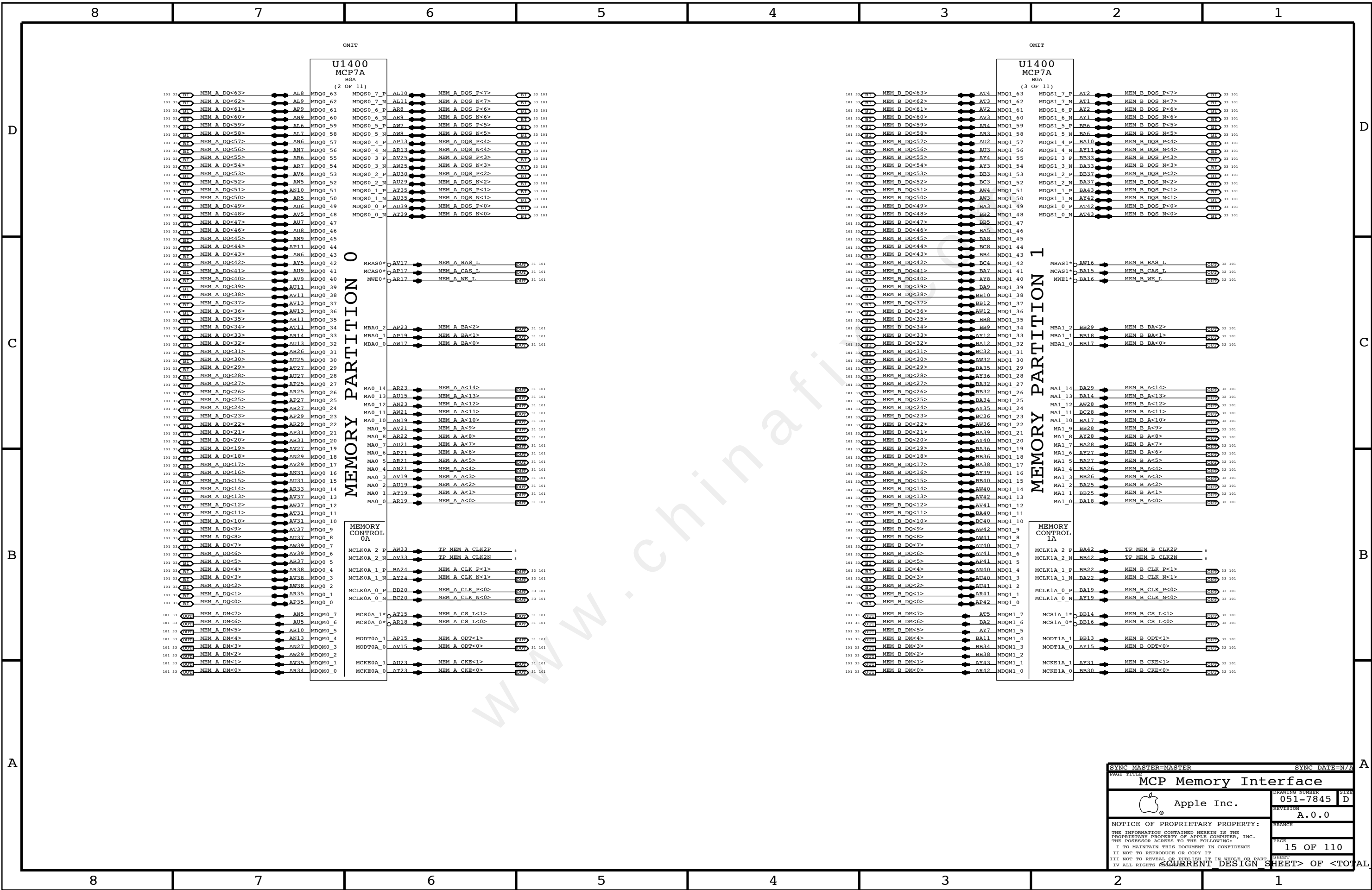


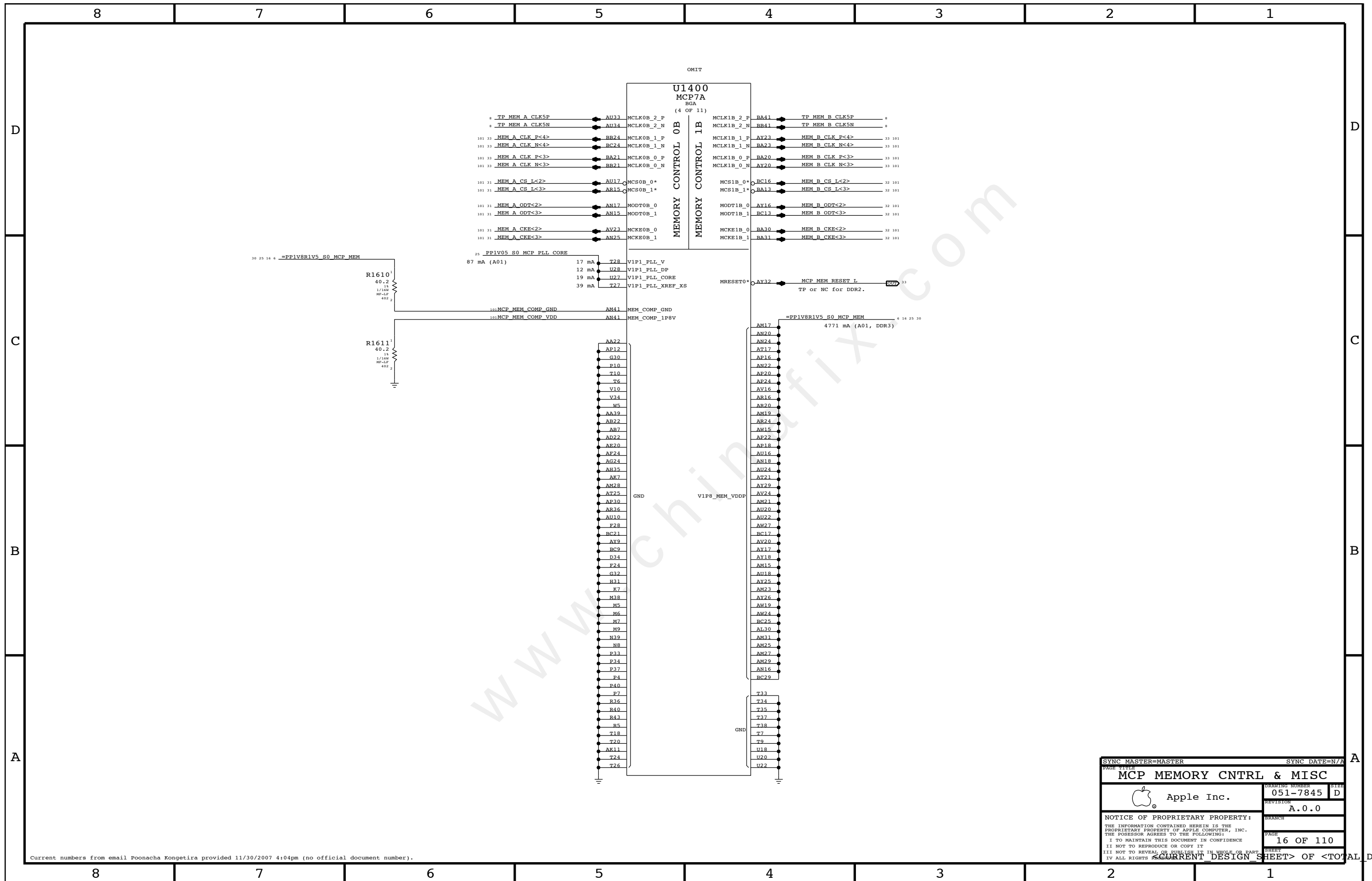


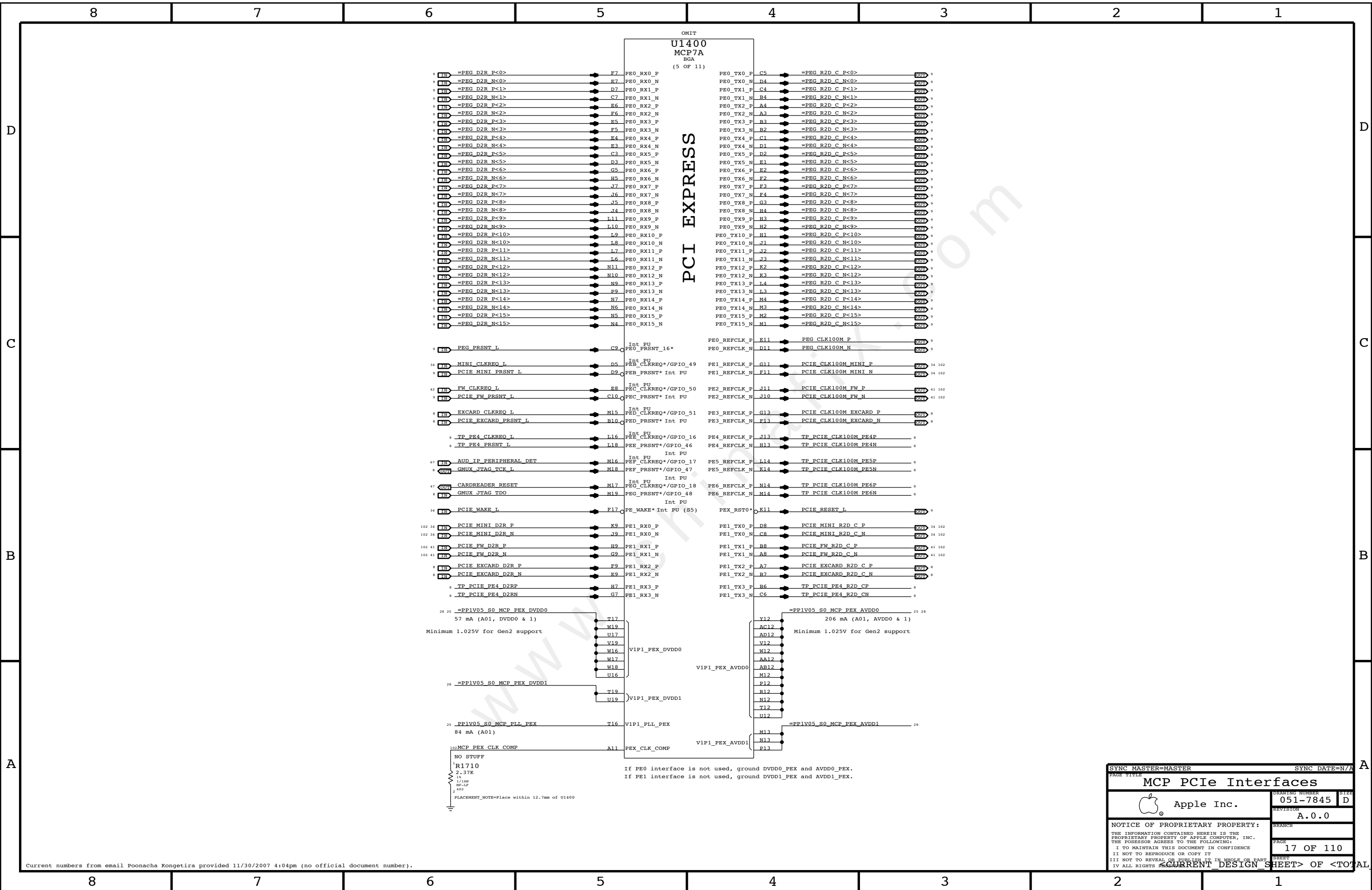


SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
Extended Debug Port (XDP)			
 Apple Inc.	DRAWING NUMBER	SIZE	
	051-7845	D	
	REVISION		
	A.0.0		
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE	
		13 OF 110	
SHEET			
SCHEMATIC DESIGN SHEET OF <TOTAL DESIGN SHEETS>			

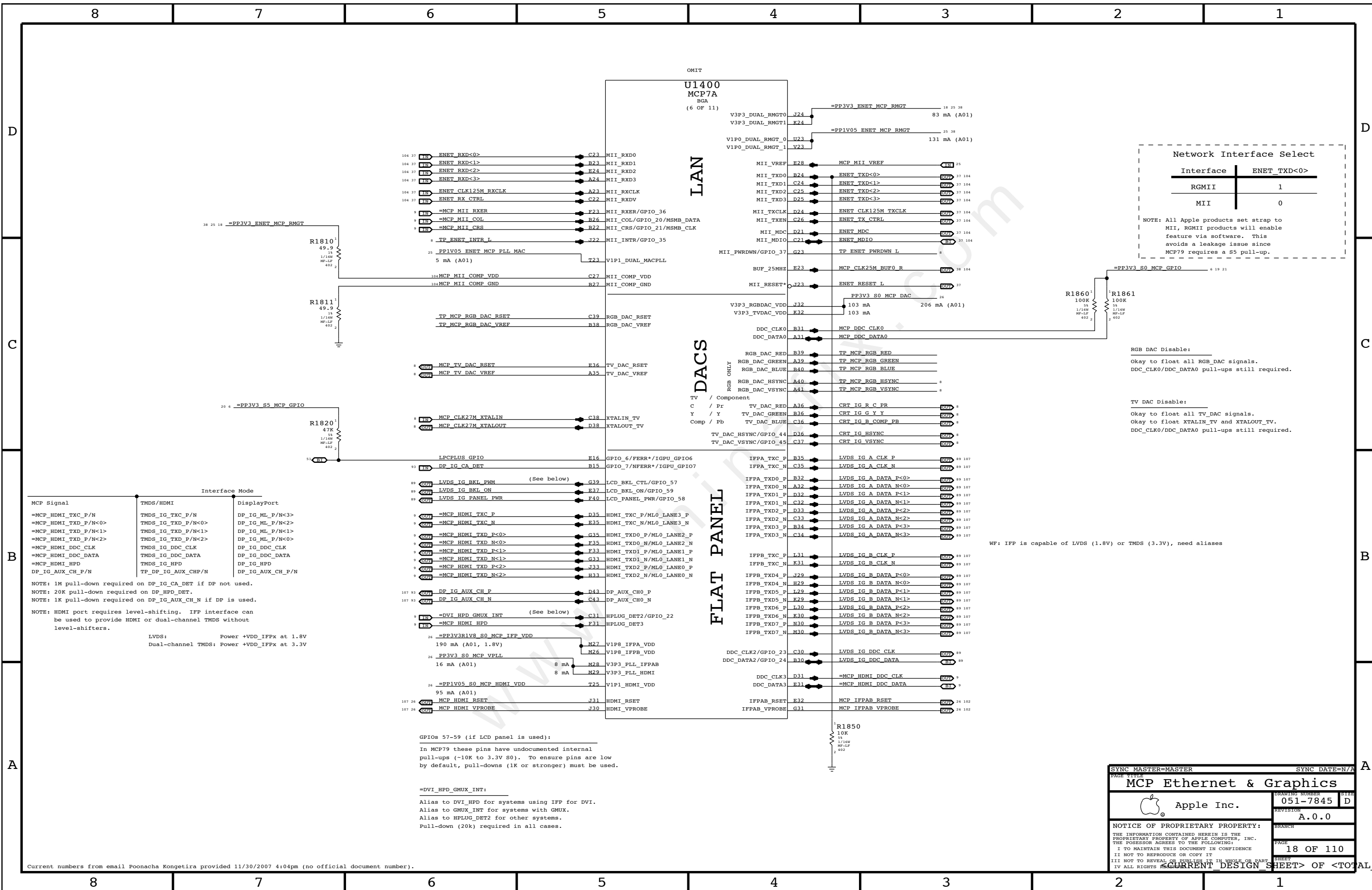


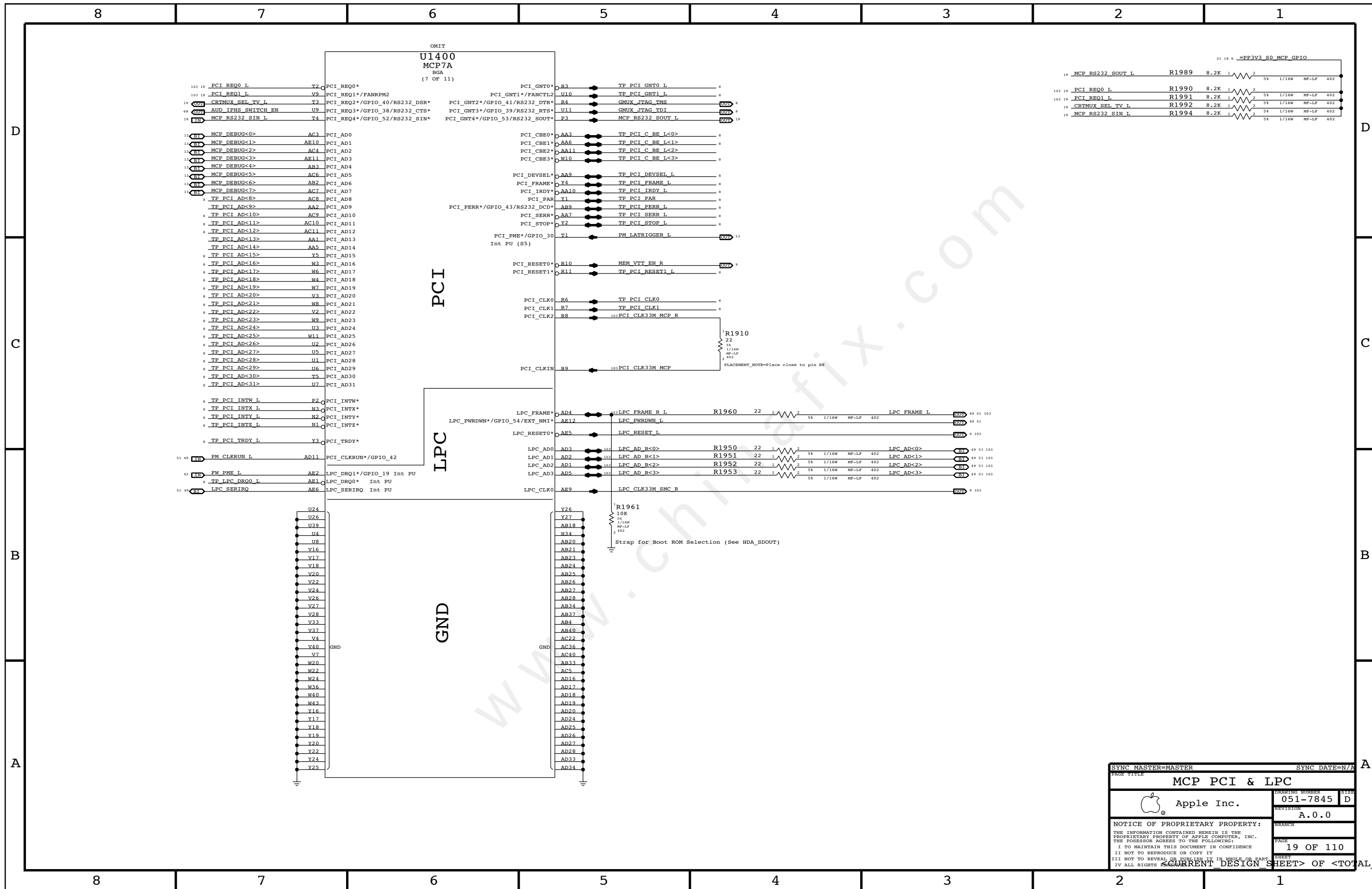


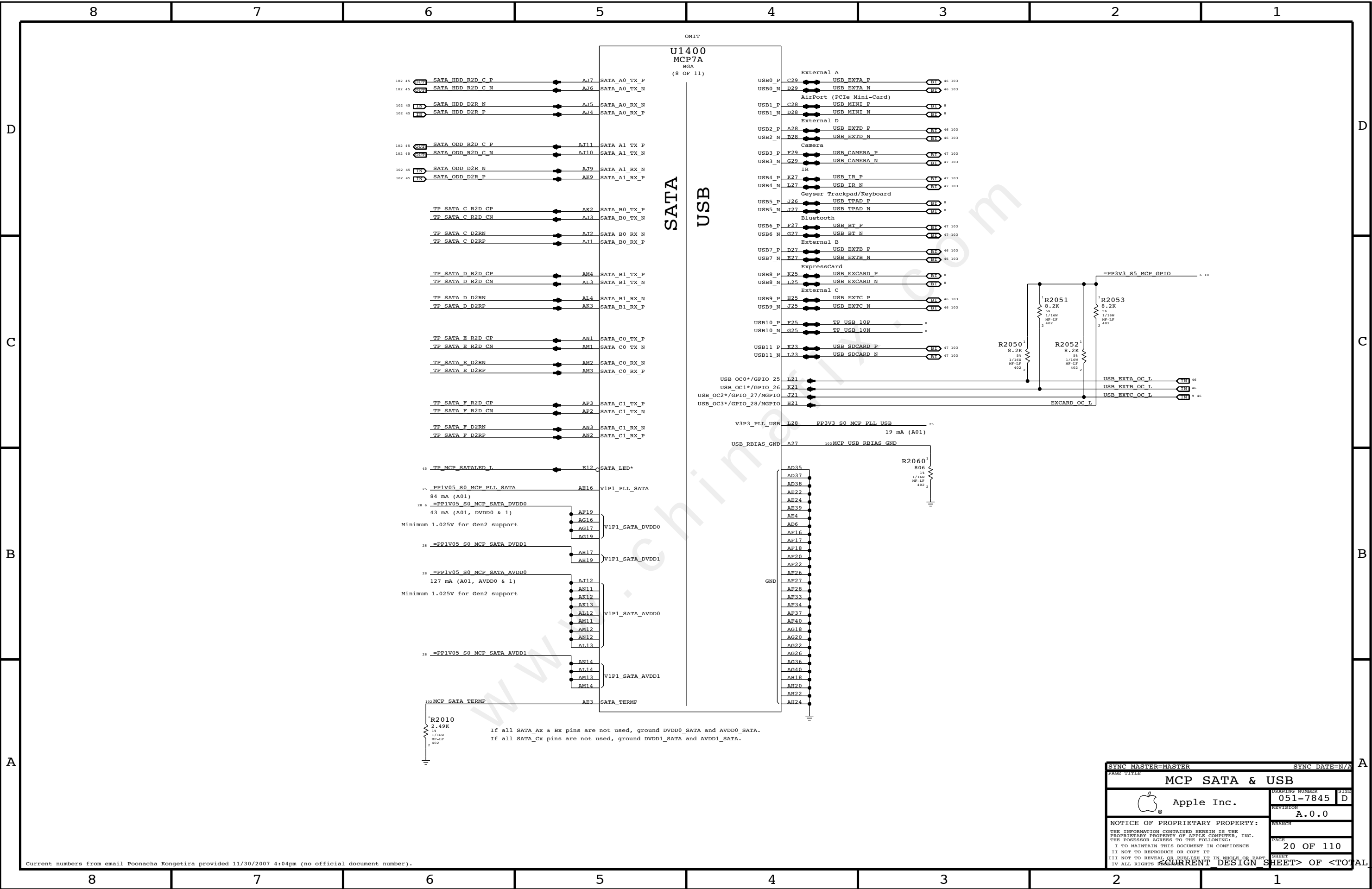




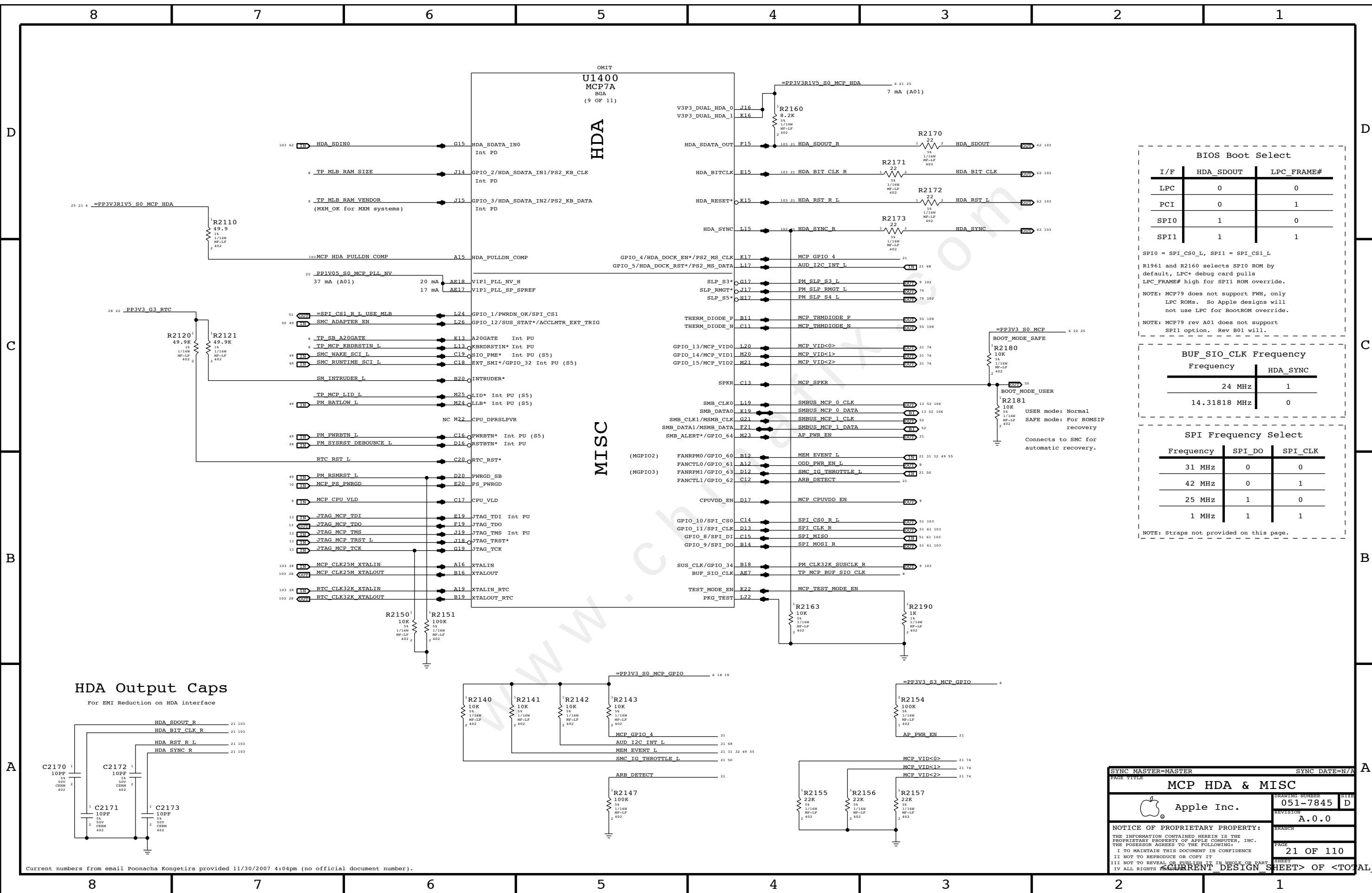
SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
MCP PCIe Interfaces			
DRAWING NUMBER		SIZE	
051-7845		D	
REVISION		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		17 OF 110	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		DESIGN SHEETS	
IV ALL RIGHTS RESERVED		CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>	







SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
MCP SATA & USB			
	Apple Inc.		DRAWING NUMBER
			051-7845
		REVISION	D
		A.0.0	
		BRANCH	
		PAGE	
		20 OF 110	
		SHEET	
		CURRENT DESIGN SHEET> OF <TOTAL>	



BIOS Boot Select

I/F	HDA_SDOUT	LPC_FRAME#
LPC	0	0
PCI	0	1
SPI0	1	0
SPI1	1	1

SPI0 = SPI_CS0_L, SPI1 = SPI_CS1_L
 R1961 and R2160 selects SPI0 ROM by default, LPC+ debug card pulls LPC_FRAME# high for SPI1 ROM override.
 NOTE: MCP79 does not support FWH, only LPC ROMs. So Apple designs will not use LPC for BootROM override.
 NOTE: MCP79 rev A01 does not support SPI1 option. Rev B01 will.

BUF_SIO_CLK Frequency

Frequency	HDA_SYNC
24 MHz	1
14.31818 MHz	0

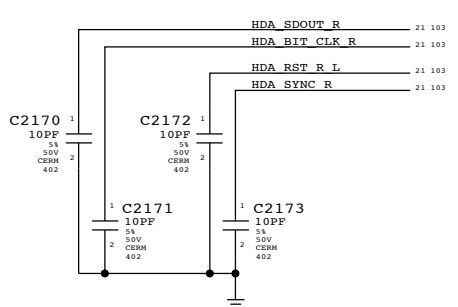
SPI Frequency Select

Frequency	SPI_DO	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

NOTE: Straps not provided on this page.

HDA Output Caps

For EMI Reduction on HDA interface



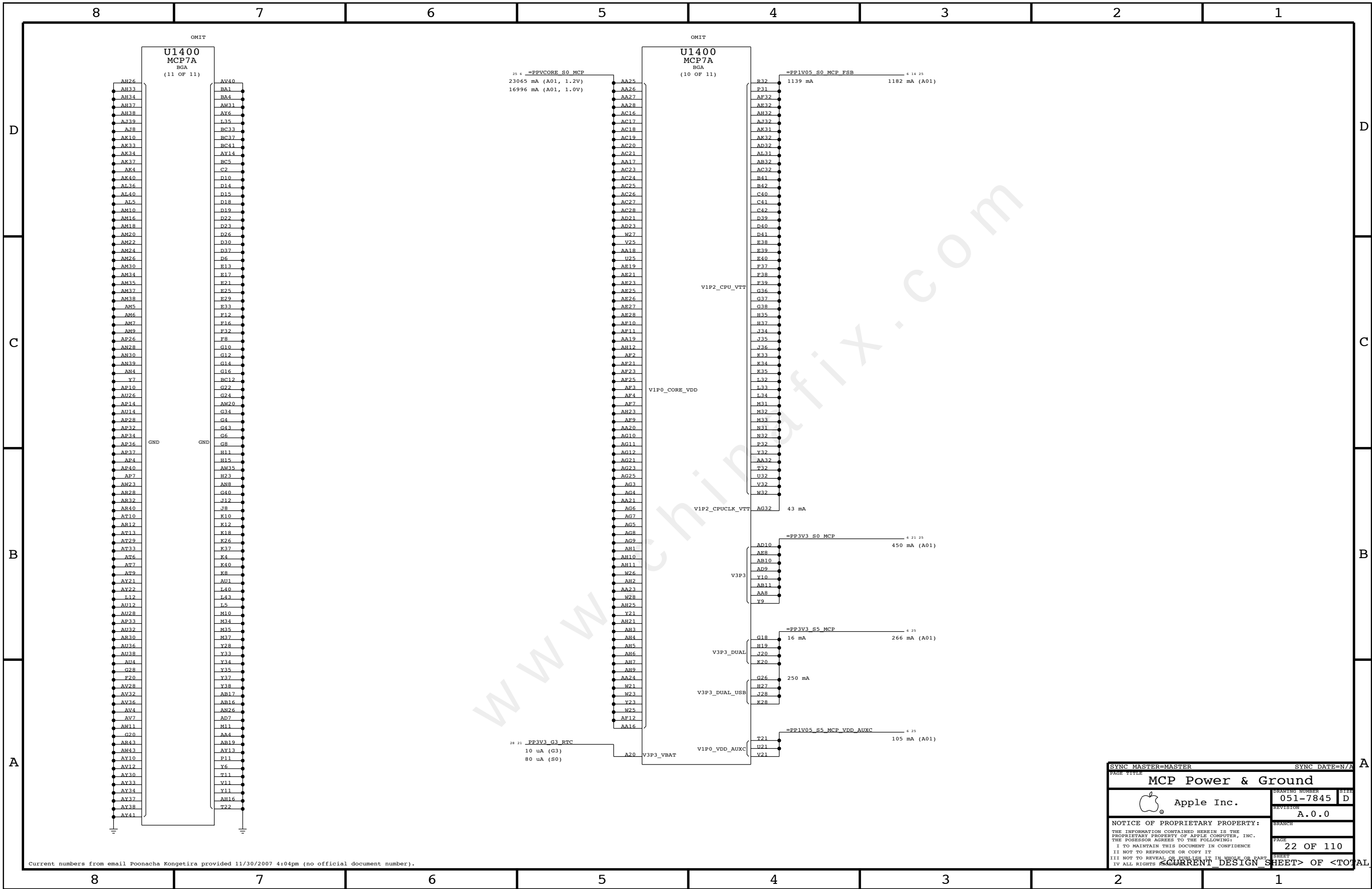
SYNC MASTER=MASTER SYNC DATE=N/A

MCP HDA & MISC

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
 THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
 I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
 II NOT TO REPRODUCE OR COPY IT
 III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
 IV ALL RIGHTS RESERVED

DRAWING NUMBER: 051-7845
 REVISION: A.0.0
 SHEET: 21 OF 110



SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
MCP Power & Ground			
 Apple Inc.		DRAWING NUMBER	SHEET
		051-7845	D
		REVISION	
		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		22 OF 110	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		DESIGN SHEET	
IV ALL RIGHTS RESERVED		<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>	



8	7	6	5	4	3	2	1
D							
C							
B							
A							
8	7	6	5	4	3	2	1

www.chinafix.com

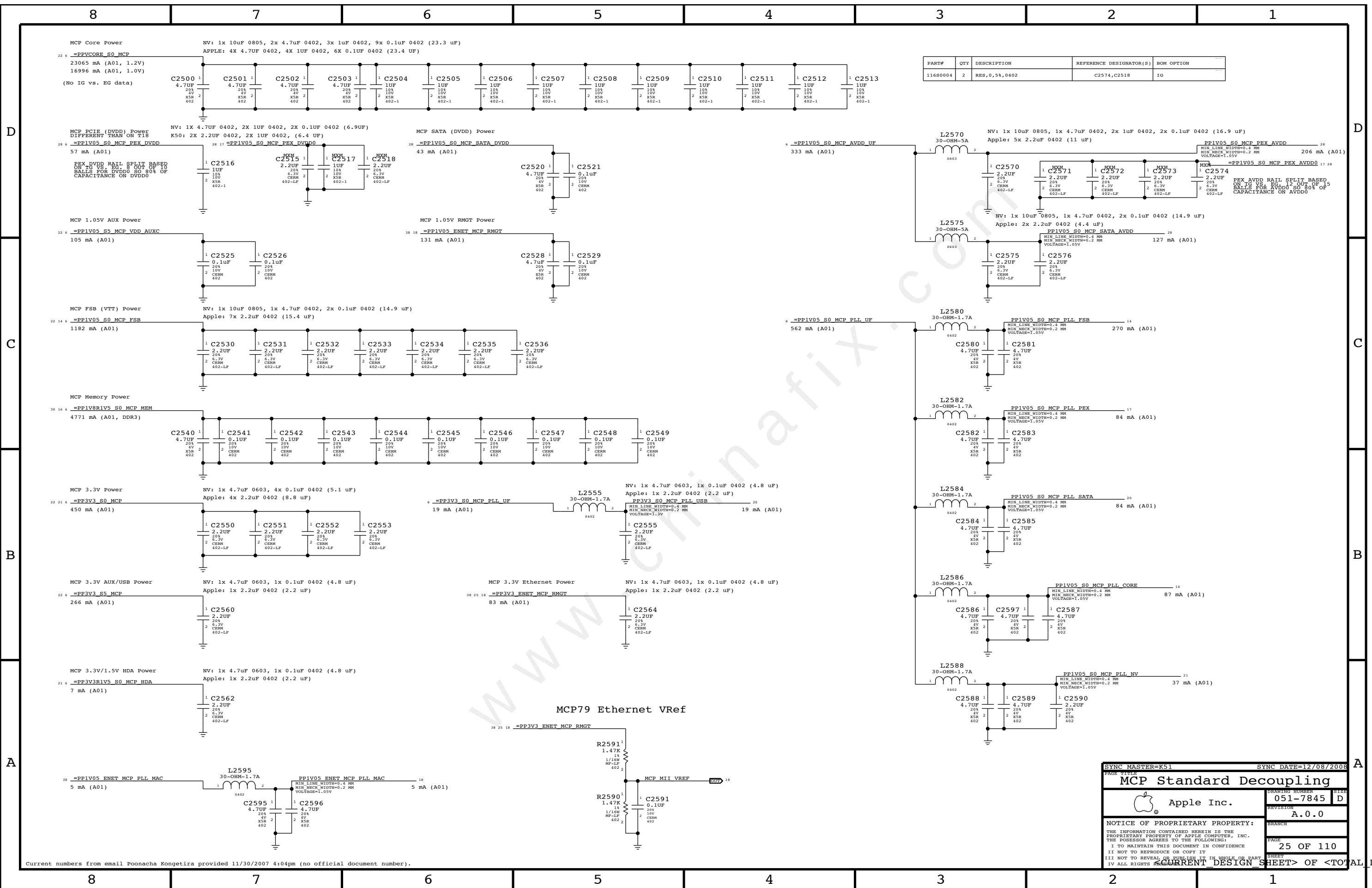
SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	BONCH
		A.0.0	
		PAGE	SHEET
		23 OF 110	
<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>			



8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BIONCH	
		PAGE	
		24 OF 110	
		SHEET	



Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
MCP Standard Decoupling			
 Apple Inc.		DRAWING NUMBER	051-7845
		D	8142
NOTICE OF PROPRIETARY PROPERTY:		REVISION	A.0.0
		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II. NOT TO REPRODUCE OR COPY IT III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV. ALL RIGHTS RESERVED		PAGE	25 OF 110
		SHEET	
SECURE CURRENT DESIGN SHEET		OF <TOT>	

CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>

D

C

B

A

D

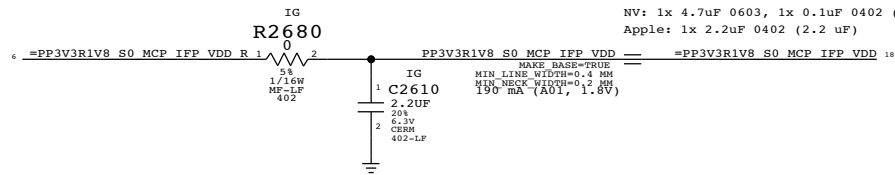
C

B

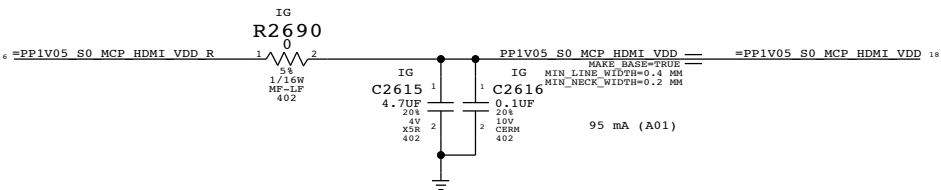
A

WF: Checklist says 0-ohm resistor placeholder for ferrite bead.

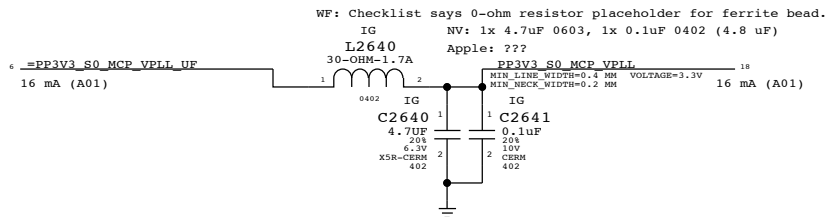
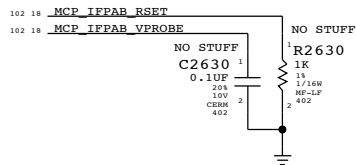
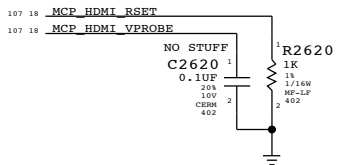
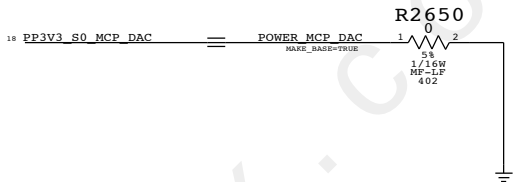
NV: 1x 4.7uF 0603, 1x 0.1uF 0402 (4.8 uF)
Apple: 1x 2.2uF 0402 (2.2 uF)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
116S0004	1	RES,0,5%,402	C2610		MXM



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
116S0004	1	RES,0,5%,402	C2616		MXM



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
116S0004	1	RES,0,5%,402	C2641		MXM

SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
MCP Graphics Support			
Apple Inc.		DRAWING NUMBER	051-7845
		REVISION	A.0.0
NOTICE OF PROPRIETARY PROPERTY:		PAGE	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		26 OF 110	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	
II NOT TO REPRODUCE OR COPY IT		26 OF 110	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		DESIGN SHEETS	
IV ALL RIGHTS RESERVED		<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>	

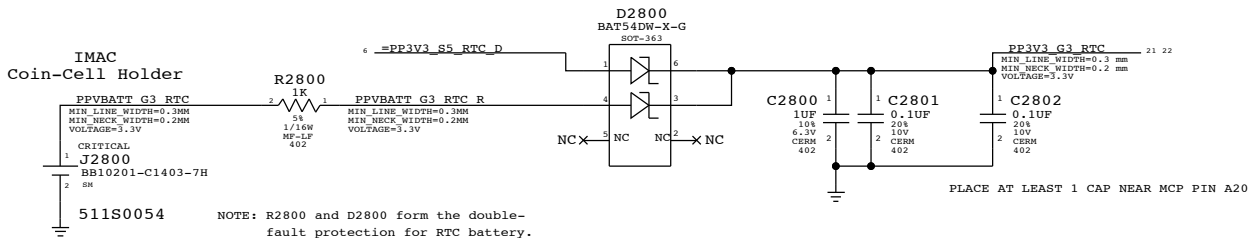


8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

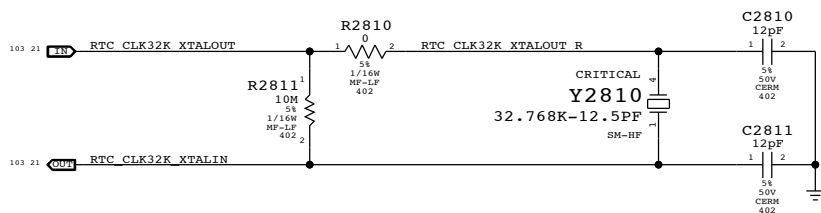
www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		27 OF 110	
		SHEET	

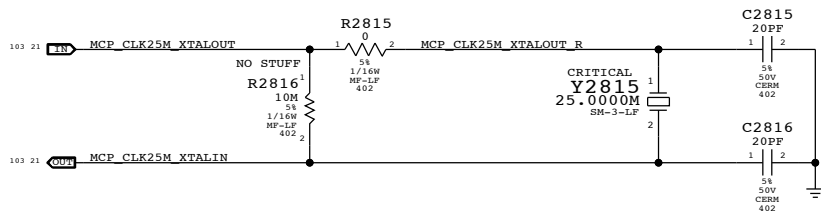
RTC Power Sources



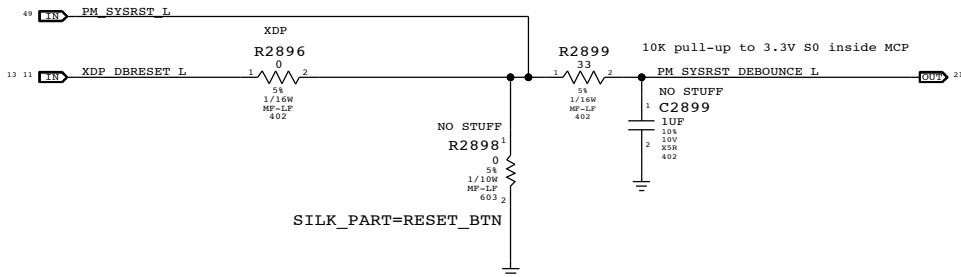
RTC Crystal



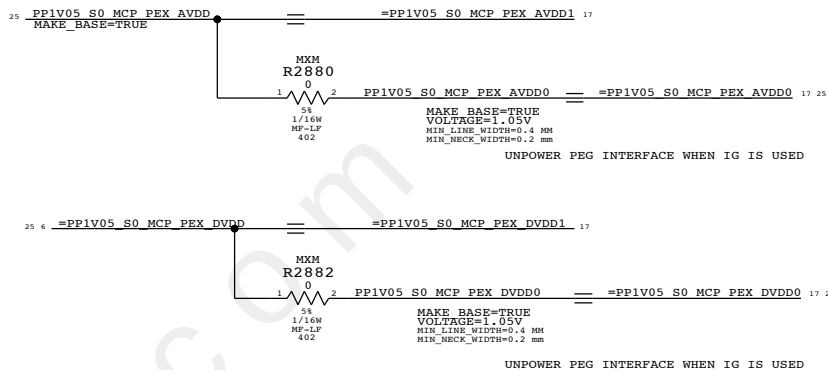
MCP 25MHz Crystal



Reset Button

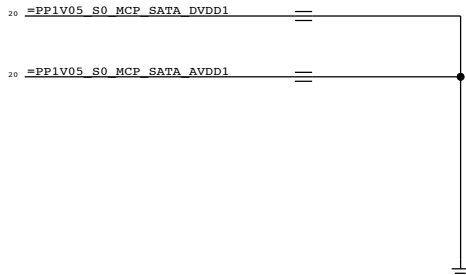


PEG POWER ALIAS/OPTION TO GND UNUSED POWER PIN



SATA ALIAS/GROUNDING UNUSED DVDD1 AND AVDD1

AVDD IS FILTERED ON P25
PP1V05 S0 MCP SATA AVDD MAKE BASE=TRUE
PP1V05 S0 MCP SATA DVDD MAKE BASE=TRUE
DVDD DOES NOT NEED FILTER





Page Notes

Power aliases required by this page:

```
- =PP3V3_S3_VREFMRGN
- =PP3V3_S5_VREFMRGN
- =PPVTT_S3_DDR_BUF
```

Signal aliases required by this page:

```
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA
```

BOM options provided by this page:

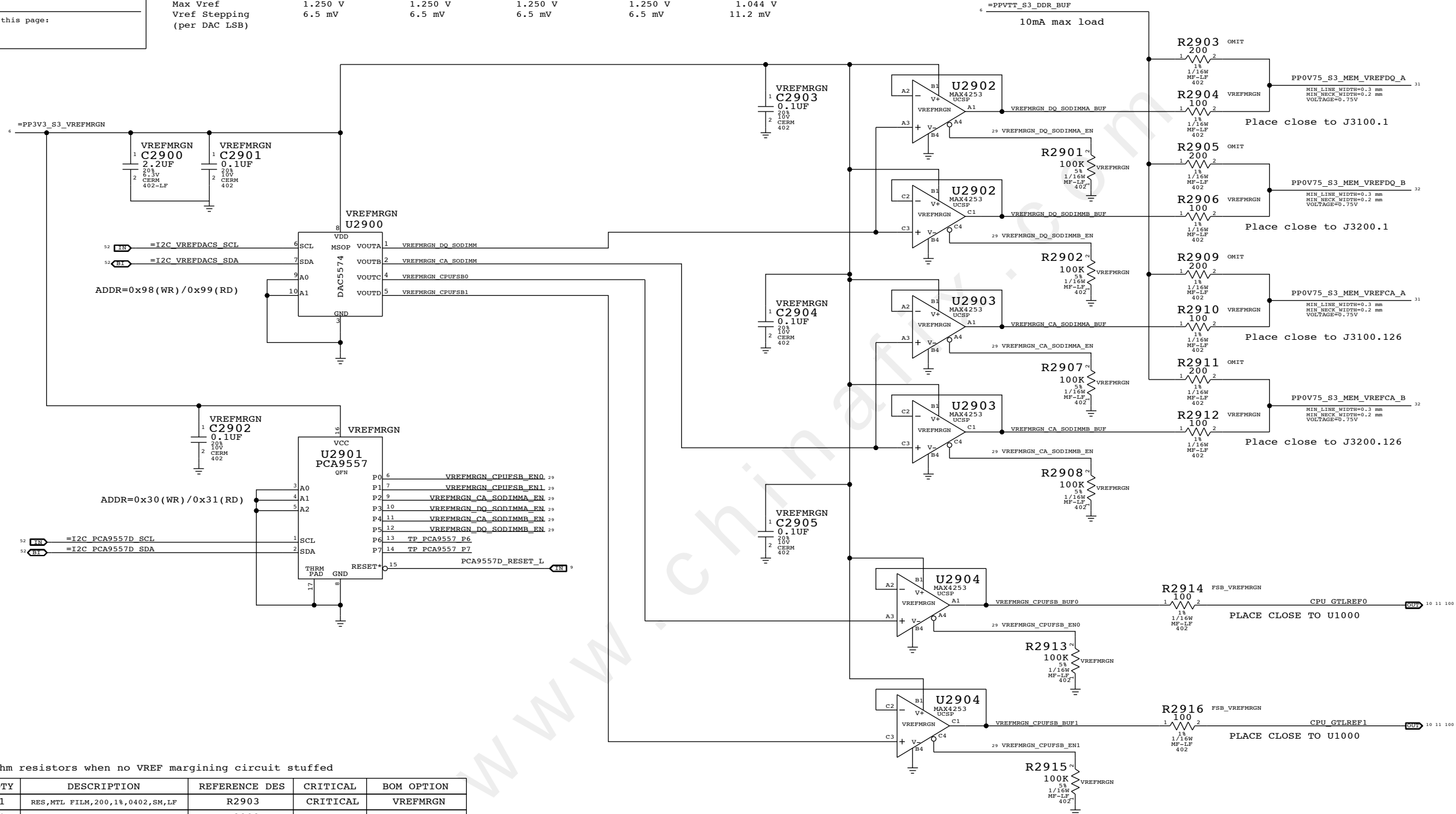
VREFMRGN

PRODUCTION

DAC channel
Min DAC code
Max DAC code
Max sink I
Max source I
Nominal Vref
Min Vref
Max Vref
Vref Stepping
(per DAC LSB)

MEM A VREF DQMEM A VREF CAMEM B VREF DMEM B VREFCPU FSB V

SO-DIMM A and SO-DIMM B Vref settings should be margined separately (i.e. not simultaneously) due to current limitation of TPS51116 regulator.

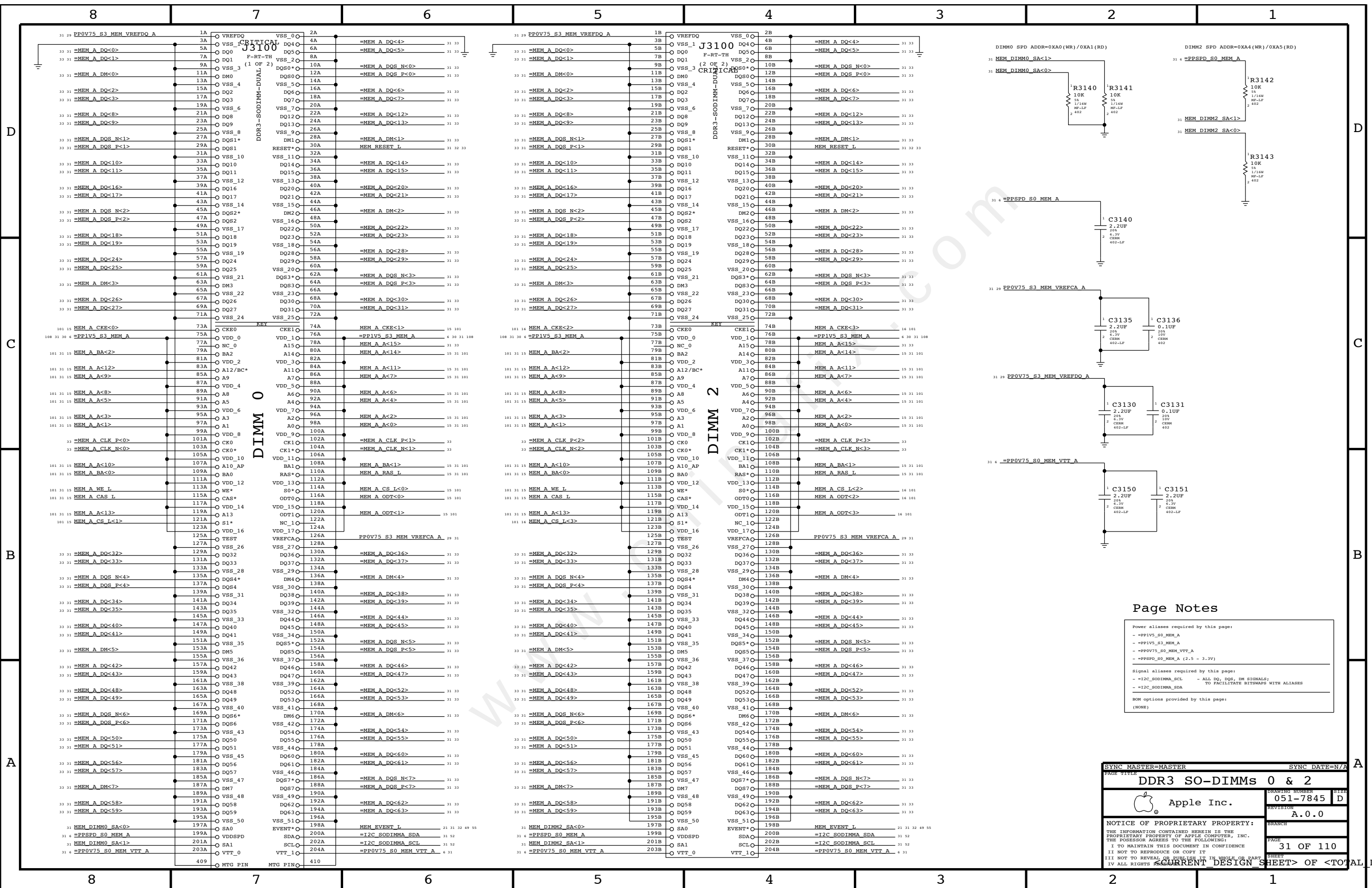


Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0149	1	RES,MTL FILM,200,18,0402,SM,LF	R2903	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,58,0402,SM,LF	R2903	CRITICAL	PRODUCTION
114S0149	1	RES,MTL FILM,200,18,0402,SM,LF	R2905	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,58,0402,SM,LF	R2905	CRITICAL	PRODUCTION
114S0149	1	RES,MTL FILM,200,18,0402,SM,LF	R2909	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,58,0402,SM,LF	R2909	CRITICAL	PRODUCTION
114S0149	1	RES,MTL FILM,200,18,0402,SM,LF	R2911	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,58,0402,SM,LF	R2911	CRITICAL	PRODUCTION

SYMC MASTER-MASTER		SYMC DATE-MASTER	
PAGE 114			
FSB/DDR3 Vref Margining			
	Apple Inc.		DRAWING NUMBER 051-7845
			SIZE D
		REVISION A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS ARE RESERVED			
CURRENT DESIGN SHEET		PAGE 29 OF 110	
		SHEET OF 40	

DESIGN SHEETS>



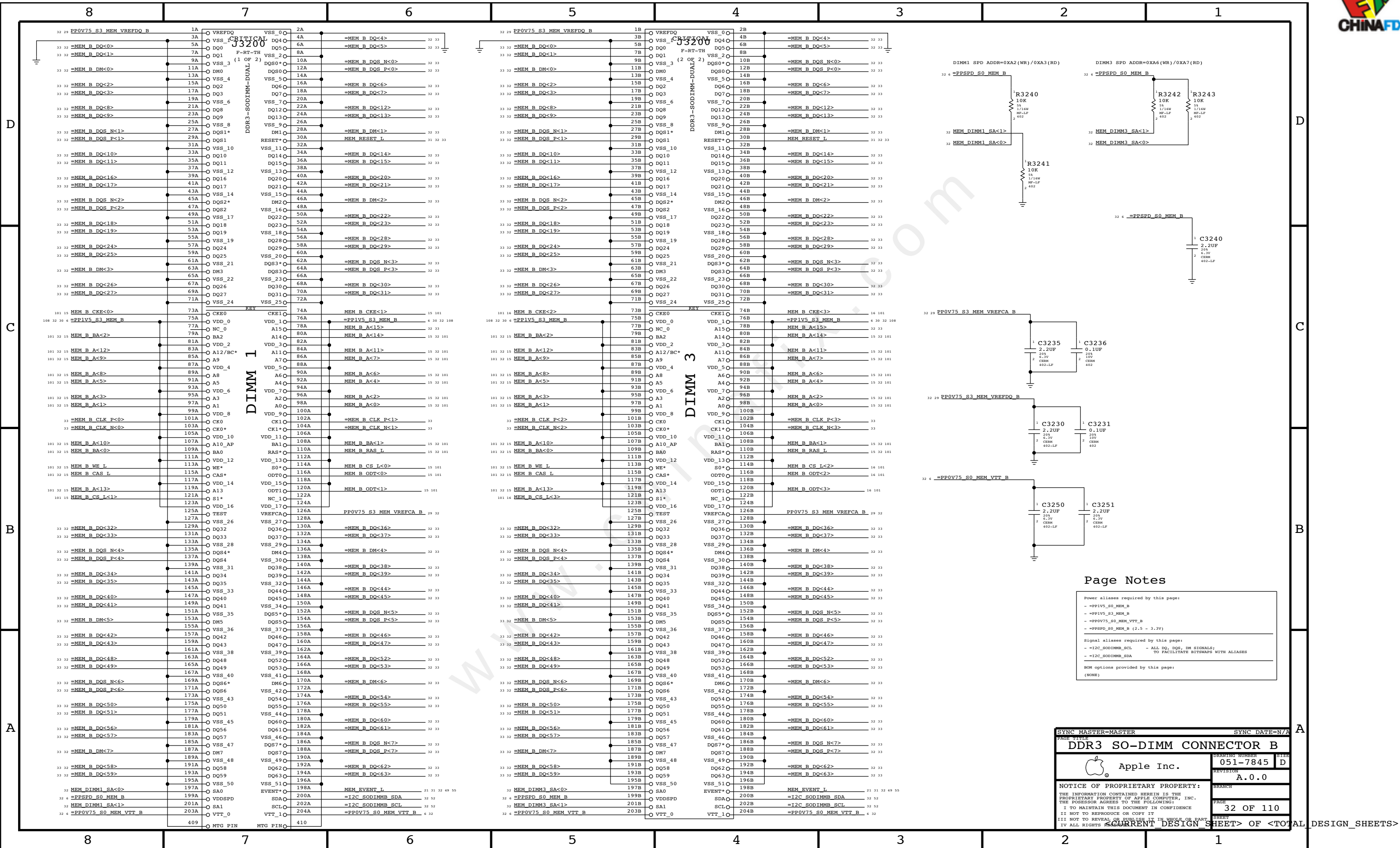
Page Notes

Power aliases required by this page:
- PP1V5_S0_MEM_A
- PP1V5_S3_MEM_A
- PP0V75_S0_MEM_VTT_A
- PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:
- I2C_SODIMMA_SCL - ALL DQ, DQS, DM SIGNALS;
TO FACILITATE BITSTREAMS WITH ALIASES
- I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)

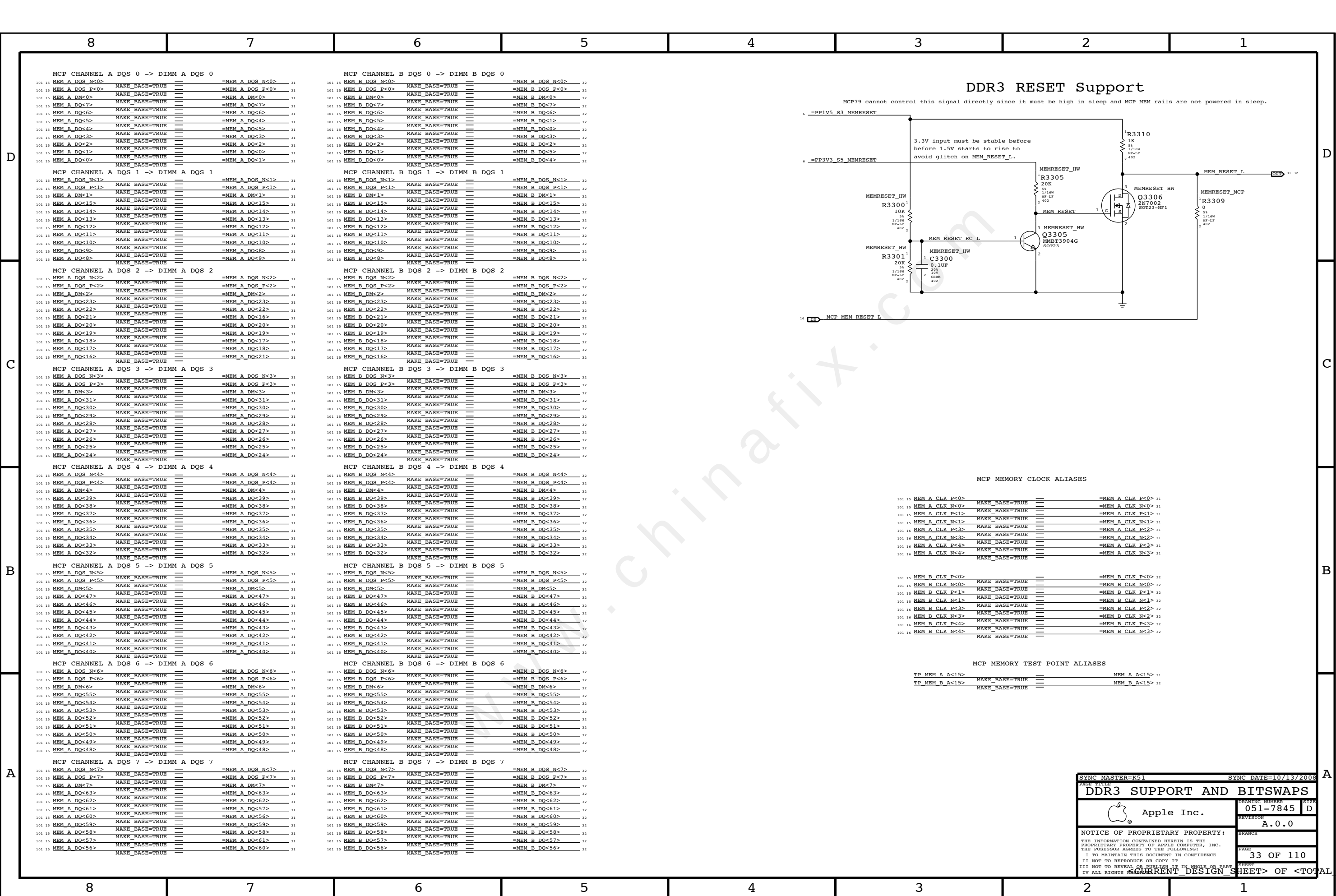
SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE		DRAWING NUMBER	
DDR3 SO-DIMMs 0 & 2		051-7845	
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I DO NOT REPRODUCE OR COPY IT I DO NOT REVEAL OR PUBLISH IT IN WHOLE OR PART I DO NOT REUSE IT IN ANY MANNER I DO NOT REUSE IT IN ANY MANNER		A.0.0	
PAGE		31 OF 110	
SHEET		SHEET	
DESIGN SHEET		DESIGN SHEET	

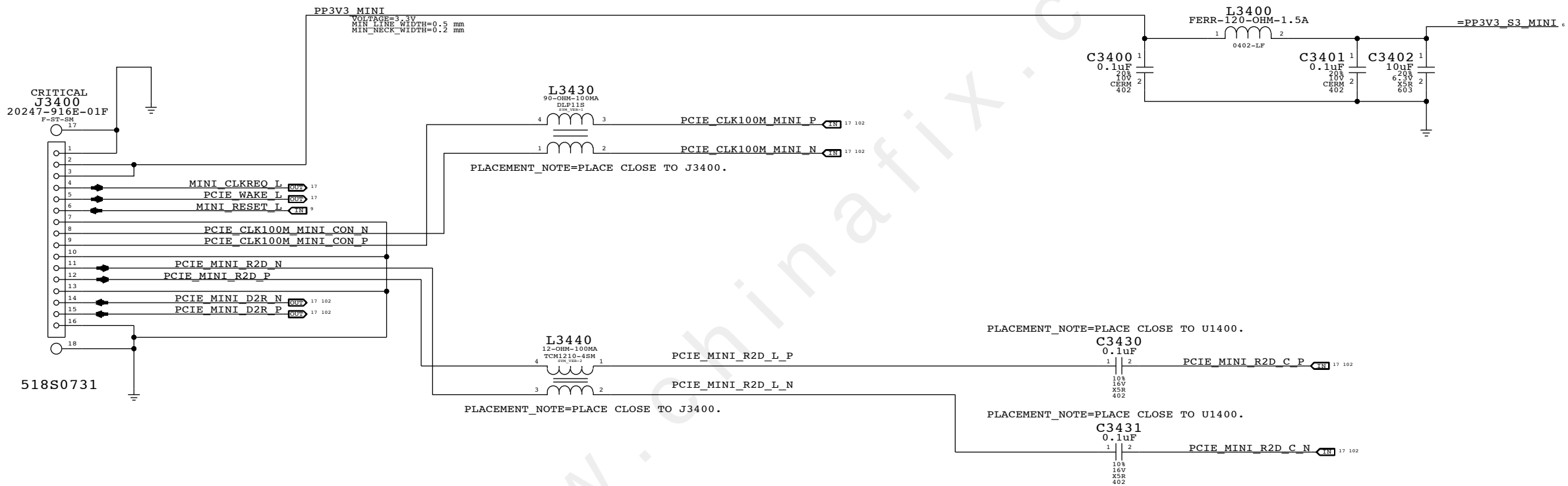


Page Notes

- Power aliases required by this page:
- =PP1V5_S0_MEM_B
 - =PP1V5_S3_MEM_B
 - =PP0V75_S0_MEM_VTT_B
 - =PPSPD_S0_MEM_B (2.5 - 3.3V)
- Signal aliases required by this page:
- =I2C_SODIMMB_SCL - ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSTREAMS WITH ALIASES
 - =I2C_SODIMMB_SDA
- BOM options provided by this page:
- (NONE)

SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
DDR3 SO-DIMM CONNECTOR B			
Apple Inc.		DRAWING NUMBER	051-7845
		REVISION	A.0.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I DO NOT REPRODUCE OR COPY IT I DO NOT REVEAL OR PUBLISH IT IN WHOLE OR PART I DO NOT REUSE OR REPRODUCE IT IN ANY MANNER I DO NOT REUSE OR REPRODUCE IT IN ANY MANNER I DO NOT REUSE OR REPRODUCE IT IN ANY MANNER		PAGE	32 OF 110
I DO NOT REPRODUCE OR COPY IT		SHEET	
I DO NOT REVEAL OR PUBLISH IT IN WHOLE OR PART			
I DO NOT REUSE OR REPRODUCE IT IN ANY MANNER			







8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

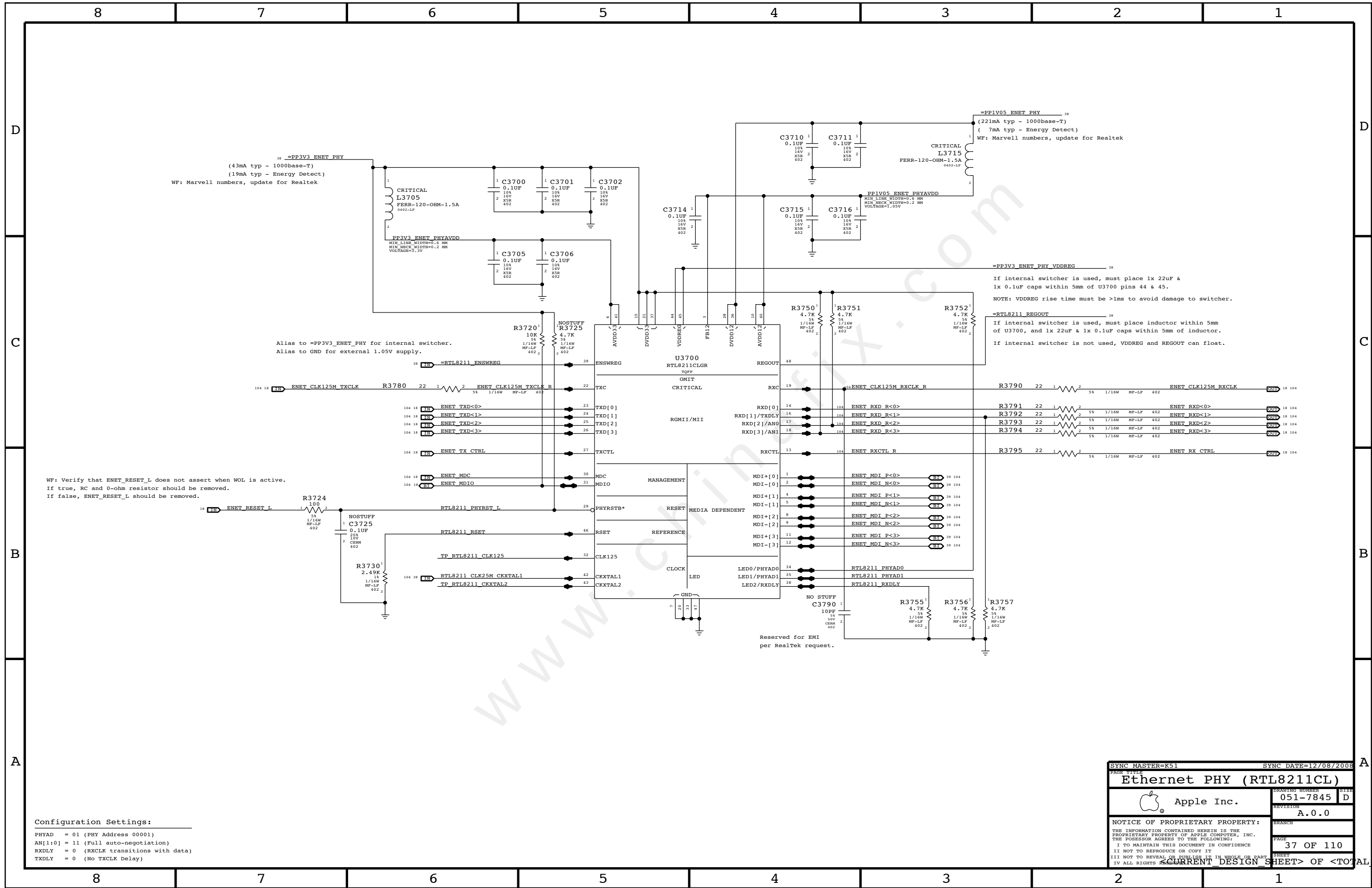
SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		35 OF 110	
		SHEET	



8	7	6	5	4	3	2	1
D							
C							
B							
A							
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		36 OF 110	
		SHEET	



D

C

B

A

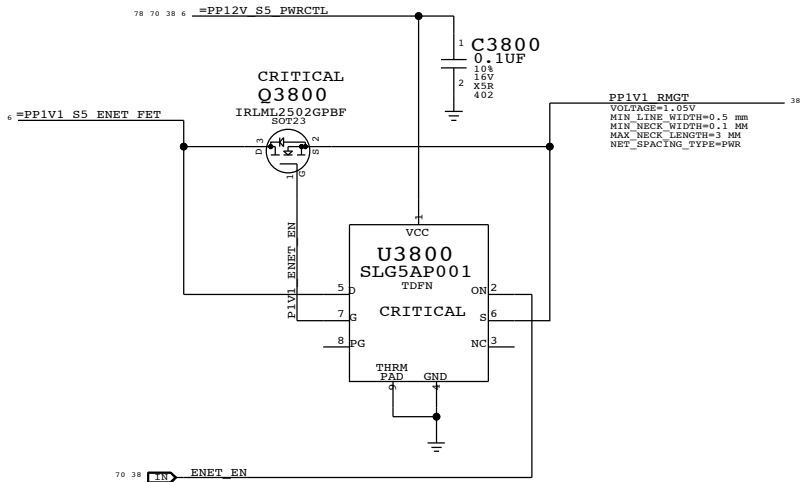
D

C

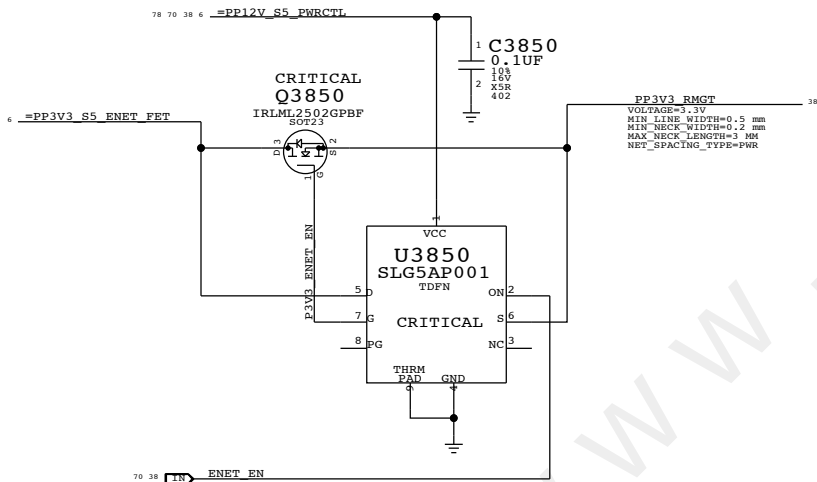
B

A

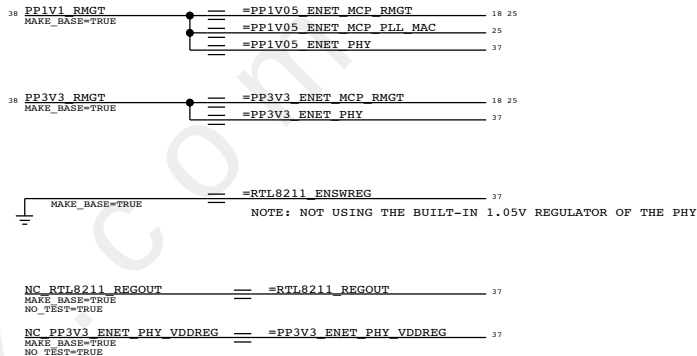
1.1V ENET FET



3.3V ENET FET

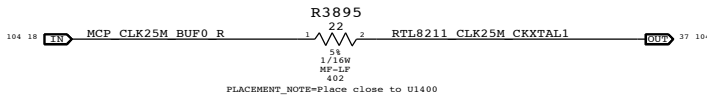


ENET ALIASES

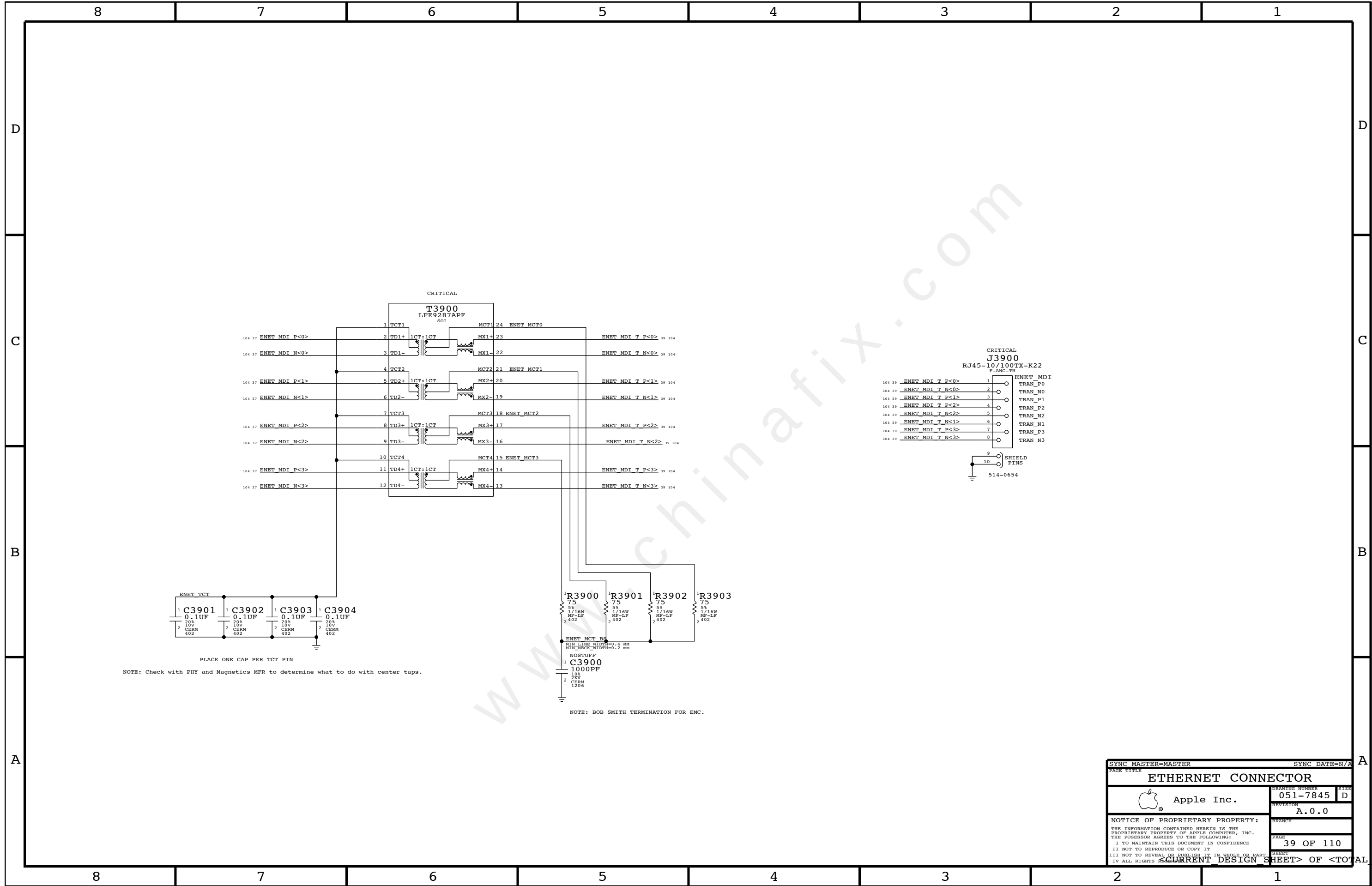


RTL8211 25MHz Clock

NOTE: MCP79 can provide 25MHz clock, but clock runs whenever RMGT rails are powered. Designs must ensure PHY is powered whenever RMGT rails are, or use separate crystal.



SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
Ethernet Support			
	Apple Inc.	DRAWING NUMBER	SIZE
		051-7845	D
		REVISION	A.0.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		38	OF 110
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		DESIGN SHEET> OF <TOTAL DESIGN SHEETS>	
IV ALL RIGHTS RESERVED			

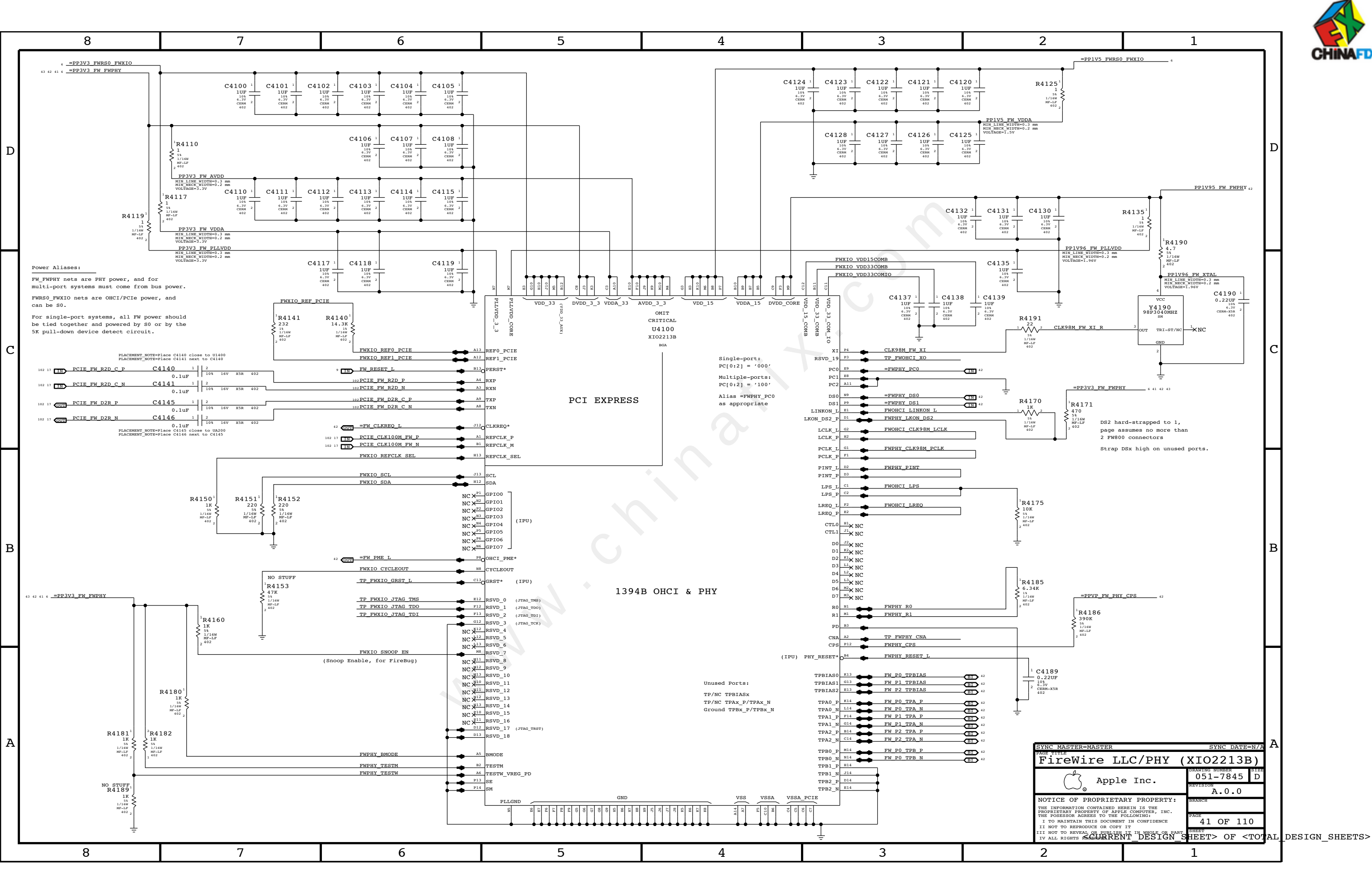




8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		40 OF 110	
		SHEET	



D

C

B

A

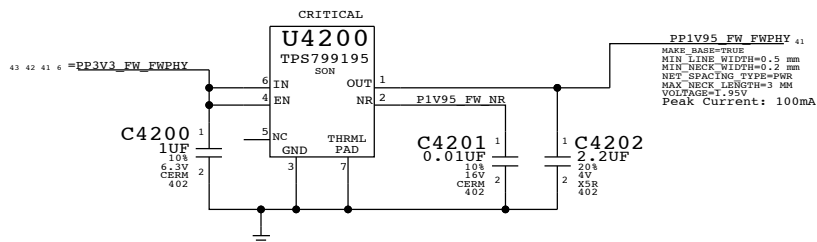
D

C

B

A

1394 PHY 1.95V SUPPLY

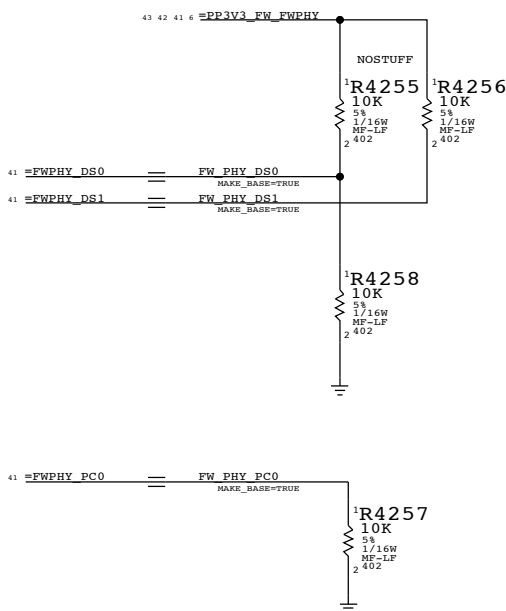


FireWire Aliases For Connectivity

```

41=FW_CLKREQ_L == FW_CLKREQ_L 17
41=FW_PME_L == FW_PME_L 19
41=PPVP_FW_PHY_CPS == PPVP_FW_PHY_CPS 43
  
```

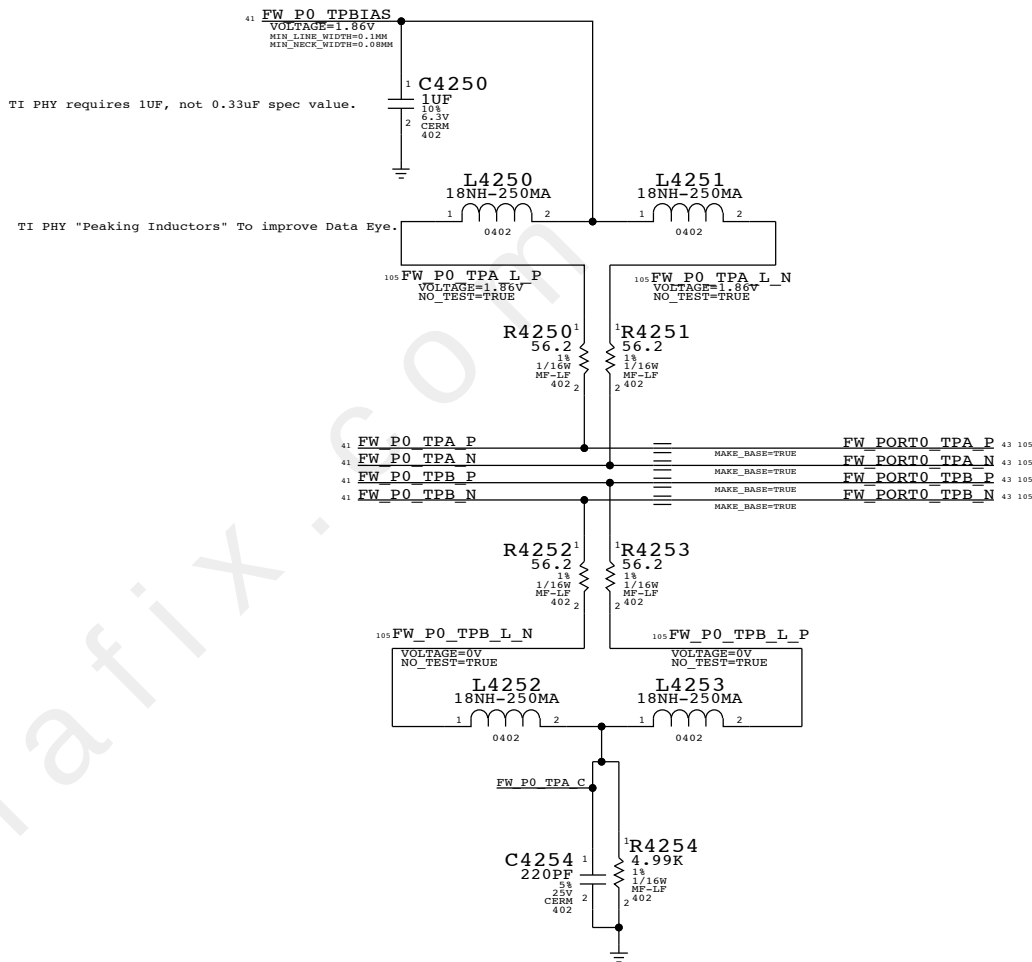
1394 PHY STRAPPING OPTIONS



THERE ARE THREE FIREWIRE PORTS, BUT ONLY ONE IS USED.NO STUFF MEANS THAT IT IS IN BILINGUL MODE PULL-UPS ASSERT/ENABLE DATA STROBE ONLY MODE.

iMacs are now one port only and have Power Code "000"

Termination Place close to FireWire PHY



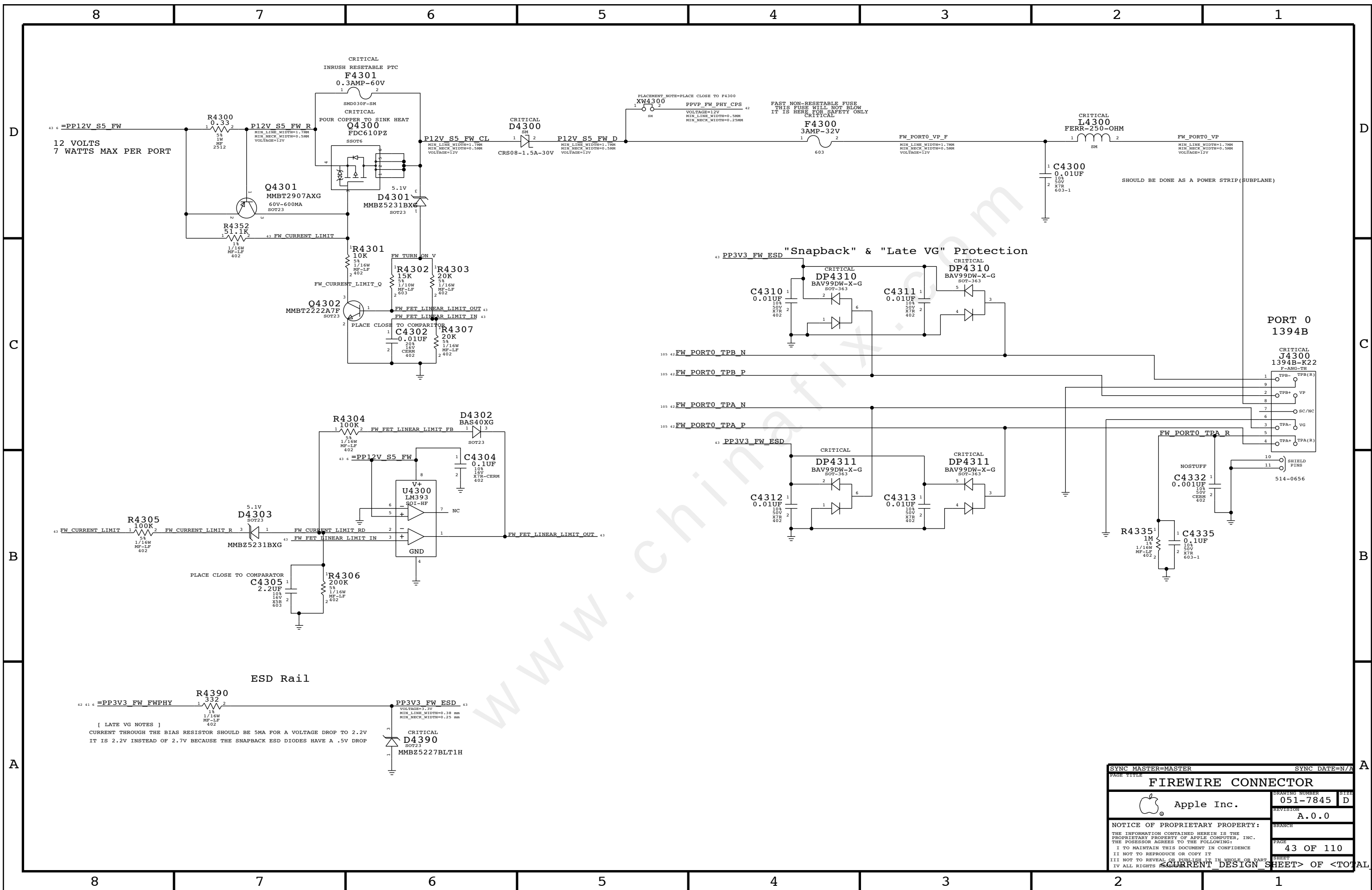
2ND & 3RD TPA/TPB PAIR UNUSED

```

41 FW_P1_TPBIA == NC_FW_PORT1_TPBIA
41 FW_P1_TPA_P == NC_FW_PORT1_TPA_P
41 FW_P1_TPA_N == NC_FW_PORT1_TPA_N

41 FW_P2_TPBIA == NC_FW_PORT2_TPBIA
41 FW_P2_TPA_P == NC_FW_PORT2_TPA_P
41 FW_P2_TPA_N == NC_FW_PORT2_TPA_N
  
```

SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE		FW: 1394B MISC	
DRAWING NUMBER		051-7845	D
REVISION		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
PAGE		42 OF 110	
SHEET			
CURRENT DESIGN SHEET		OF TOTAL DESIGN SHEETS	





8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

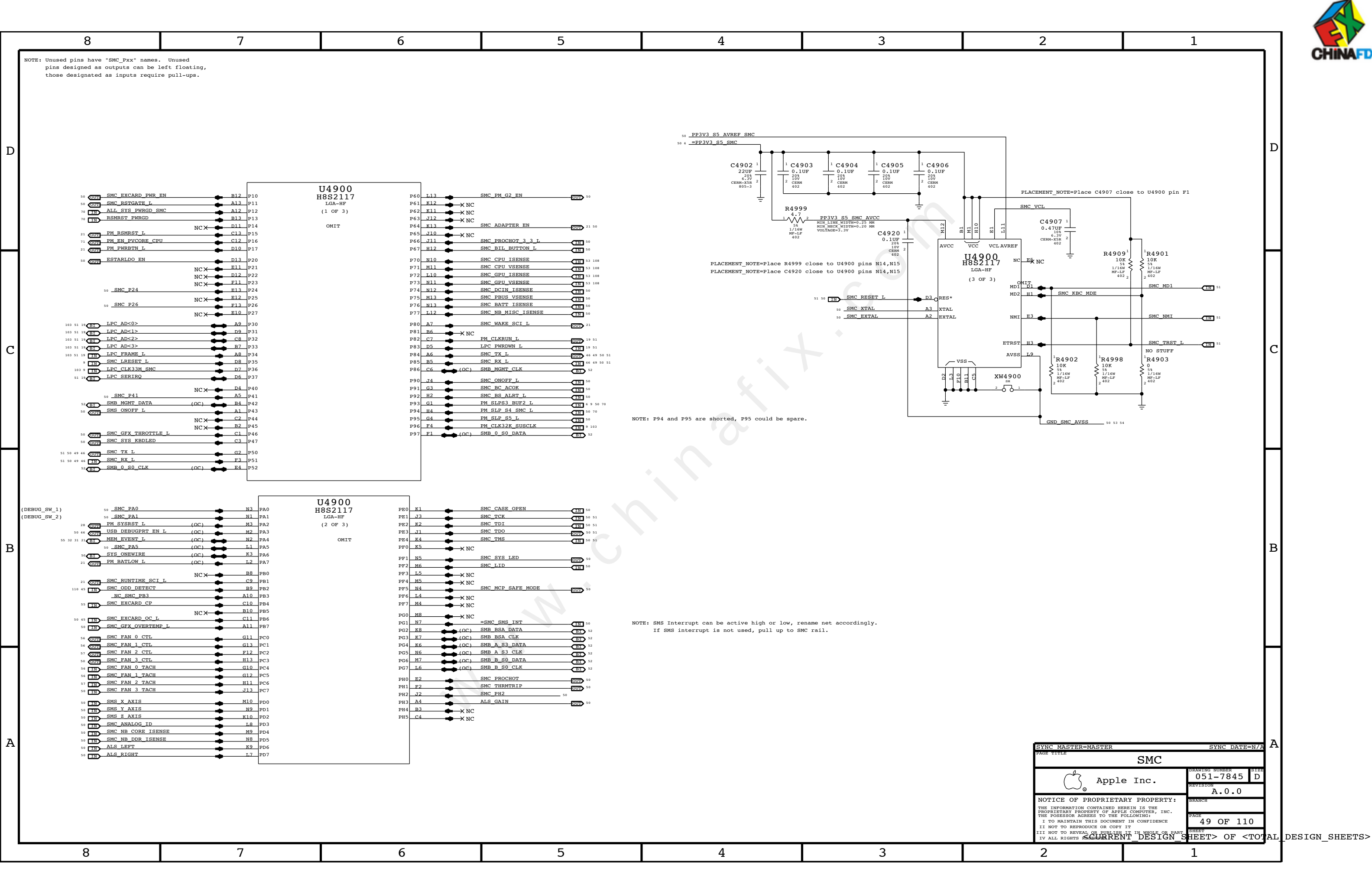
SYNC MASTER=K51		SYNC DATE=10/13/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		44 OF 110	
		SHEET	

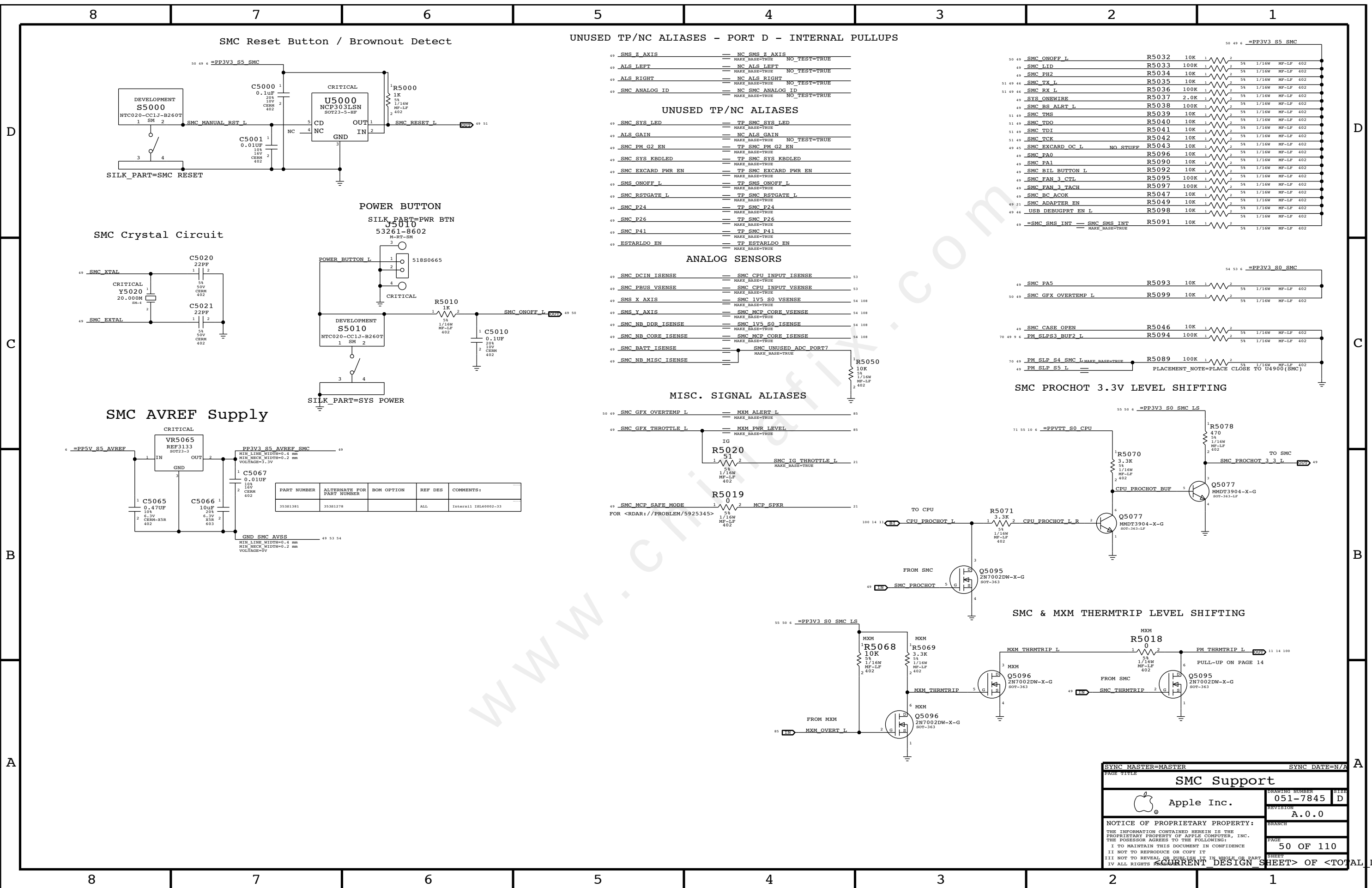


8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=10/13/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	051-7845
		SIZE	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	A.0.0
		BONCH	
		PAGE	48 OF 110
		SHEET	





UNUSED TP/NC ALIASES - PORT D - INTERNAL PULLUPS

49 SMC_Z_AXIS == NC_SMC_Z_AXIS
MAKE_BASE=TRUE NO_TEST=TRUE
49 ALS_LEFT == NC_ALS_LEFT
MAKE_BASE=TRUE NO_TEST=TRUE
49 ALS_RIGHT == NC_ALS_RIGHT
MAKE_BASE=TRUE NO_TEST=TRUE
49 SMC_ANALOG_ID == NC_SMC_ANALOG_ID
MAKE_BASE=TRUE NO_TEST=TRUE

UNUSED TP/NC ALIASES

49 SMC_SYS_LED == TP_SMC_SYS_LED
MAKE_BASE=TRUE
49 ALS_GAIN == NC_ALS_GAIN
MAKE_BASE=TRUE NO_TEST=TRUE
49 SMC_PM_G2_EN == TP_SMC_PM_G2_EN
MAKE_BASE=TRUE
49 SMC_SYS_KBDLED == TP_SMC_SYS_KBDLED
MAKE_BASE=TRUE
49 SMC_EXCARD_PWR_EN == TP_SMC_EXCARD_PWR_EN
MAKE_BASE=TRUE
49 SMC_ONOFF_L == TP_SMC_ONOFF_L
MAKE_BASE=TRUE
49 SMC_RSTGATE_L == TP_SMC_RSTGATE_L
MAKE_BASE=TRUE
49 SMC_P24 == TP_SMC_P24
MAKE_BASE=TRUE
49 SMC_P26 == TP_SMC_P26
MAKE_BASE=TRUE
49 SMC_P41 == TP_SMC_P41
MAKE_BASE=TRUE
49 ESTARLDO_EN == TP_ESTARLDO_EN
MAKE_BASE=TRUE

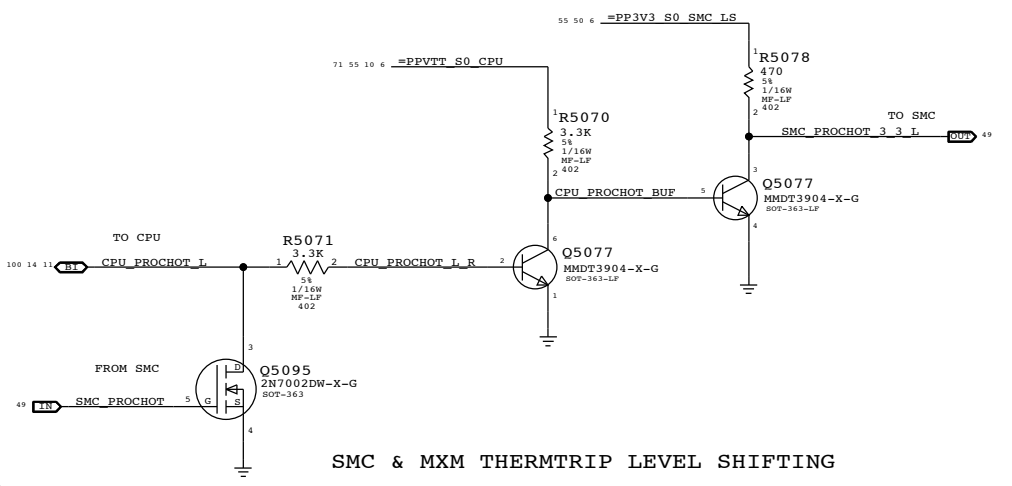
ANALOG SENSORS

49 SMC_DCIN_ISENSE == SMC_CPU_INPUT_ISENSE
MAKE_BASE=TRUE
49 SMC_PBUS_VSENSE == SMC_CPU_INPUT_VSENSE
MAKE_BASE=TRUE
49 SMC_X_AXIS == SMC_IVS_S0_VSENSE
MAKE_BASE=TRUE
49 SMC_Y_AXIS == SMC_MCP_CORE_VSENSE
MAKE_BASE=TRUE
49 SMC_NB_DDR_ISENSE == SMC_IVS_S0_ISENSE
MAKE_BASE=TRUE
49 SMC_NB_CORE_ISENSE == SMC_MCP_CORE_ISENSE
MAKE_BASE=TRUE
49 SMC_BATT_ISENSE == SMC_UNUSED_ADC_PORT7
MAKE_BASE=TRUE
49 SMC_NB_MISC_ISENSE == SMC_UNUSED_ADC_PORT7
MAKE_BASE=TRUE

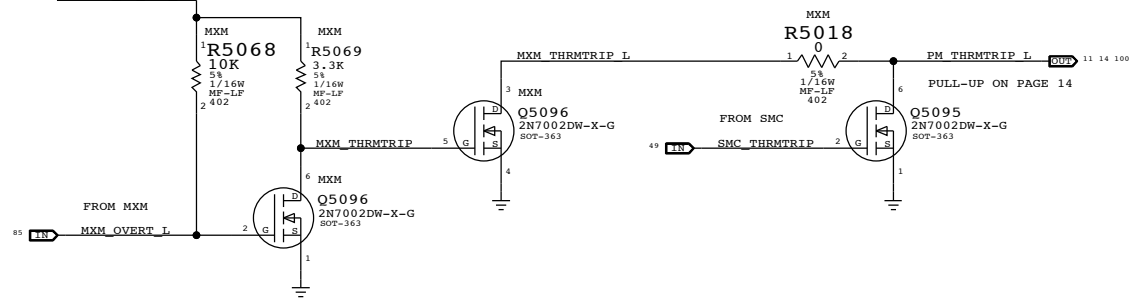
MISC. SIGNAL ALIASES

49 SMC GFX_OVERTEMP_L == MXM_ALERT_L
MAKE_BASE=TRUE
49 SMC GFX_THROTTLE_L == MXM_PWR_LEVEL
MAKE_BASE=TRUE
49 SMC_IG_THROTTLE_L == SMC_IG_THROTTLE_L
MAKE_BASE=TRUE
49 SMC_MCP_SAFE_MODE == MCP_SPKR
FOR <RDAR://PROBLEM/5925345>

SMC PROCHOT 3.3V LEVEL SHIFTING



SMC & MXM THERMTRIP LEVEL SHIFTING



SYNC MASTER=MASTER

SYNC DATE=N/A

Apple Inc.

051-7845

A.0.0

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

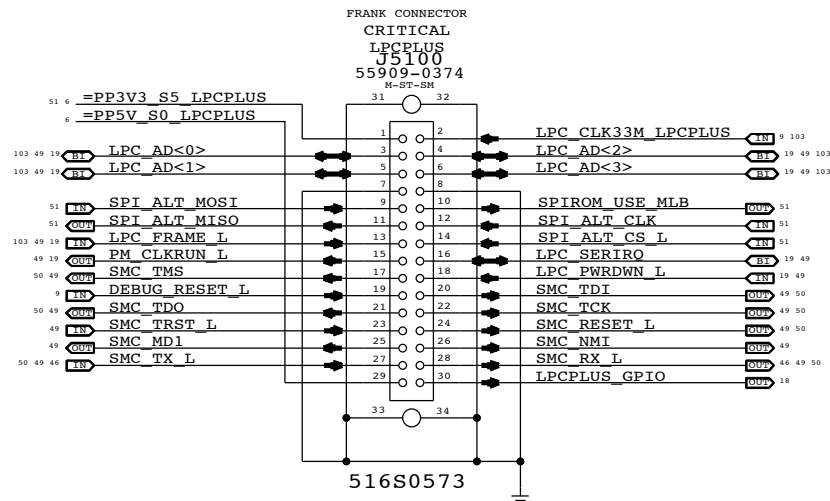
50 OF 110

SHEET

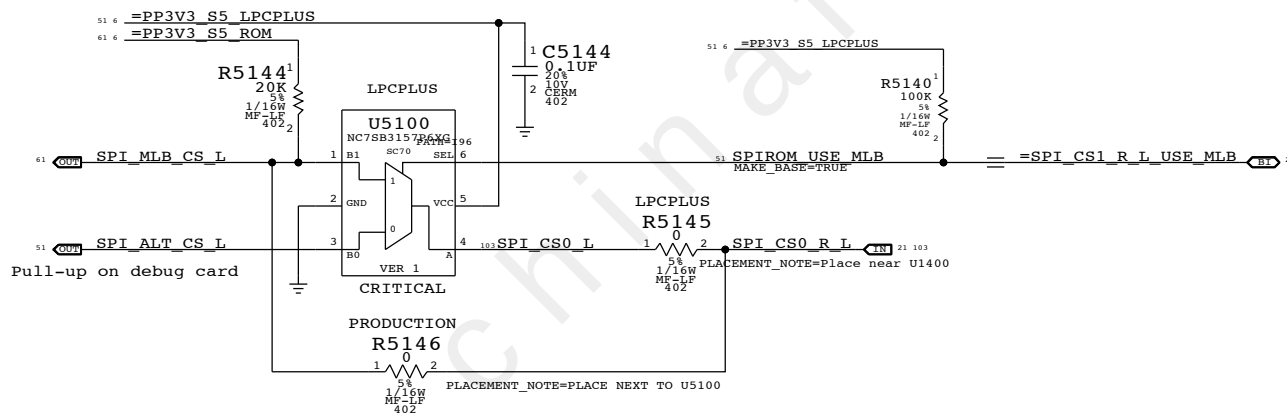
<CURRENT DESIGN SHEET>

OF <TOTAL DESIGN SHEETS>

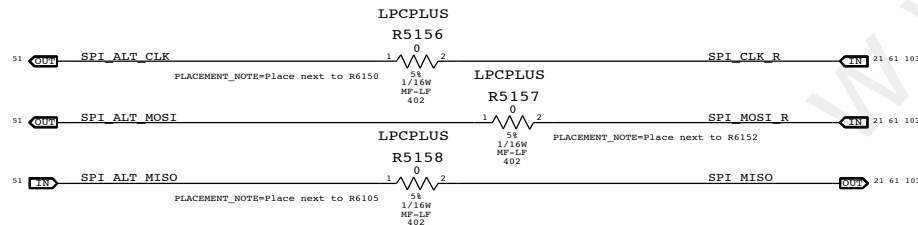
LPC+SPI Connector



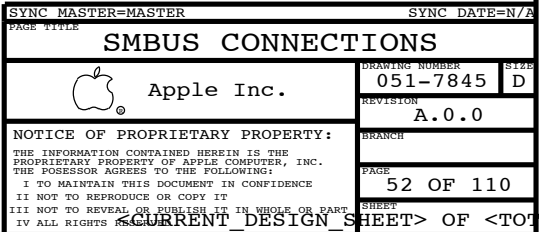
Alternate SPI ROM Support

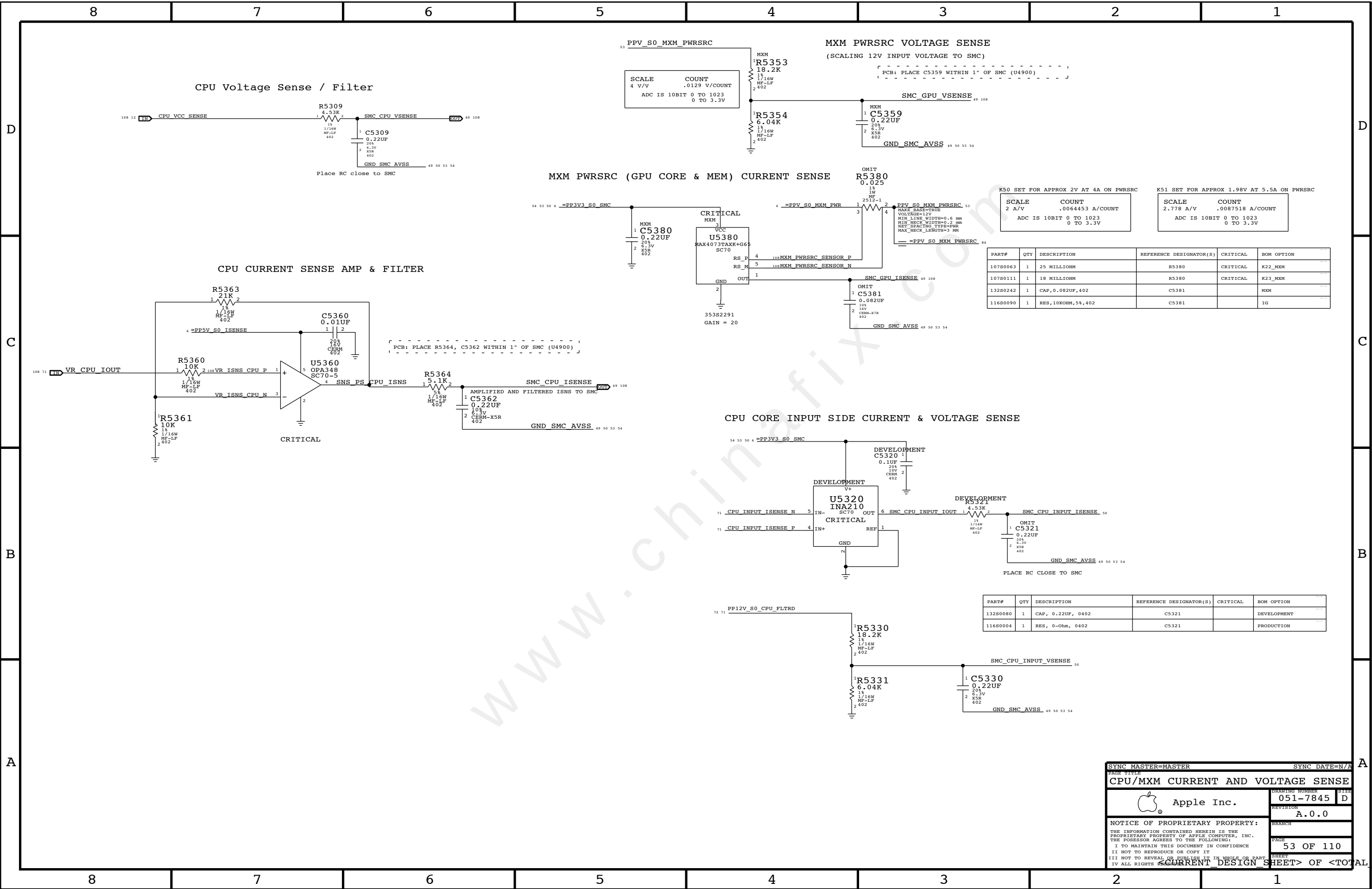


SPI Bus Series Resistance Option



SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE LPC+SPI Debug Connector			
DRAWING NUMBER 051-7845		SIZE D	
REVISION A.0.0		BRANCH	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			
PAGE 51 OF 110		SHEET	
CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>			





D

C

B

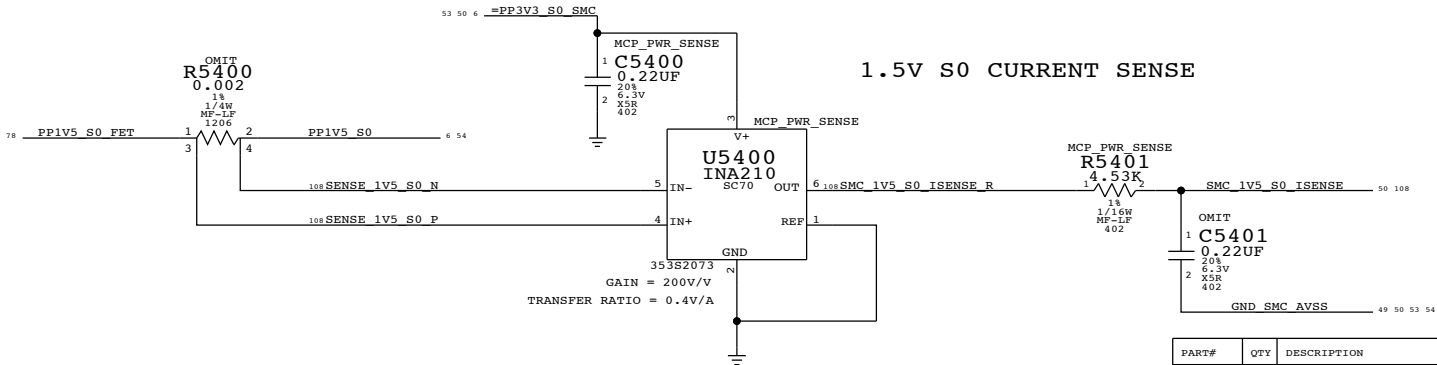
A

D

C

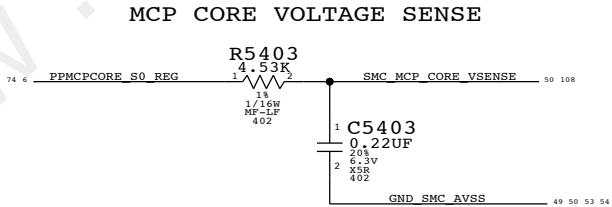
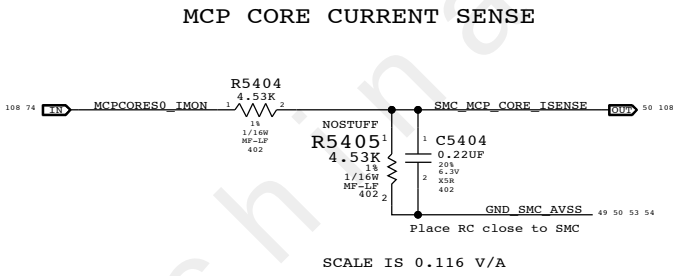
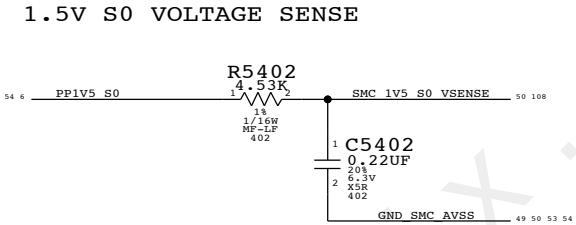
B

A



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
104S0018	1	RES,2 MILLIOHM,1206	R5400	CRITICAL	MCP_PWR_SENSE
101S0414	1	RES,0 OHM,1206,20MILLIOHM MAX	R5400	CRITICAL	PRODUCTION

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
132S0080	1	CAP, 0.22UF, 0402	C5401		MCP_PWR_SENSE
116S0004	1	RES, 0 OHM, 0402	C5401		PRODUCTION



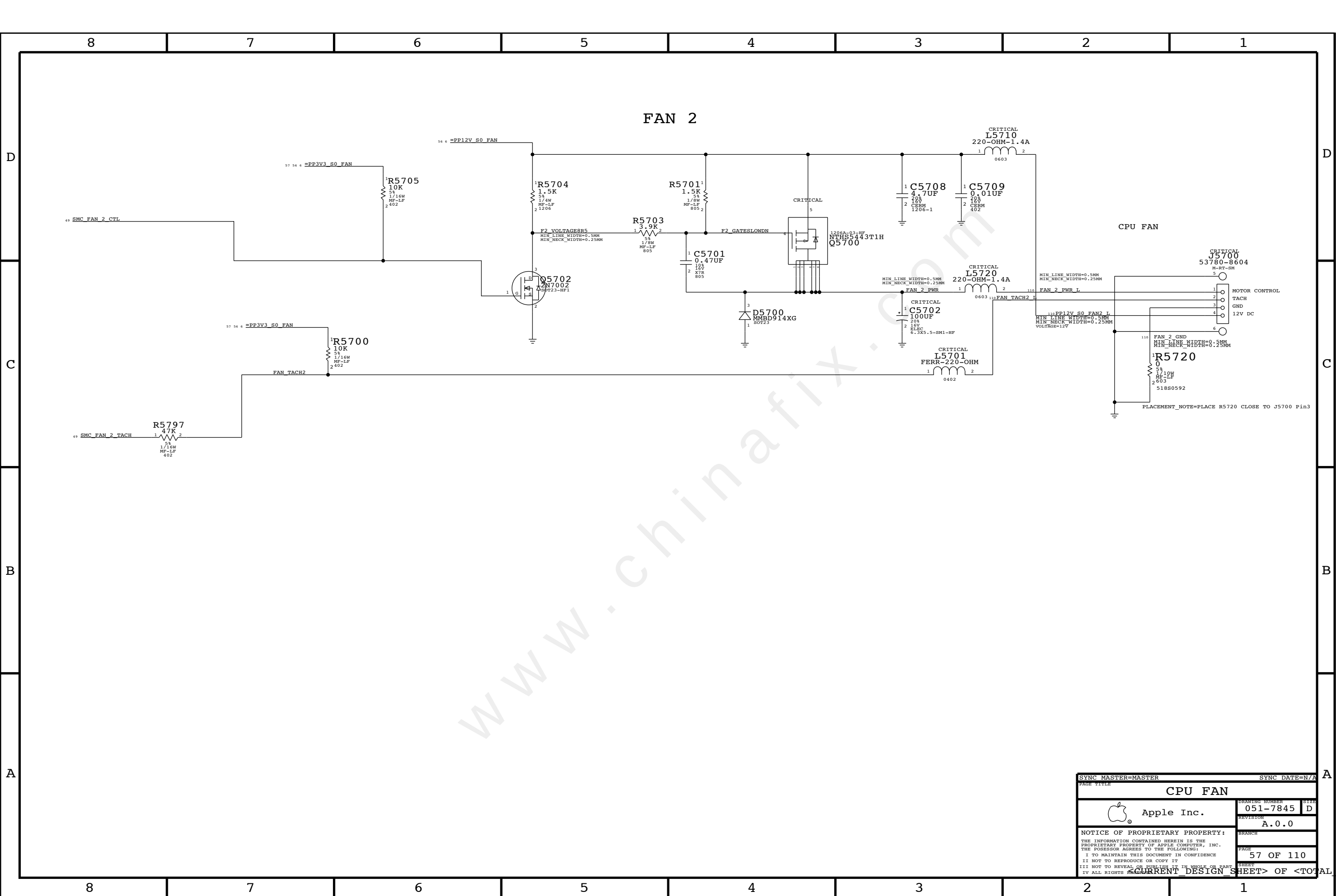
SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
MCP CURRENT AND VOLTAGE SENSE			
 Apple Inc.		DRAWING NUMBER	051-7845
		SIZE	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	A.0.0
		BRANCH	
CURRENT DESIGN SHEET		PAGE	54 OF 110
		SHEET	
		SHEET	



C



A





8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	051-7845
		SIZE	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	A.0.0
		BIONCH	
		PAGE	58 OF 110
		SHEET	



8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

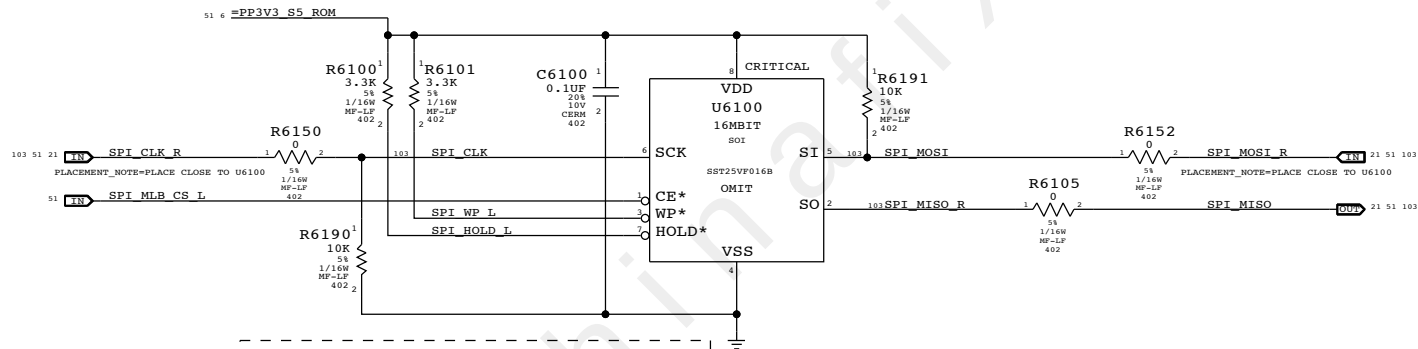
SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BIONCH	
		PAGE	
		59 OF 110	
		SHEET	



8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BIONCH	
		PAGE	
		60 OF 110	
		SHEET	



MCP79 SPI Frequency Select		
Frequency	SPI_MOSI	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

NOTE: MCP79 only issues 'READ' (0x03) commands not 'READ_FAST' (0x0B). Limits SPI bus frequency and part selection.

SST25VF016B max speed for READ command is 25MHz.

SYNC MASTER=K51

SYNC DATE=12/08/2008

SPI ROM

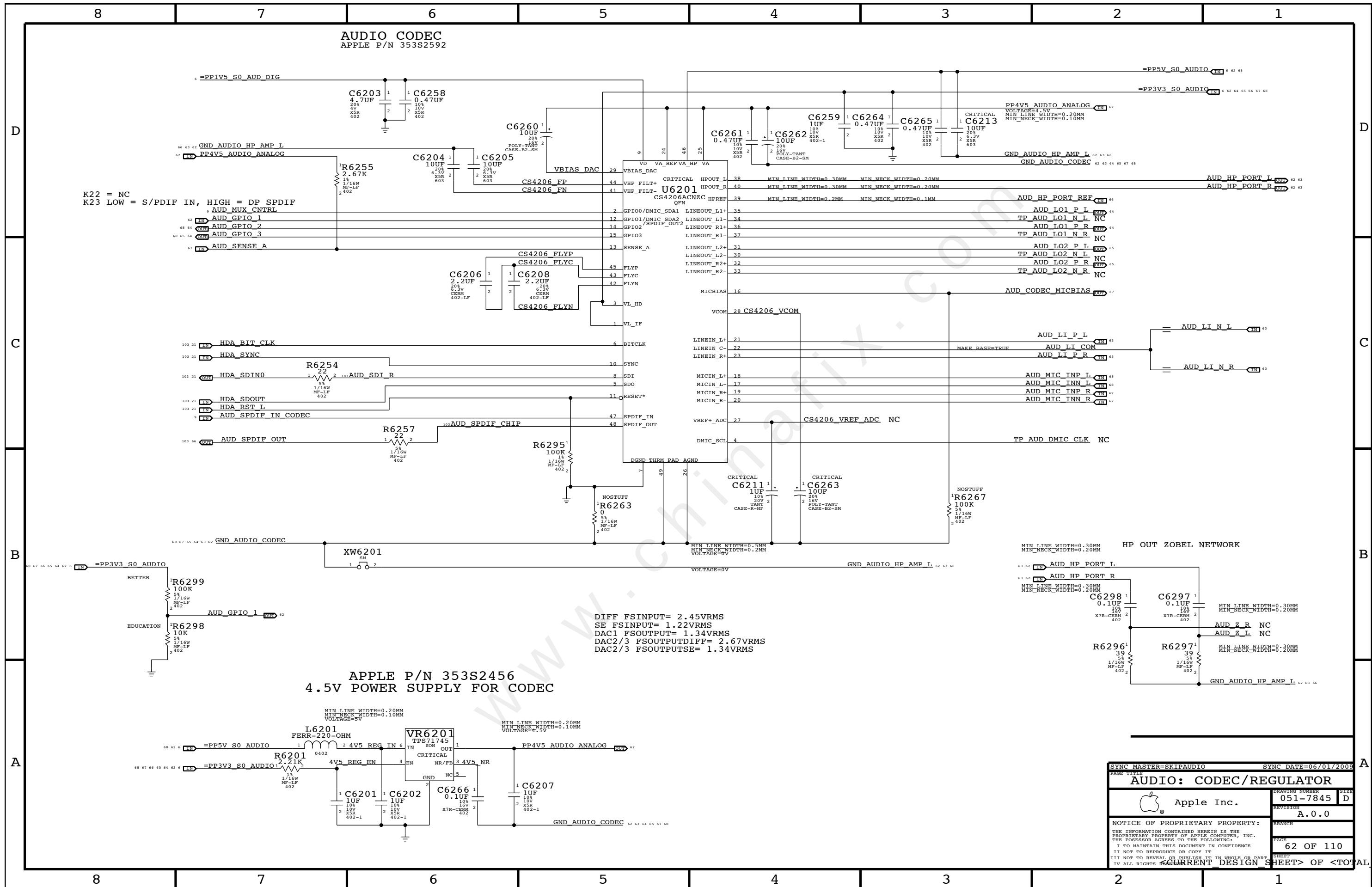
Apple Inc.

051-7845

A.0.0

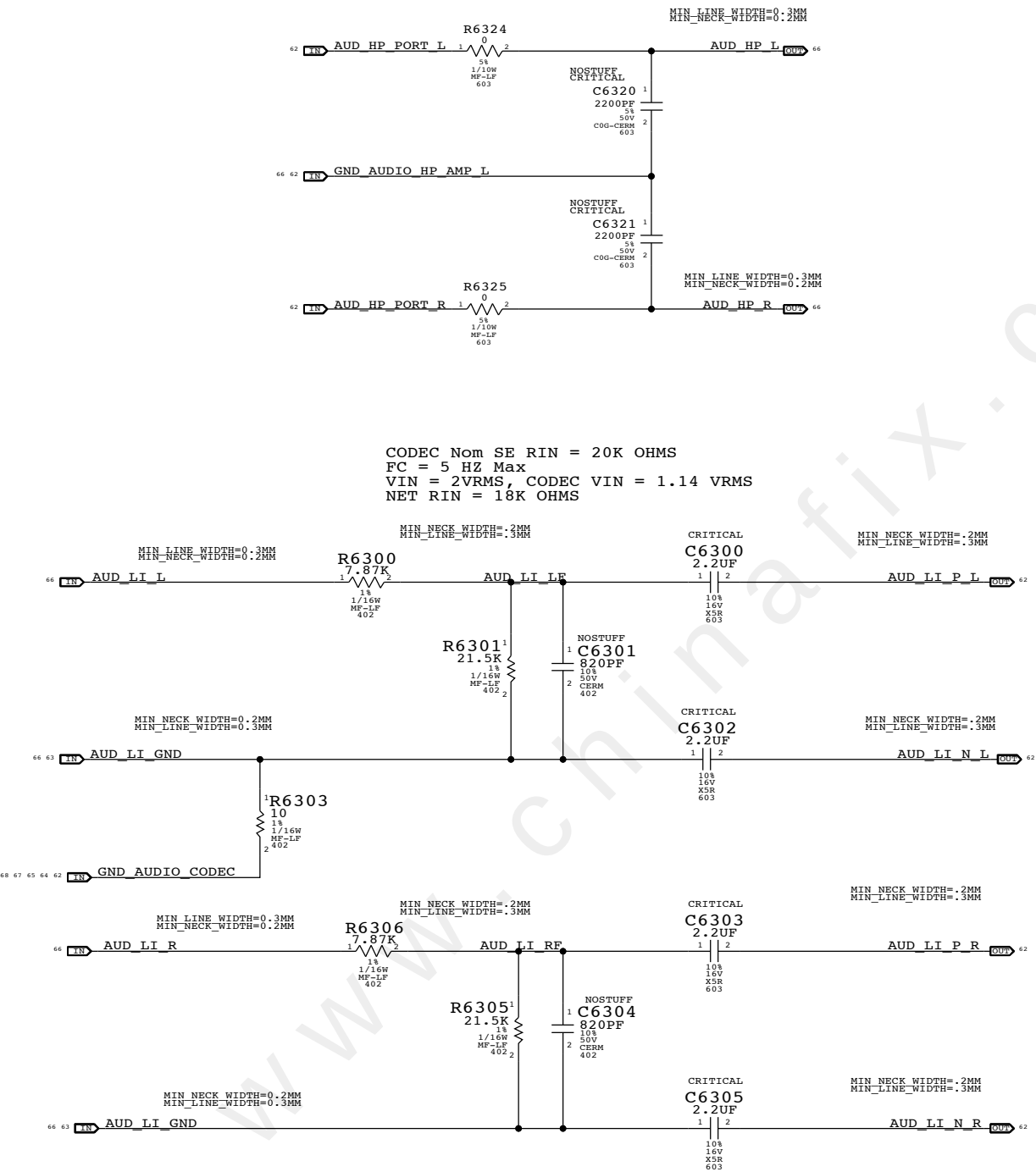
61 OF 110

DESIGN_SHEETS>



PAGE TITLE		PAGE NUMBER	
AUDIO: CODEC/REGULATOR		051-7845	
Apple Inc.		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		62 OF 110	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		SHEET	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		DESIGN SHEETS	
II NOT TO REPRODUCE OR COPY IT		OF <TOTAL DESIGN SHEETS>	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

1ST ORDER DAC FILTER PLACEHOLDER



SYNC MASTER=SKIPAUDIO		SYNC DATE=06/01/2009	
PAGE TITLE			
AUDIO: FILTER/BUFFER			
 Apple Inc.		DRAWING NUMBER	051-7845
		REVISION	A.0.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	63 OF 110
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
CURRENT DESIGN SHEET> OF <TOTAL>			

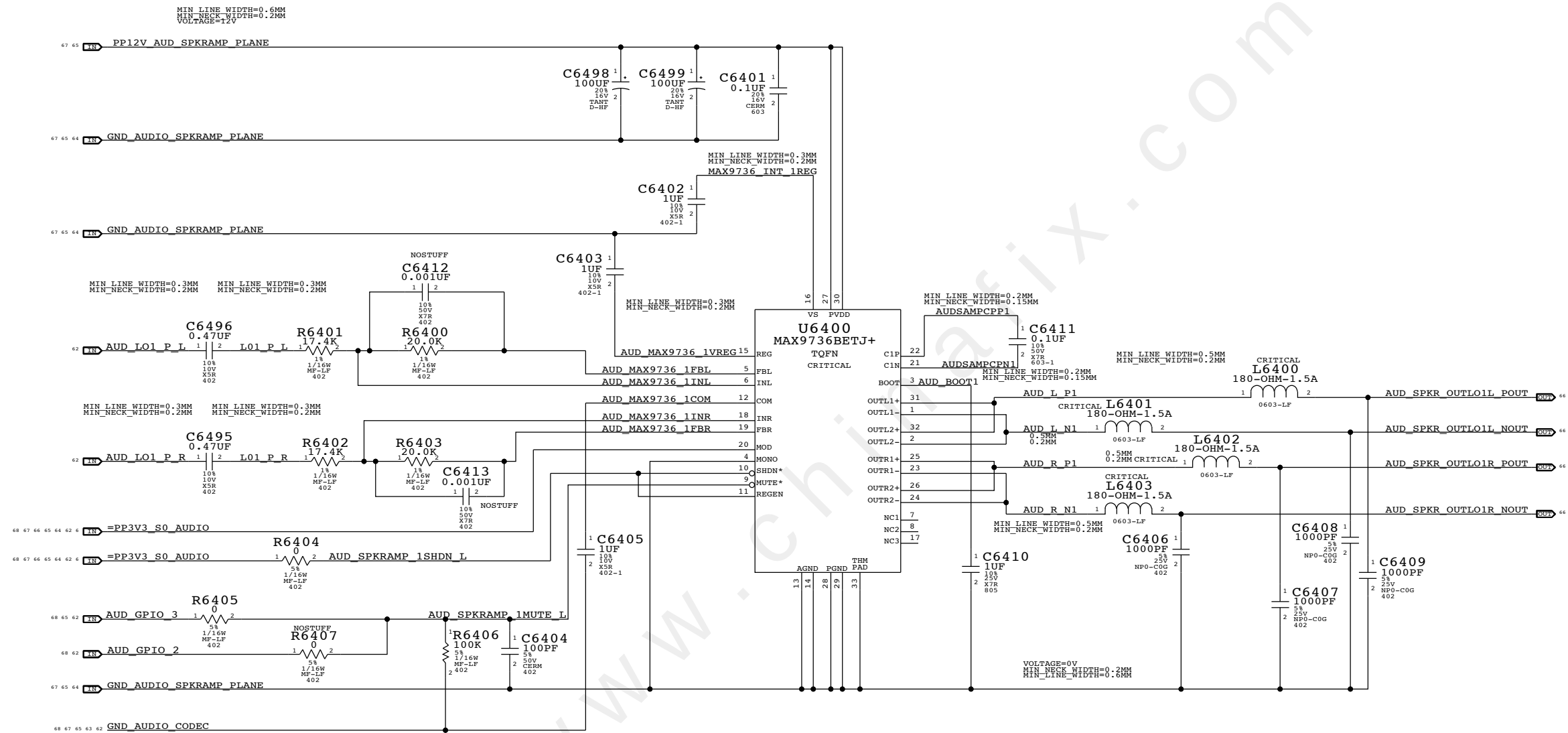
D

TWEETER SPEAKER AMPLIFIER

MAX9736B APN:353S2042

GAIN = -4.8(20K/17.4K) TURN ON TIME: 110MS
CODEC OUT = 1.335VRMS TURN ON DELAY: 150MS
AMP VOUT = 7.355VRMS RIN = 17.4 OHMS
FC = 19.5 HZ
POUT = 6.76 W INTO 8 OHMS @ 1% THD+N

D



B

B

A

A |

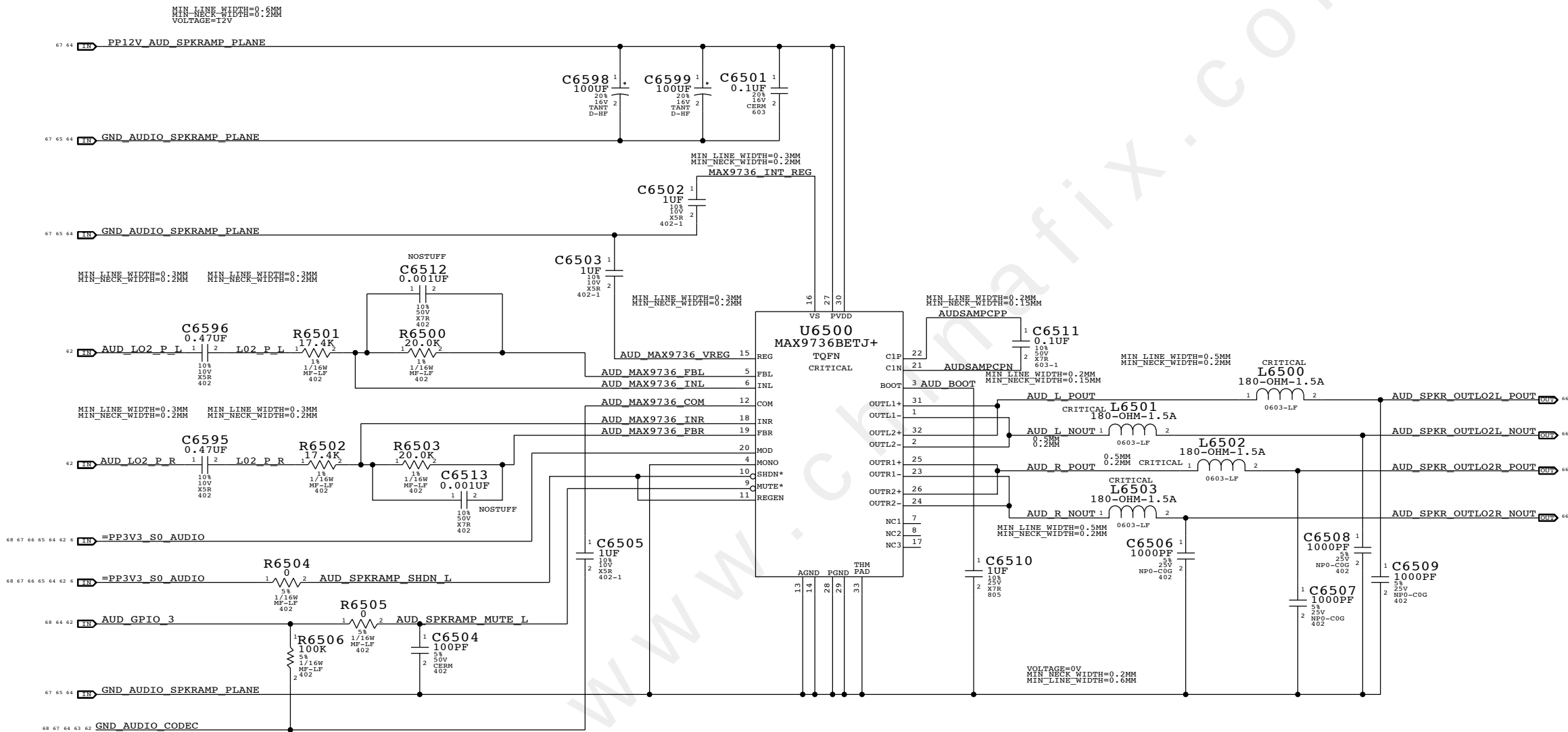
SYNC MASTER=SKIPAUDIO		SYNC DATE=06/01/2005	
PAGE TITLE			
AUDIO: SPEAKER AMP			
 Apple Inc.	DURATION NUMBER	SIZE	
	051-7845	D	
	REVISION		
A.0.0			
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			
BRANCH			
PAGE		64 OF 110	
SHEET		OF <TOTAL>	
CURRENT DESIGN SHEET>			

DESIGN_SHEETS>

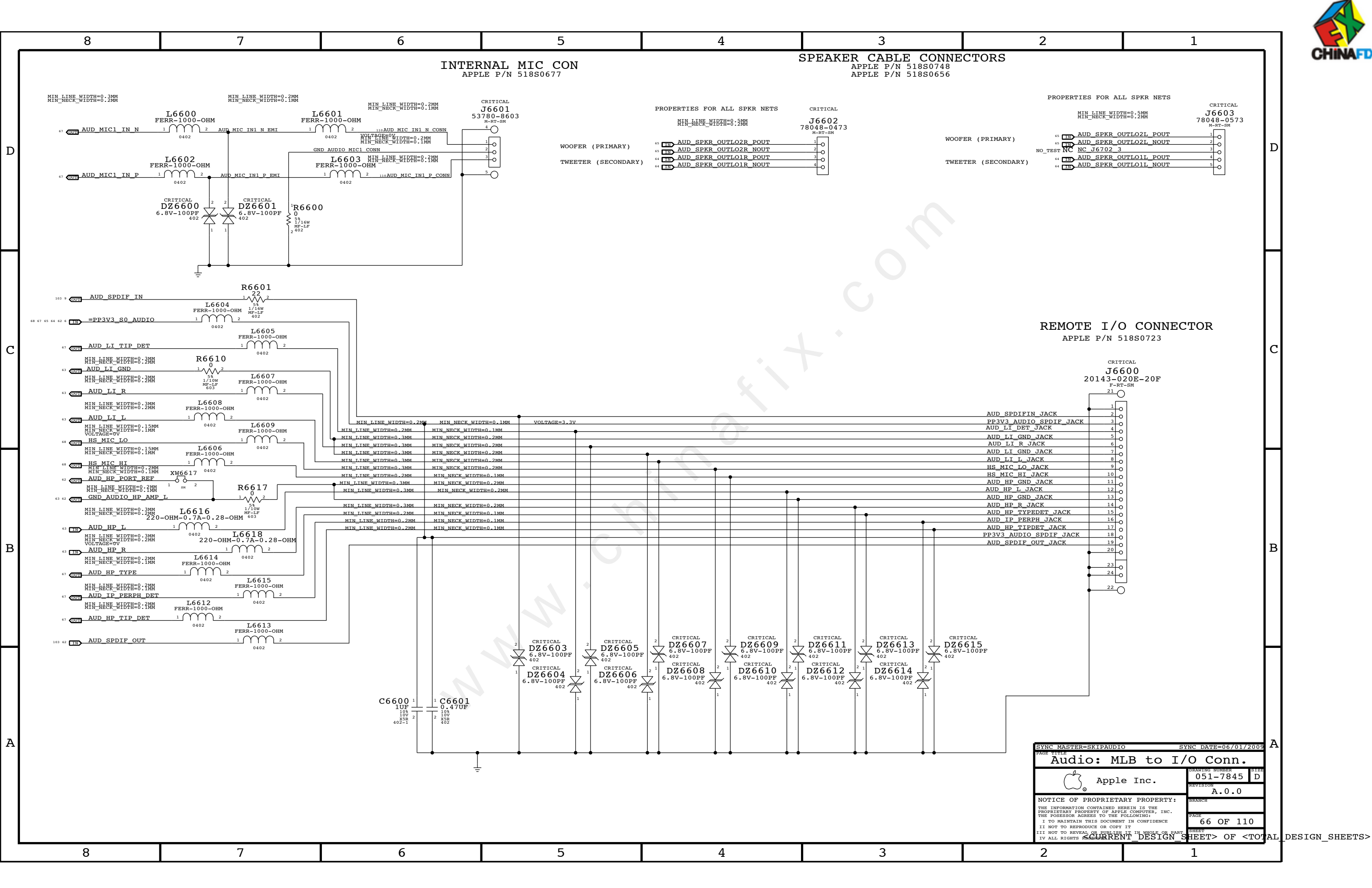
WOOFER SPEAKER AMPLIFIER

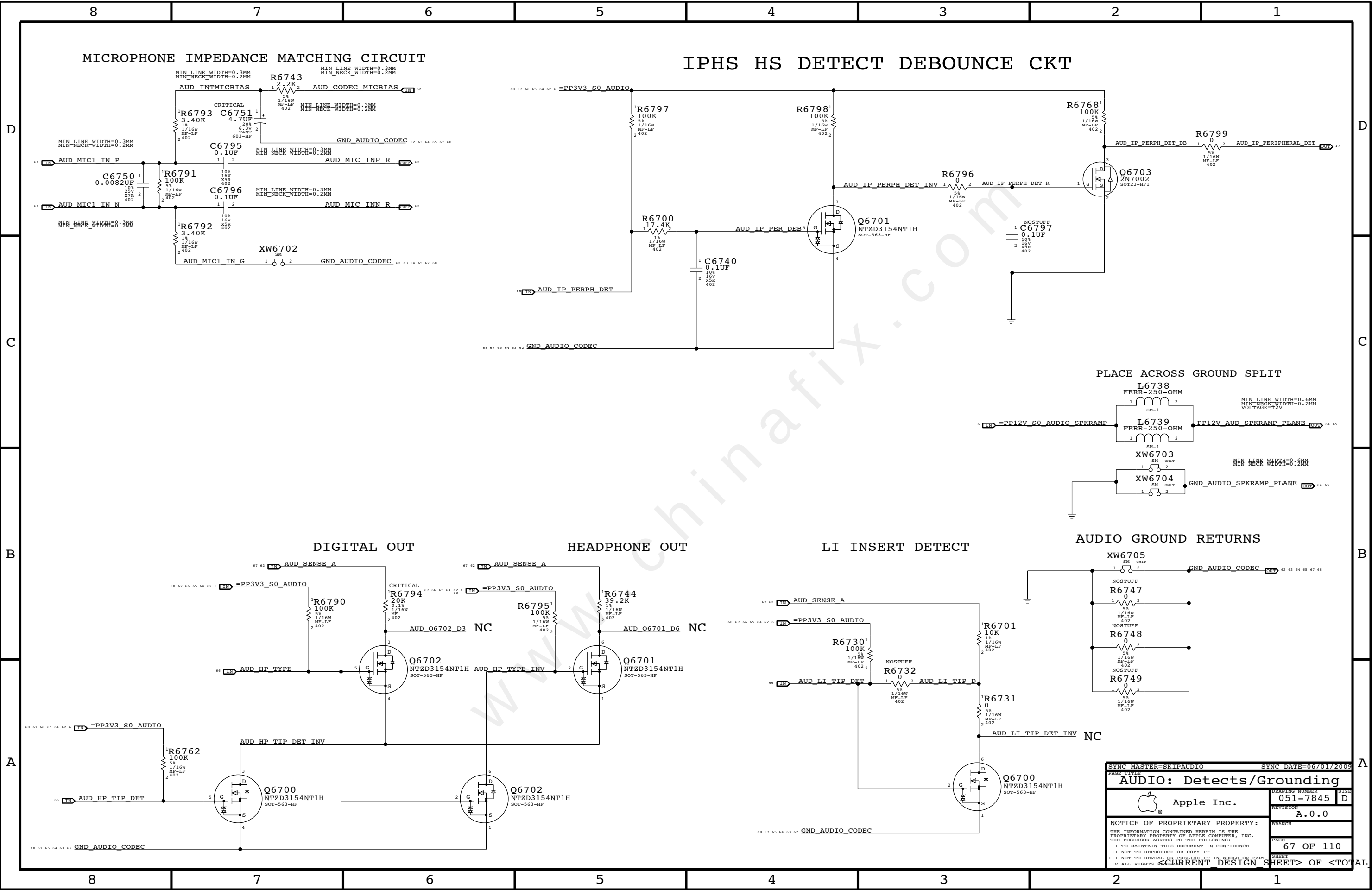
MAX9736B APN:353S2042

GAIN = -4.8(20K/17.4K) TURN ON TIME: 110MS
CODEC OUT = 1.335VRMS TURN ON DELAY: 150MS
AMP VOUT = 7.355VRMS RIN = 17.4 OHMS
POUT = 6.76 W INTO 8 OHMS @ 1% THD+N FC = 19.5 HZ



SYNC MASTER=SKIPAUDIO		SYNC DATE=06/01/2009	
PAGE TITLE			
AUDIO: SPEAKER AMP			
 Apple Inc.		DRAWING NUMBER	051-7845
		REVISION	A.0.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	65 OF 110
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	
II NOT TO REPRODUCE OR COPY IT		CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			





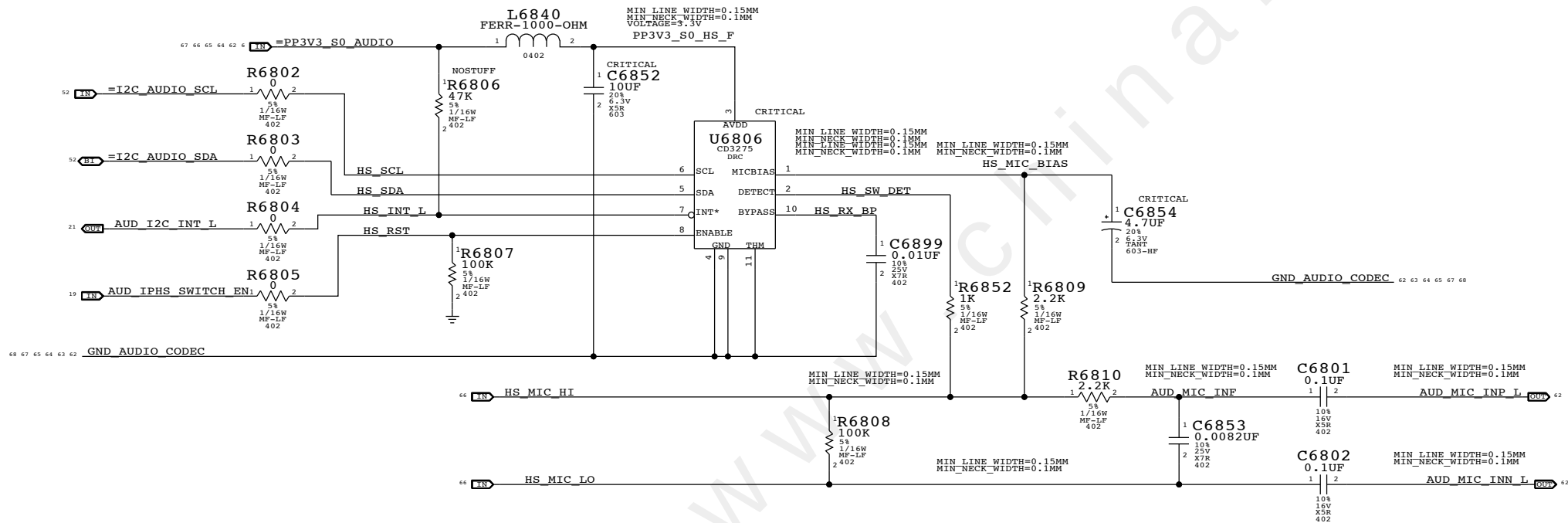
SYNC MASTER=SKIPAUDIO		SYNC DATE=06/01/2009	
PAGE TITLE		AUDIO: Detects/Grounding	
DRAWING NUMBER		051-7845	D
REVISION		A.0.0	
BRANCH			
PAGE		67 OF 110	
SHEET			
NOTICE OF PROPRIETARY PROPERTY:		THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		II NOT TO REPRODUCE OR COPY IT	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		IV ALL RIGHTS RESERVED	

<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>

FUNCTION	PIN	CONVERTER	VOLUME	ENABLE/ CNTRL TYPE	DETECT/ INTERRUPT
PRIMARY	0X0B	0X04	0X04	GPIO 3	N/A
SECONDARY	0X0A	0X03	0X03	GPIO 3	N/A
HEADPHONES	0X09	0X02	0X02	N/A	0X09 (A)
LINE INPUT	0X0C	0X05	0X05	N/A	LINE IN
BUILT-IN MICROPHONE	0X0D (13,B,RIGHT)	0X06	0X06	MICBIAS 80%	N/A
HEADSET MICROPHONE	0X0D (13,V22,B,LEFT)	0X06	0X06	MIKEY	MIKEY
SPDIF OUT	0X10	0X08	N/A	N/A	0X0C (B)
SPDIF IN	0X0F	0X07	N/A	N/A	N/A
MIKEY	N/A	N/A	N/A	MCP GPIO_38	MCP GPIO_5

MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2256



FLP = 8.82 KHZ
FHP = 80 HZ

SYNC MASTER=SKIPAUDIO		SYNC DATE=06/01/2009	
PAGE TITLE			
AUDIO: Mikey			
DRAWING NUMBER		SIZE	
051-7845		D	
REVISION		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		68 OF 110	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		DESIGN SHEETS	
IV ALL RIGHTS RESERVED		DESIGN SHEETS	

D

D

C

C

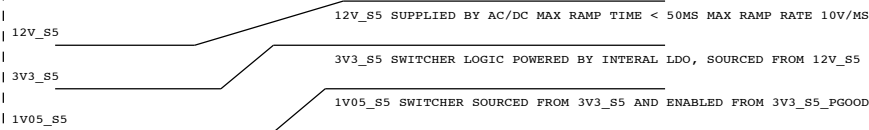
B

B

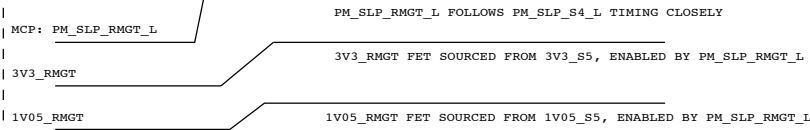
A

A

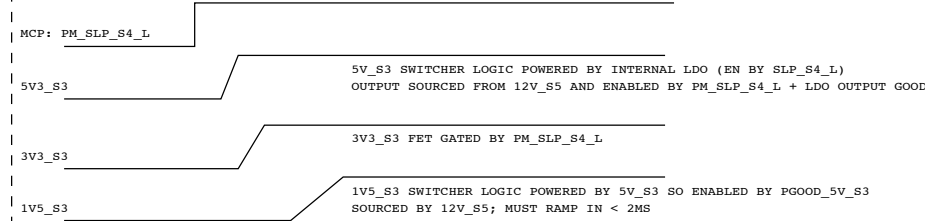
S5 POWER RAIL SEQUENCING



RMGT POWER RAIL SEQUENCING

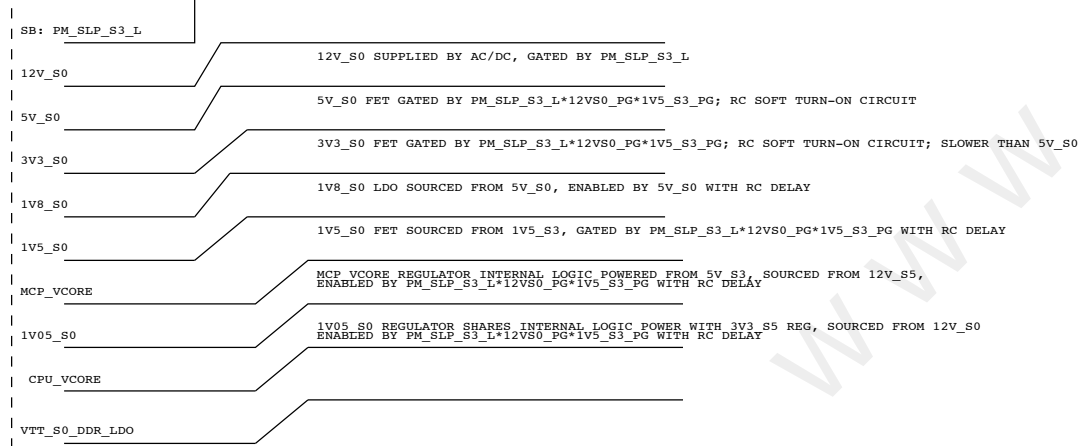


S3 POWER RAIL SEQUENCING

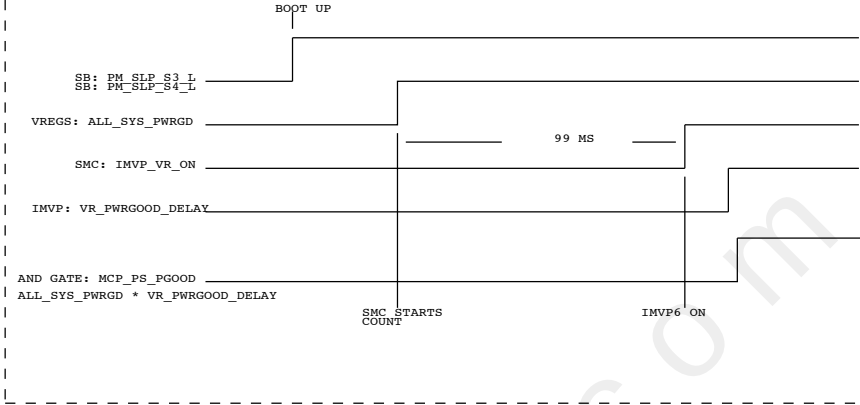


NOTE: NO SEQUENCING REQUIREMENTS FOR THESE 3 RAILS

S0 POWER RAIL SEQUENCING

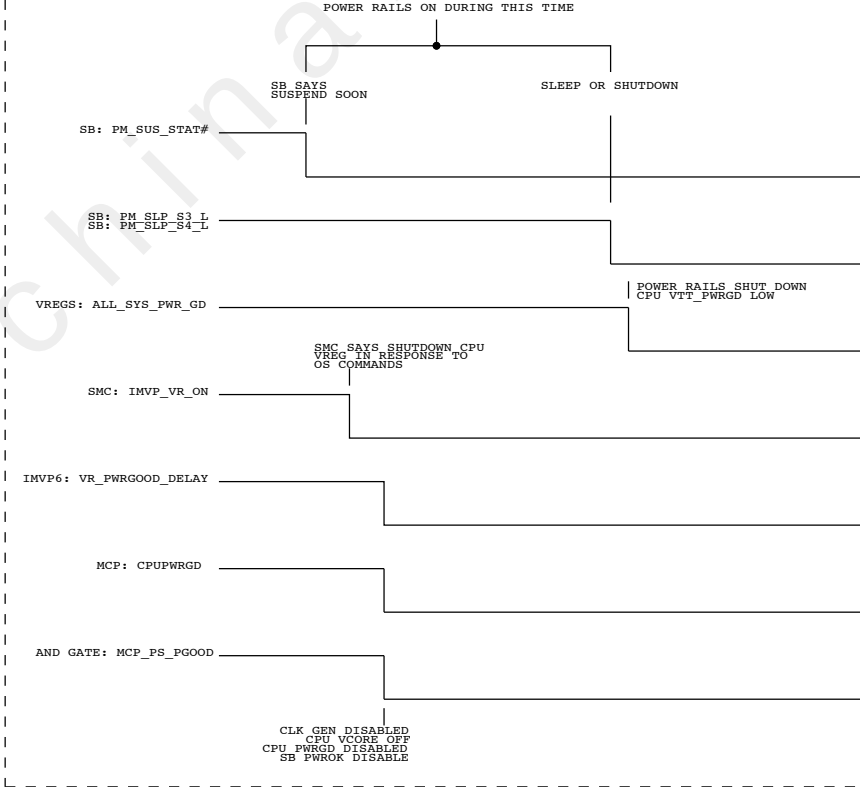


STARTUP (BOOT OR WAKE) TIMING



State	Manageability	SMC_PM_G2_ENABLE	PM_S4_STATE_L	PM_SLP_S3_L	PM_SLP_S4_L	PM_SLP_M_L
Run (S0/M0)	N/A	1	1	1	1	1
Sleep (S3/M1)	On	1	1	0	1	1
Soft-Off (S5/M1)	On	1	0	0	1	1
Sleep (S3/M-Off)	Off	1	1	0	1	0
Soft-Off (S5/M-Off)	Off	1	0	0	0	0
Battery Off (G3Hot)	N/A	0	0	0	0	0

SHUT DOWN (SHUTDOWN OR SLEEP) TIMING



SYNC MASTER=K51 SYNC DATE=12/08/2008

PAGE TITLE

POWER SEQUENCING BLOCK DIAGRAM

DRAWING NUMBER 051-7845 D

REVISION A.0.0

BRANCH

PAGE 69 OF 110

SHEET

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

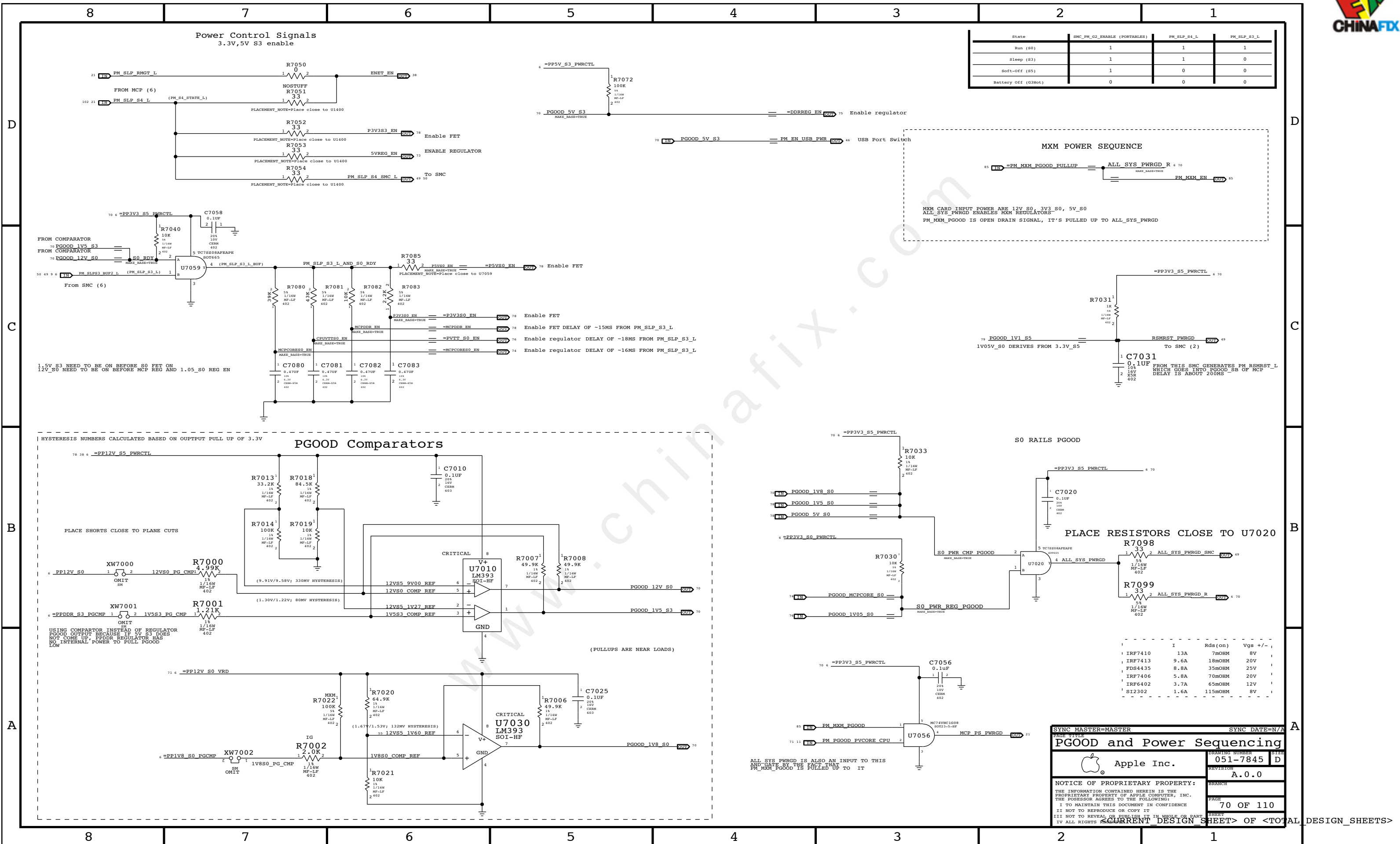
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

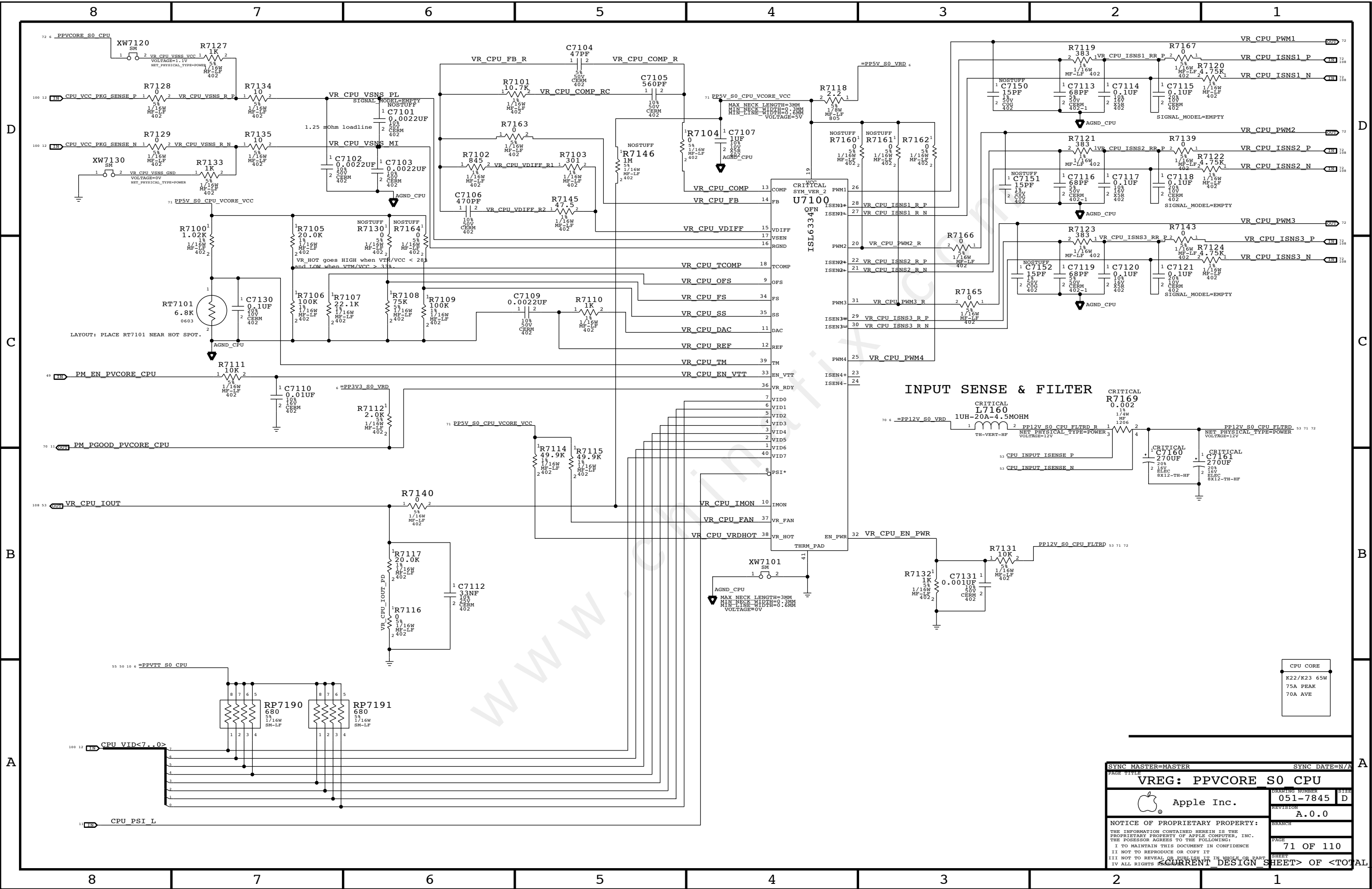
II NOT TO REPRODUCE OR COPY IT

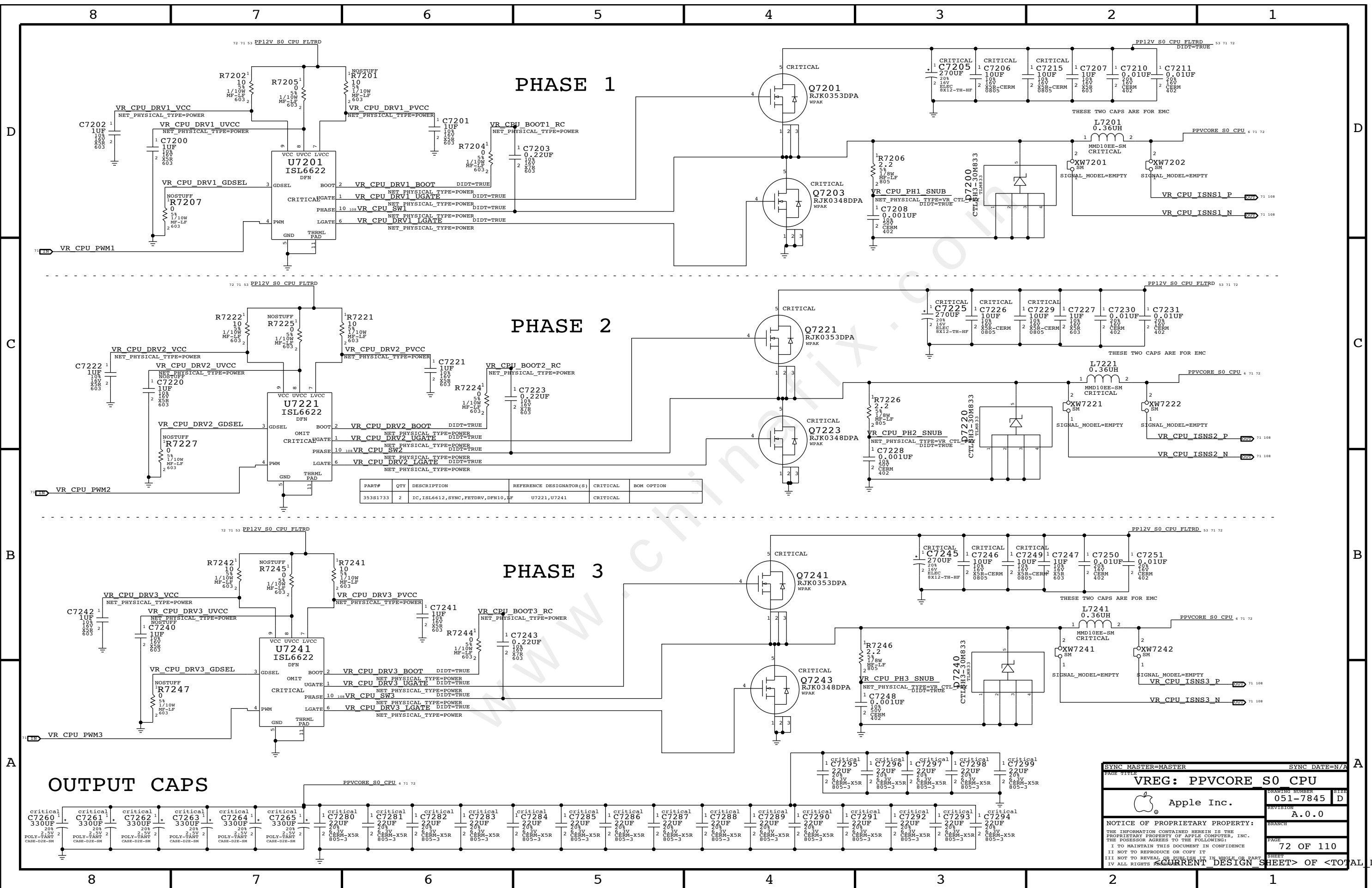
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>







PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S1733	2	IC, ISL6612, SYNC, FETDRV, DFN10, 4P	U7221, U7241	CRITICAL	

SYNC MASTER=MASTER

SYNC DATE=N/A

VREG: PPVCORE S0 CPU

Apple Inc.

051-7845

A.0.0

72 OF 110

DESIGN SHEETS

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

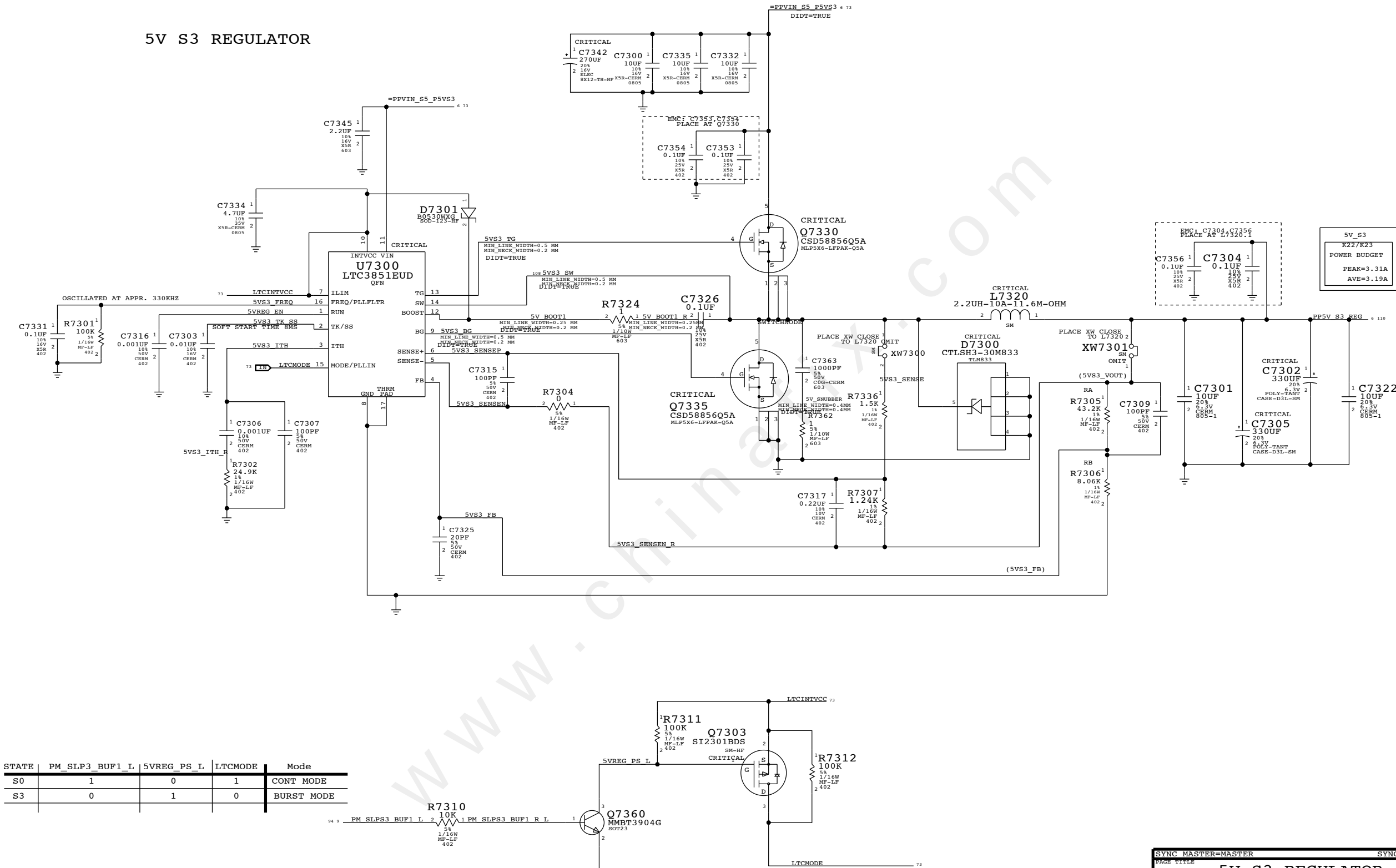
I DO NOT REPRODUCE OR COPY IT

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

5V S3 REGULATOR



STATE	PM_SLP3_BUF1_L	5VREG_PS_L	LTCMODE	Mode
S0	1	0	1	CONT MODE
S3	0	1	0	BURST MODE

SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
5V S3 REGULATOR			
Apple Inc.		DRAWING NUMBER	051-7845
		REVISION	A.0.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		73 OF 110	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		DESIGN SHEETS	
IV ALL RIGHTS RESERVED		OF <TOTAL>	

C

D

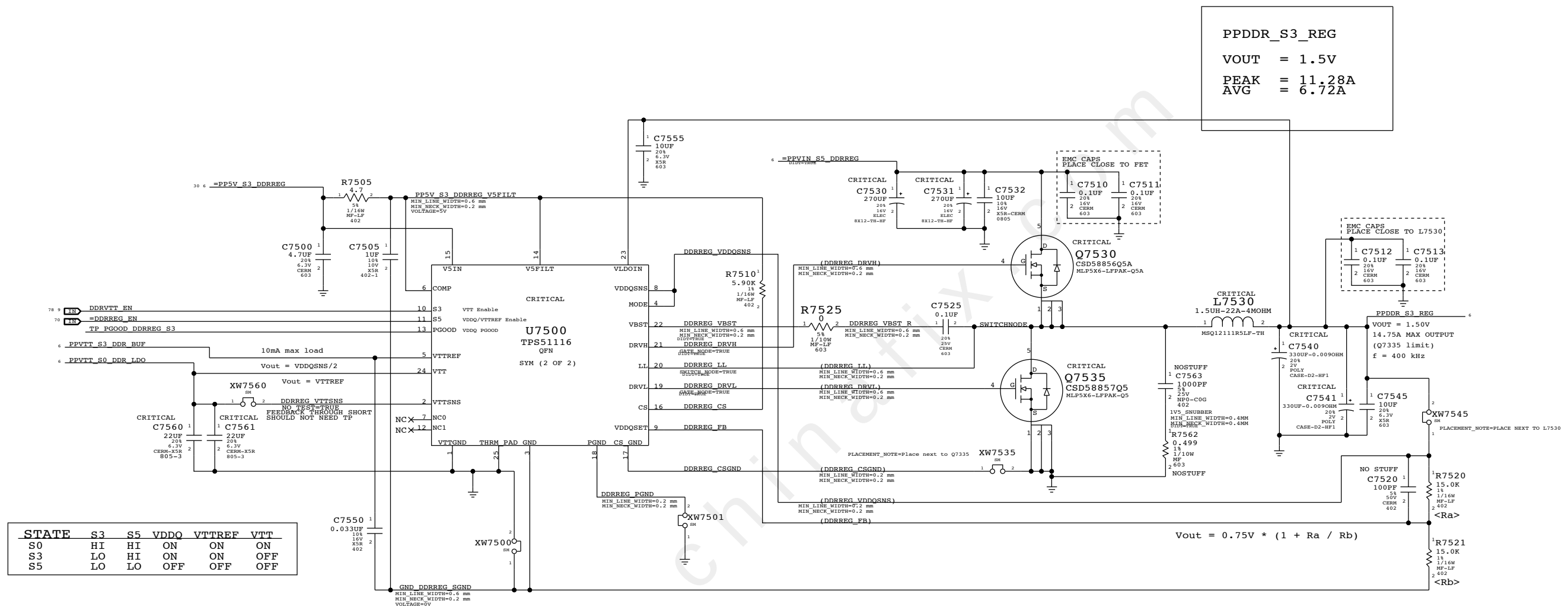
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S2497	1	INTERSIL ISL9563A	U7401	CRITICAL	MCP_ISL9563A
353S2303	1	INTERSIL ISL6263D	U7401	CRITICAL	MCP_ISL6263D

1.1V DEFAULT, OTHER VALUES TBD

VID<2:0>	Voltage
000	+1.100V

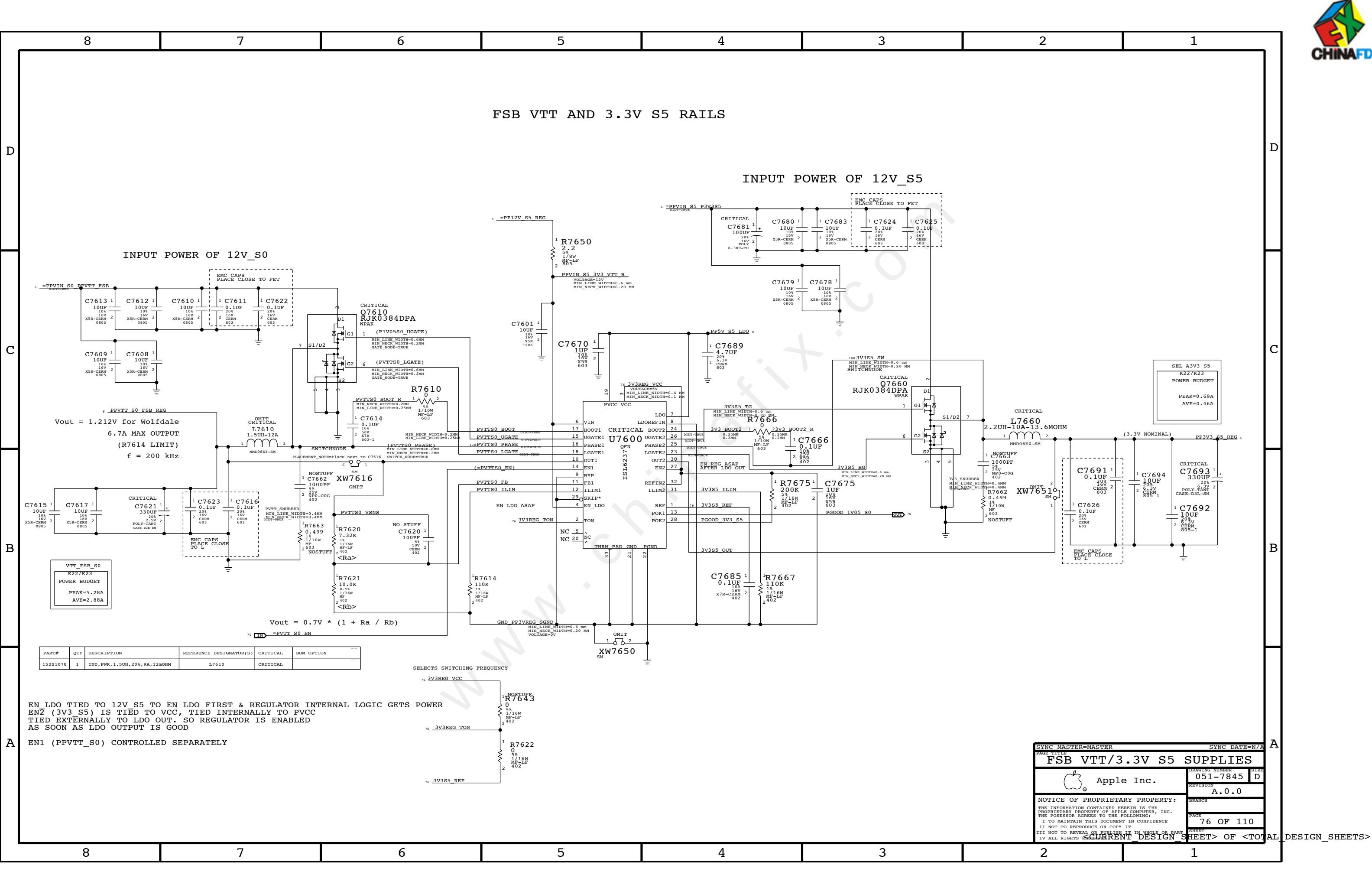
SYNC MASTER-MASTER		SYNC DATE-N/A	
PAGE TITLE			
MCP CORE REGULATOR			
	Apple Inc.		DRAWING NUMBER 051-7845
			SIZE D
		REVISION A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I 1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS ARE RESERVED		PAGE 74 OF 110	
CURRENT DESIGN SHEET		SHEET SHEET> OF <TOTAL	

1.5 V DDR SUPPLY



STATE	S3	S5	VDDO	VTTREF	VTT
S0	HI	HI	ON	ON	ON
S3	LO	HI	ON	ON	OFF
S5	LO	LO	OFF	OFF	OFF

SYNC MASTER-MASTER		SYNC DATE=N/A	
PAGE TITLE			
1.5V DDR SUPPLY			
	Apple Inc.		DRAWING NUMBER 051-7845
			SIZE D
			REVISION A.0.0
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I - I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II - NOT REPRODUCE OR COPY IT III - NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			
BRANCH PAGE		75 OF 110	
SHEET <CURRENT DESIGN SHEET>		OF <TOTAL SHEETS>	



```

EN LDO TIED TO 12VS5 TO EN LDO FIRST & REGULATOR INTERNAL LOGIC GETS POWER
EN2 (3V3 S5) IS TIED TO VCC, TIED INTERNALLY TO PVCC
TIED EXTERNALLY TO LDO OUT. SO REGULATOR IS ENABLED
AS SOON AS LDO OUTPUT IS GOOD

```

EN1 (PPVTT_S0) CONTROLLED SEPARATELY

SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
FSB VTT/3.3V S5 SUPPLIES			
 Apple Inc.		DRAWING NUMBER 051-7845	
		SIZE D	
		REVISION A.0.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I. TO RETAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH PAGE 76 OF 110	
CURRENT DESIGN SHEET		SHEET OF <TOTAL>	



8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		77 OF 110	
		SHEET	



D

C

B

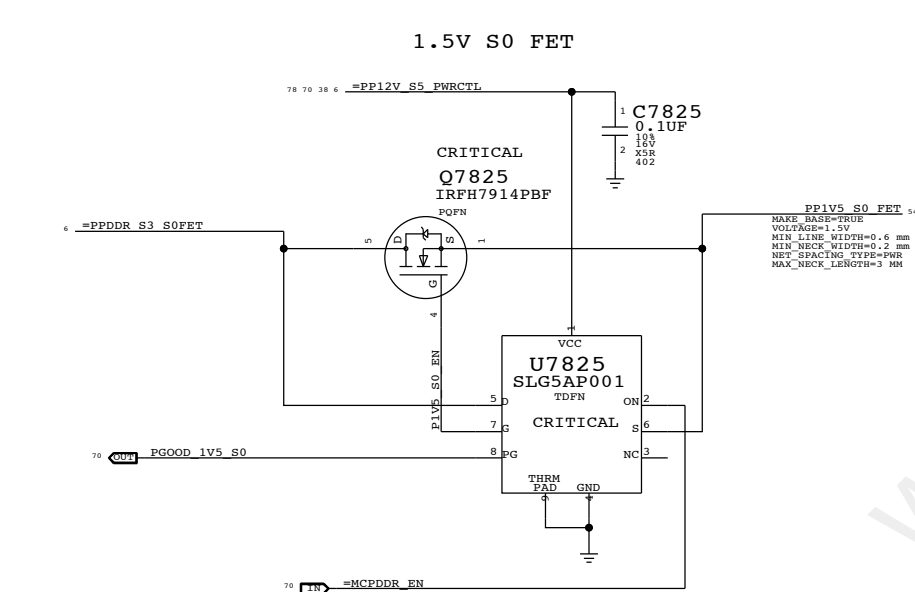
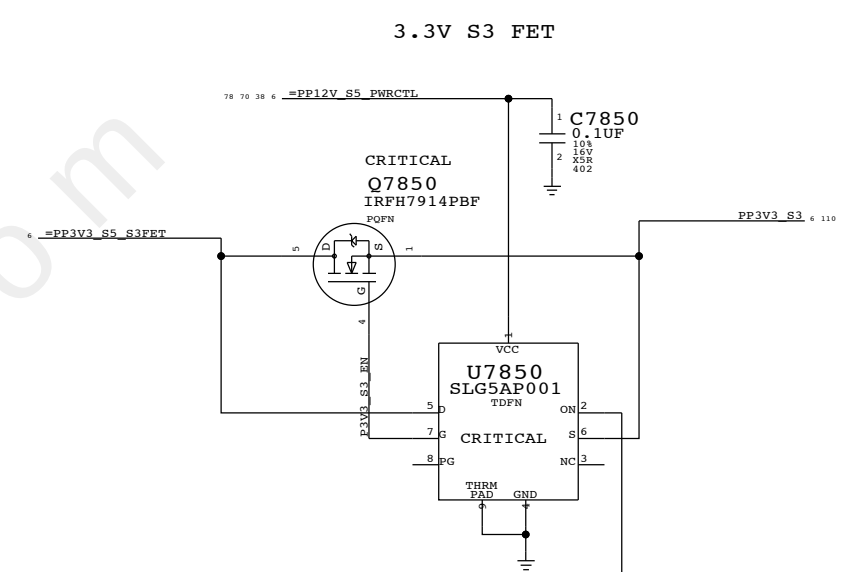
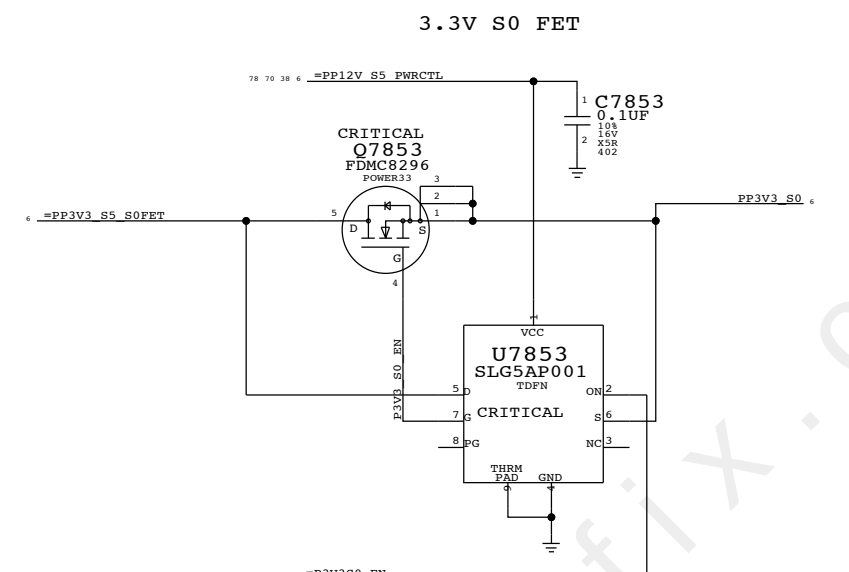
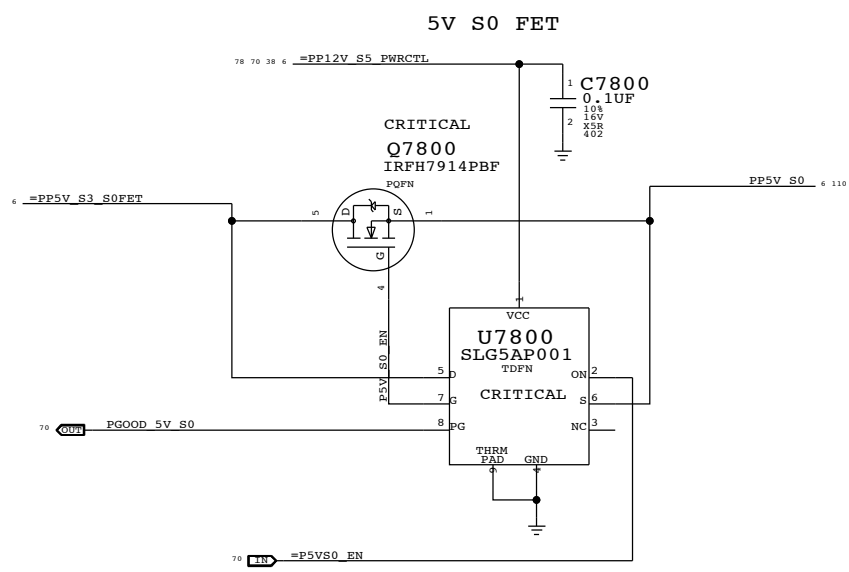
A

D

C

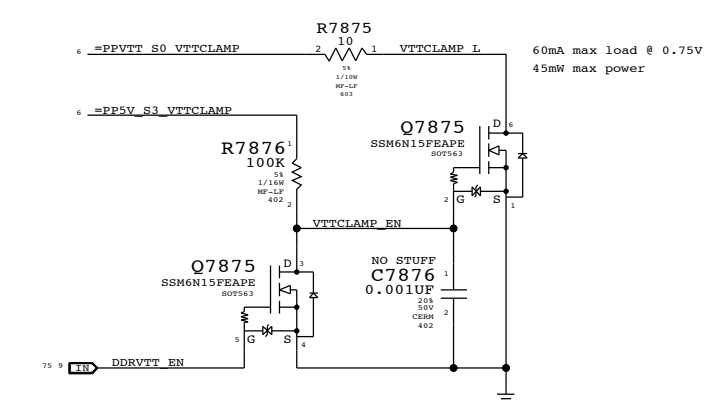
B

A



MCP79 DDRVTT FET

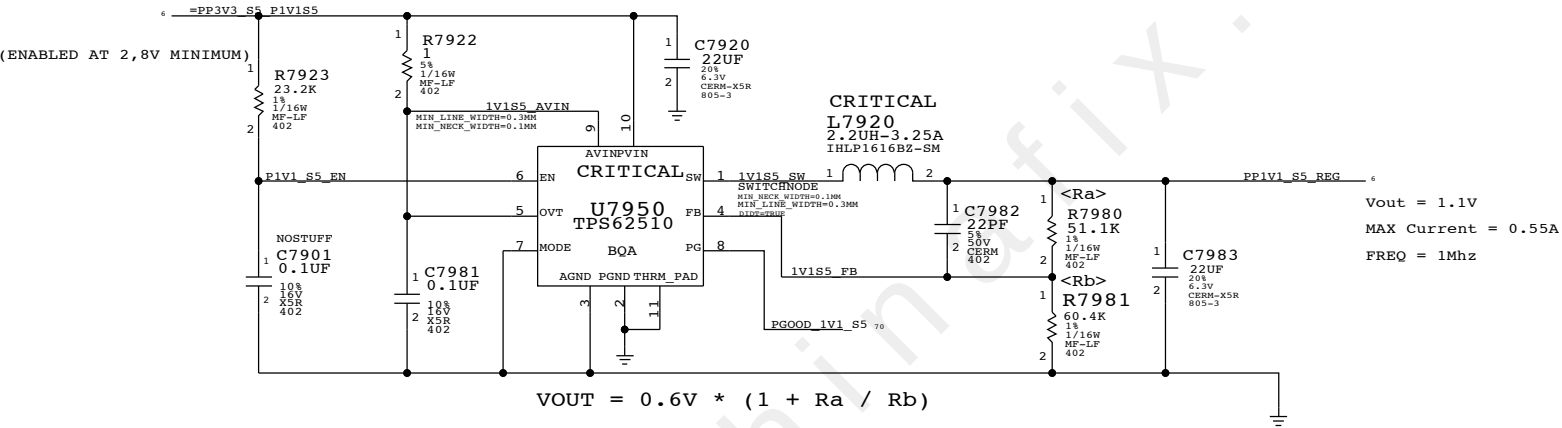
MCP79 DDR PAD LEAKAGE IS HIGH ENOUGH THAT NVIDIA RECOMMENDS UNPOWERING DURING SLEEP. IN ORDER TO SUPPORT UNPOWERING RAIL, HARDWARE MUST GUARANTEE MEM_CKE SIGNALS ARE LOW BEFORE RAIL IS TURNED OFF, AND REMAINS LOW UNTIL AFTER RAIL TURNS BACK ON OR DIMMS WILL EXIT SELF-REFRESH PREMATURELY. MEM_VTT_EN OUTPUT FROM MCP79 USED TO ENABLE CLAMP ON VTT RAIL, WHICH PULLS ALL CKE SIGNALS LOW THROUGH VTT TERMINATION RESISTORS.

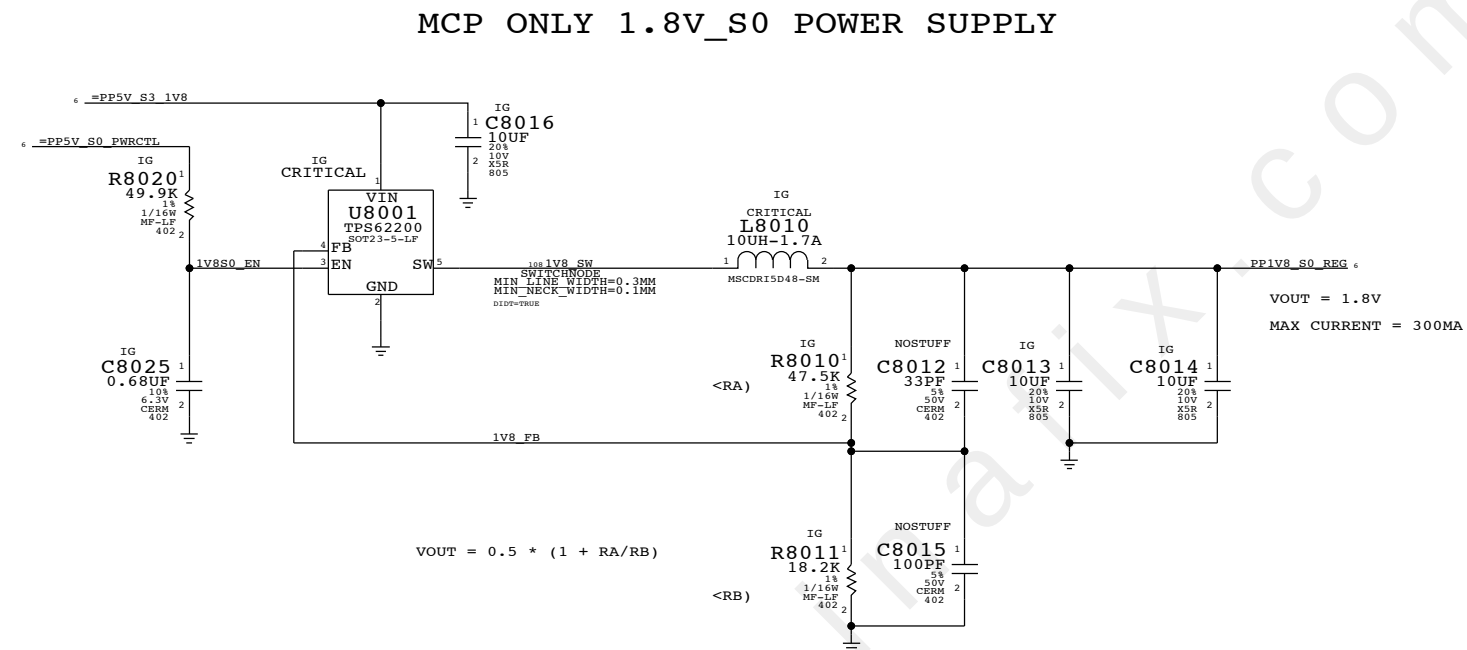


SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
S3 & S0 FETs			
 Apple Inc.		DRAWING NUMBER	051-7845
		D	
		REVISION	A.0.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
SCURRENT DESIGN S		PAGE	78 OF 110
		SHEET	
		SHEET> OF <TOT	

<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>

MCP 1.1V_S5 AUXC SUPPLY





SYNC MASTER-MASTER		SYNC DATE=N/A		
PAGE TITLE				
1V8 POWER SUPPLY				
	Apple Inc.		DRAWING NUMBER	SID
			051-7845	
			REVISION	
			A.0.0	
NOTICE OF PROPRIETARY PROPERTY:			BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:				
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			PAGE	
II NOT REPRODUCE OR COPY IT			80	OF 110
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			SHEET	
IV ALL RIGHTS RESERVED			SHEET> OF <TOT	



	8	7	6	5	4	3	2	1
D								
C								
B								
A								
	8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		81 OF 110	
		SHEET	



8	7	6	5	4	3	2	1
D							
C							
B							
A							
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BIONCH	
		PAGE	
		82 OF 110	
		SHEET	



8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		83 OF 110	
		SHEET	

Page Notes

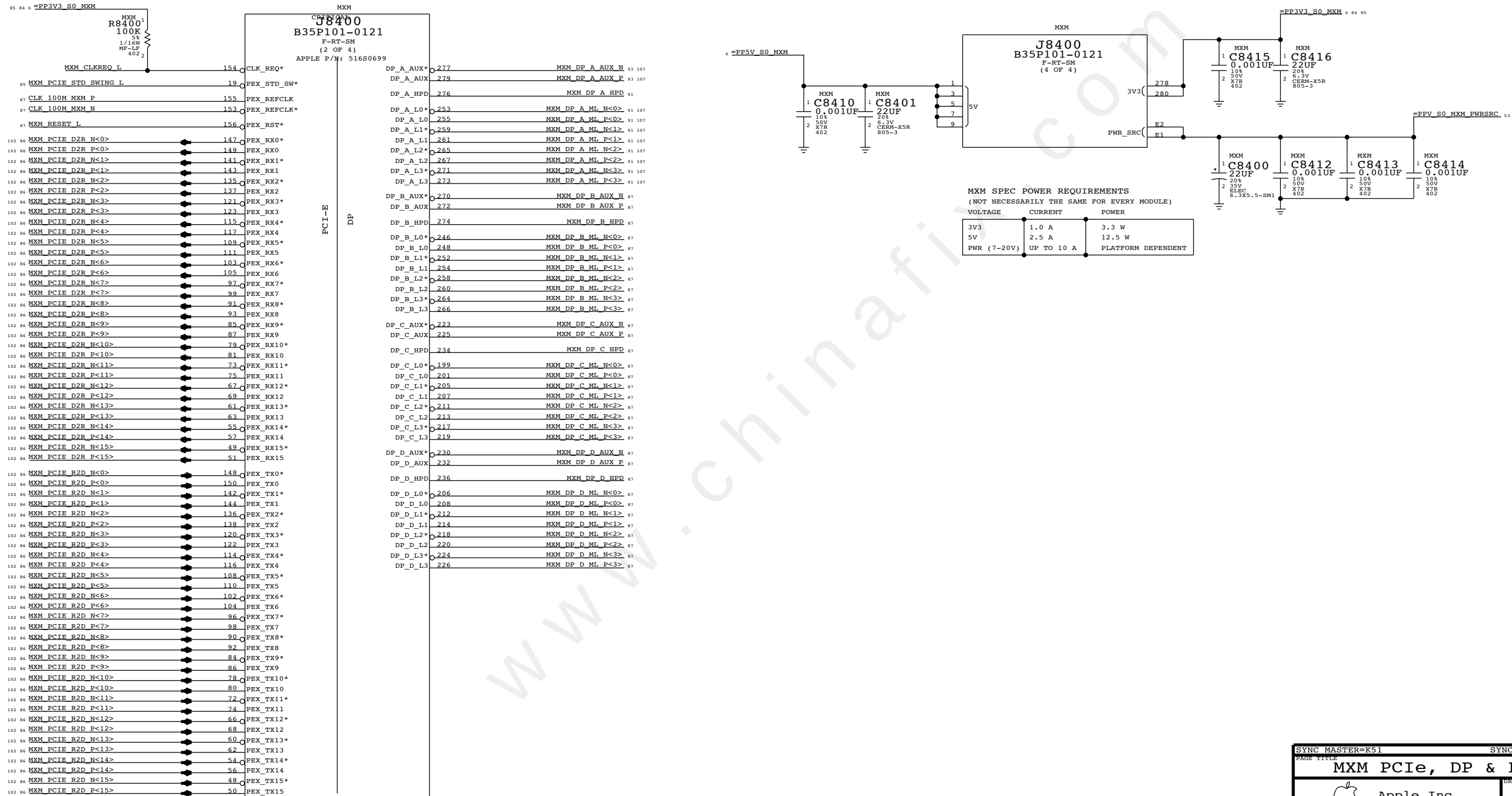
Power aliases required by this page:

- =PP3V3_S0_MXM
- =PP5V_S0_MXM
- =PPV_S0_MXM_PWSRC

Signal aliases required by this page:
(NONE)

BOM options provided by this page:

- MXM



SYNC MASTER=K51		SYNC DATE=10/31/2008	
PAGE TITLE			
MXM PCIe, DP & Power			
DRAWING NUMBER		SIZE	
051-7845		D	
REVISION		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		84 OF 110	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		CURRENT DESIGN SHEET	
IV ALL RIGHTS RESERVED		OF <TOTAL DESIGN SHEETS>	

Page Notes

Power aliases required by this page:

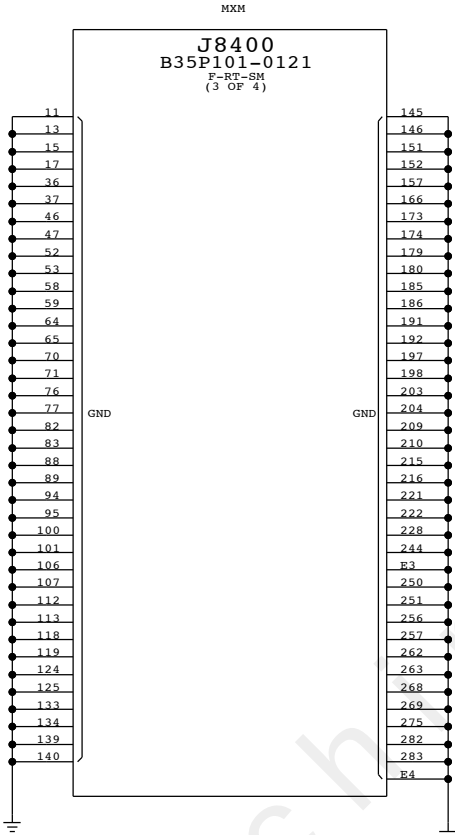
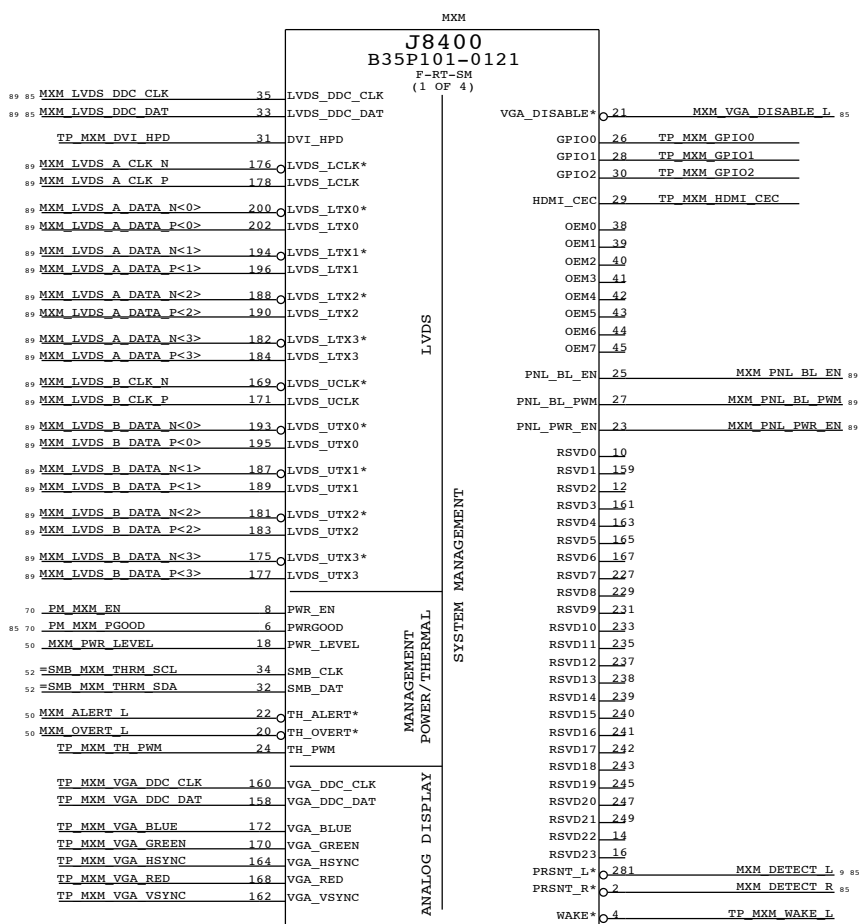
- =PP3V3_S0_MXM

Signal aliases required by this page:

- =SMB_MXM_THRM_DATA - =PM_MXM_PGOOD_PULLUP
- =SMB_MXM_THRM_CLK

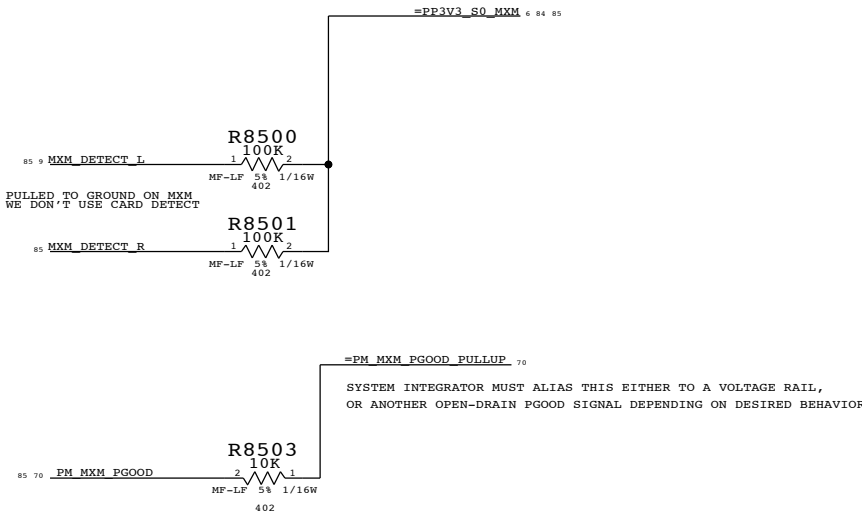
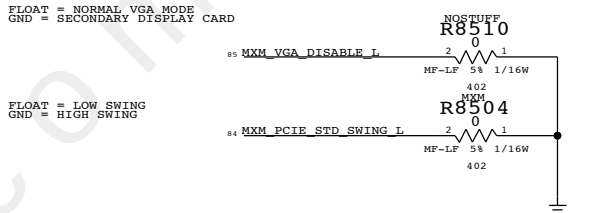
BOM options provided by this page:

PULLUPS & PULLDOWNS AT MXM CONNECTOR



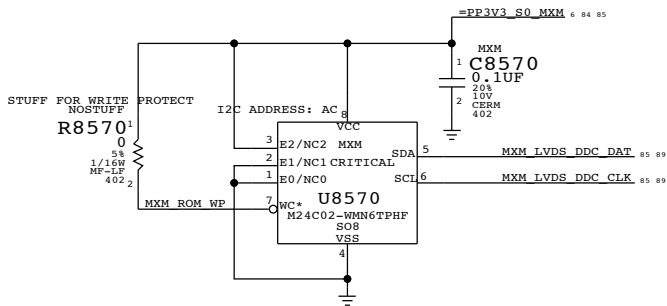
FLOAT = NORMAL VGA MODE
GND = SECONDARY DISPLAY CARD

FLOAT = LOW SWING
GND = HIGH SWING



MXM SYSTEM INFORMATION ROM

PLACE CLOSE TO J7800



SYNC MASTER=K51		SYNC DATE=10/31/2008	
PAGE TITLE			
MXM I/O			
 Apple Inc.		DRAWING NUMBER	SHEET
		051-7845	D
		REVISION	
		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		85	OF 110
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		CURRENT DESIGN SHEET>	OF <TOTAL DESIGN_SHEETS>

MXM TX CAPS

102 9	PEG R2D C N<0>	MXM C8600 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<15>	84 102
102 9	PEG R2D C P<0>	MXM C8601 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<15>	84 102
102 9	PEG R2D C P<1>	MXM C8602 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<14>	84 102
102 9	PEG R2D C N<1>	MXM C8603 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<14>	84 102
102 9	PEG R2D C N<2>	MXM C8604 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<13>	84 102
102 9	PEG R2D C P<2>	MXM C8605 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<13>	84 102
102 9	PEG R2D C N<3>	MXM C8606 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<12>	84 102
102 9	PEG R2D C P<3>	MXM C8607 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<12>	84 102
102 9	PEG R2D C N<4>	MXM C8608 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<11>	84 102
102 9	PEG R2D C P<4>	MXM C8609 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<11>	84 102
102 9	PEG R2D C N<5>	MXM C8610 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<10>	84 102
102 9	PEG R2D C P<5>	MXM C8611 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<10>	84 102
102 9	PEG R2D C N<6>	MXM C8612 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<9>	84 102
102 9	PEG R2D C P<6>	MXM C8613 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<9>	84 102
102 9	PEG R2D C N<7>	MXM C8614 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<8>	84 102
102 9	PEG R2D C P<7>	MXM C8615 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<8>	84 102
102 9	PEG R2D C N<8>	MXM C8616 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<7>	84 102
102 9	PEG R2D C P<8>	MXM C8617 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<7>	84 102
102 9	PEG R2D C N<9>	MXM C8618 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<6>	84 102
102 9	PEG R2D C P<9>	MXM C8619 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<6>	84 102
102 9	PEG R2D C N<10>	MXM C8620 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<5>	84 102
102 9	PEG R2D C P<10>	MXM C8621 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<5>	84 102
102 9	PEG R2D C P<11>	MXM C8622 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<4>	84 102
102 9	PEG R2D C N<11>	MXM C8623 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<4>	84 102
102 9	PEG R2D C N<12>	MXM C8624 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<3>	84 102
102 9	PEG R2D C P<12>	MXM C8625 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<3>	84 102
102 9	PEG R2D C N<13>	MXM C8626 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<2>	84 102
102 9	PEG R2D C P<13>	MXM C8627 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<2>	84 102
102 9	PEG R2D C N<14>	MXM C8628 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<1>	84 102
102 9	PEG R2D C P<14>	MXM C8629 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<1>	84 102
102 9	PEG R2D C N<15>	MXM C8630 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D P<0>	84 102
102 9	PEG R2D C P<15>	MXM C8631 0.1UF 1	2 10% 16V X5R 402	MXM PCIE R2D N<0>	84 102

MXM RX CAPS

102 84	MXM PCIE D2R P<15>	MXM C8632 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<0>	9 102
102 84	MXM PCIE D2R N<15>	MXM C8633 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<0>	9 102
102 84	MXM PCIE D2R P<14>	MXM C8634 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<1>	9 102
102 84	MXM PCIE D2R N<14>	MXM C8635 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<1>	9 102
102 84	MXM PCIE D2R P<13>	MXM C8636 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<2>	9 102
102 84	MXM PCIE D2R N<13>	MXM C8637 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<2>	9 102
102 84	MXM PCIE D2R P<12>	MXM C8638 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<3>	9 102
102 84	MXM PCIE D2R N<12>	MXM C8639 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<3>	9 102
102 84	MXM PCIE D2R P<11>	MXM C8640 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<4>	9 102
102 84	MXM PCIE D2R N<11>	MXM C8641 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<4>	9 102
102 84	MXM PCIE D2R P<10>	MXM C8642 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<5>	9 102
102 84	MXM PCIE D2R N<10>	MXM C8643 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<5>	9 102
102 84	MXM PCIE D2R P<9>	MXM C8644 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<6>	9 102
102 84	MXM PCIE D2R N<9>	MXM C8645 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<6>	9 102
102 84	MXM PCIE D2R P<8>	MXM C8646 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<7>	9 102
102 84	MXM PCIE D2R N<8>	MXM C8647 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<7>	9 102
102 84	MXM PCIE D2R P<7>	MXM C8648 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<8>	9 102
102 84	MXM PCIE D2R N<7>	MXM C8649 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<8>	9 102
102 84	MXM PCIE D2R P<6>	MXM C8650 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<9>	9 102
102 84	MXM PCIE D2R N<6>	MXM C8651 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<9>	9 102
102 84	MXM PCIE D2R P<5>	MXM C8652 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<10>	9 102
102 84	MXM PCIE D2R N<5>	MXM C8653 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<10>	9 102
102 84	MXM PCIE D2R P<4>	MXM C8654 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<11>	9 102
102 84	MXM PCIE D2R N<4>	MXM C8655 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<11>	9 102
102 84	MXM PCIE D2R P<3>	MXM C8656 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<12>	9 102
102 84	MXM PCIE D2R N<3>	MXM C8657 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<12>	9 102
102 84	MXM PCIE D2R P<2>	MXM C8658 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<13>	9 102
102 84	MXM PCIE D2R N<2>	MXM C8659 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<13>	9 102
102 84	MXM PCIE D2R P<1>	MXM C8662 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<14>	9 102
102 84	MXM PCIE D2R N<1>	MXM C8663 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<14>	9 102
102 84	MXM PCIE D2R P<0>	MXM C8660 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<15>	9 102
102 84	MXM PCIE D2R N<0>	MXM C8661 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<15>	9 102

SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
MXM PCIE CAPS			
 Apple Inc.		DRAWING NUMBER	SHEET
		051-7845	D
		REVISION	
		A.0.0	
		BRANCH	
		PAGE	86 OF 110
		SHEET	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II. NOT TO REPRODUCE OR COPY IT III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV. ALL RIGHTS RESERVED			
CURRENT DESIGN SHEET> OF <TOTAL DESIGN_SHEETS>			



8	7	6	5	4	3	2	1
Page Notes Power aliases required by this page: - =PP5V_DP_AUX Signal aliases required by this page: (NONE) BOM options provided by this page: (NONE)							
MCP CONNECTIONS 84 CLK_100M_MXM_P == GPU_CLK100M_PCIE_P 9 102 MAKE_BASE=TRUE 84 CLK_100M_MXM_N == GPU_CLK100M_PCIE_N 9 102 MAKE_BASE=TRUE 84 MXM_RESET_L == PEG_RESET_L 9 90 MAKE_BASE=TRUE							
UNUSED DP INTERFACES 84 MXM_DP_C_ML_N<0..3> == NC_MXM_DP_C_ML_N<0..3> MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_C_ML_P<0..3> == NC_MXM_DP_C_ML_P<0..3> MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_C_AUX_N == NC_MXM_DP_C_AUX_N MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_C_AUX_P == NC_MXM_DP_C_AUX_P MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_C_HPD == TP_MXM_DP_C_HPD MAKE_BASE=TRUE 84 MXM_DP_B_ML_N<0..3> == NC_MXM_DP_B_ML_N<0..3> MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_B_ML_P<0..3> == NC_MXM_DP_B_ML_P<0..3> MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_B_AUX_N == NC_MXM_DP_B_AUX_N MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_B_AUX_P == NC_MXM_DP_B_AUX_P MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_B_HPD == TP_MXM_DP_B_HPD MAKE_BASE=TRUE 84 MXM_DP_D_ML_N<0..3> == NC_MXM_DP_D_ML_N<0..3> MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_D_ML_P<0..3> == NC_MXM_DP_D_ML_P<0..3> MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_D_AUX_N == NC_MXM_DP_D_AUX_N MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_D_AUX_P == NC_MXM_DP_D_AUX_P MAKE_BASE=TRUE NO_TEST=TRUE 84 MXM_DP_D_HPD == TP_MXM_DP_D_HPD MAKE_BASE=TRUE							
www.chinax.com							
SYNC MASTER=MASTER SYNC DATE=N/A PAGE TITLE MXM ALIASES Apple Inc. 051-7845 D A.0.0 NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED 87 OF 110 87 OF 110 87 OF 110							
8	7	6	5	4	3	2	1

8

7

6

5

4

3

2

1

SYNC MASTER=MASTER

SYNC DATE=N/A

MXM ALIASES

Apple Inc.

051-7845

A.0.0

87 OF 110

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>



8	7	6	5	4	3	2	1
D							
C							
B							
A							
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51

SYNC DATE=10/01/2008

PAGE TITLE

BLANK PAGE



Apple Inc.

DRAWING NUMBER

051-7845

SIZE

D

REVISION

A.0.0

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

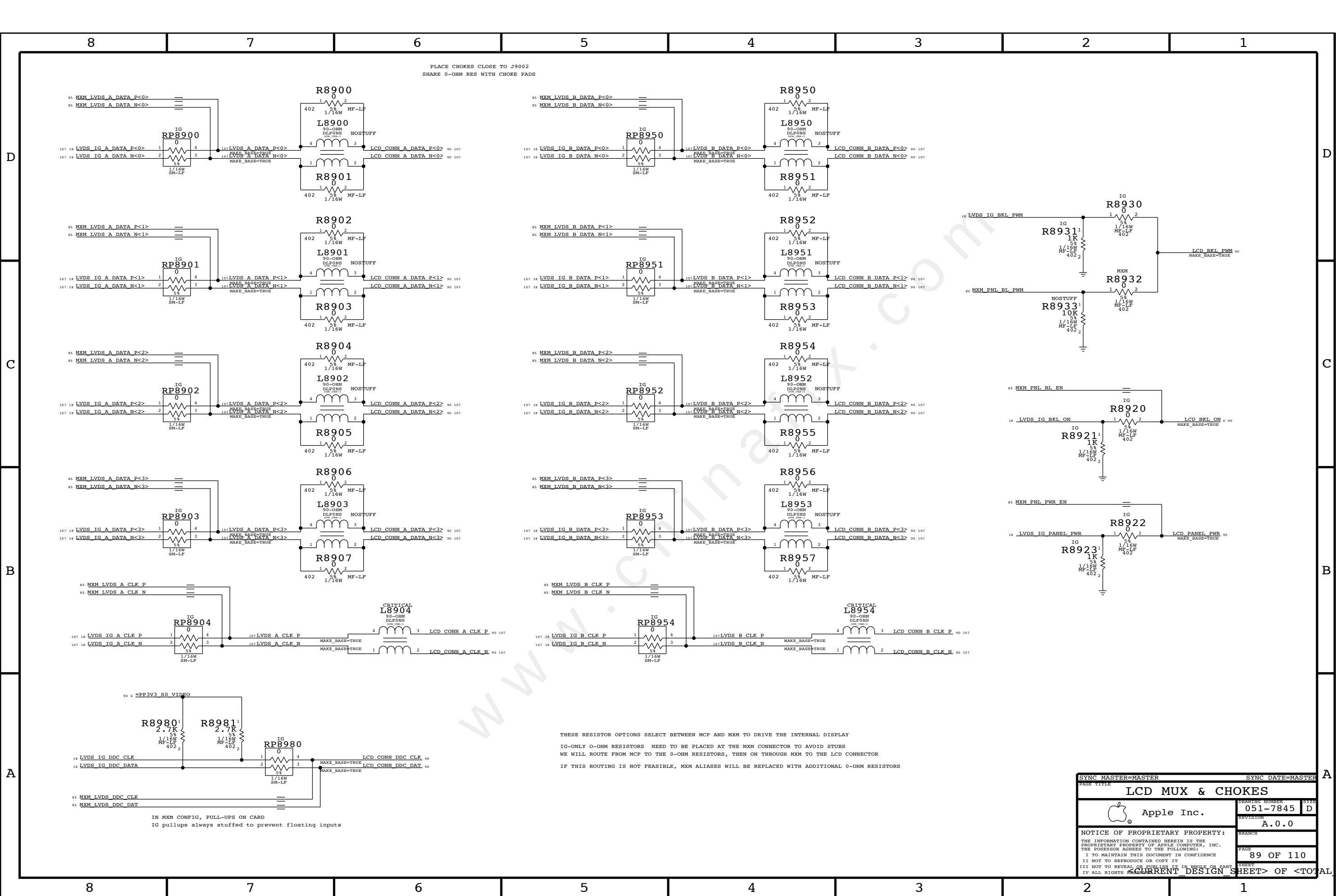
BONCH

PAGE

88 OF 110

SHEET

<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>



Page Notes

Power aliases required by this page:
- =PP12V_S0_LCD
- =PP3V3_S0_VIDEO

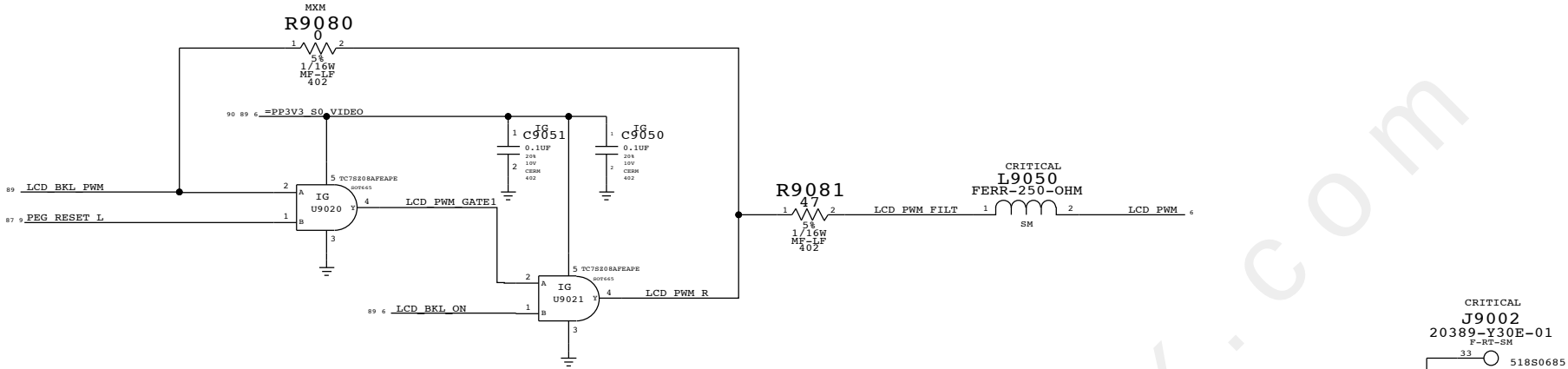
Signal aliases required by this page:
(NONE)

BOM options provided by this page:
IG, MXM

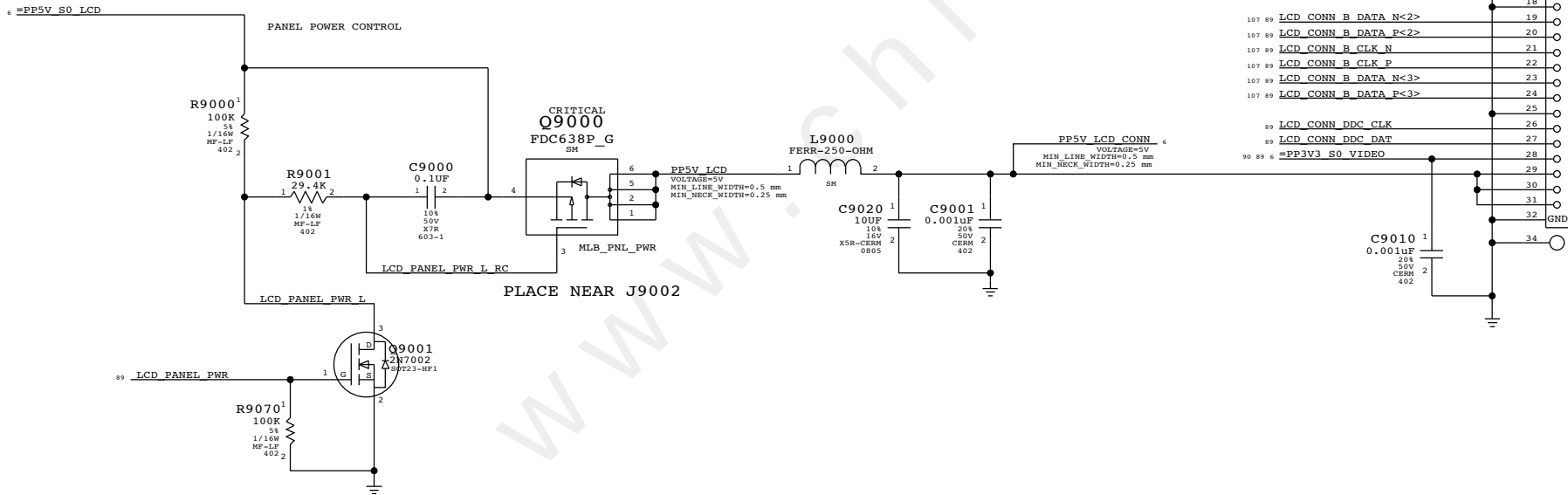
BACKLIGHT CONTROL SUPPORT

THIS AND GATE CIRCUIT PROVIDES BACKLIGHT GLITCH PREVENTION WHEN MCP GLITCHES GPIOs ON POWERUP
IT MAY BE BYPASSED IF THE PWM SOURCE IS THE MXM

IF NOT BYPASSED, THIS CAN BE USED TO FORCE THE USE OF THE BACKLIGHT ENABLE SIGNAL EVEN IF THE INVERTER DOES NOT TAKE THIS AS AN INPUT



INTERNAL LCD INTERFACE



SYNC MASTER=MASTER		SYNC DATE=MASTER	
PAGE TITLE			
INTERNAL DISPLAY			
	DRAWING NUMBER		SIZE
	051-7845		D
REVISION		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		90 OF 110	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		DESIGN SHEETS	
IV ALL RIGHTS RESERVED		<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>	

D

C

B

A

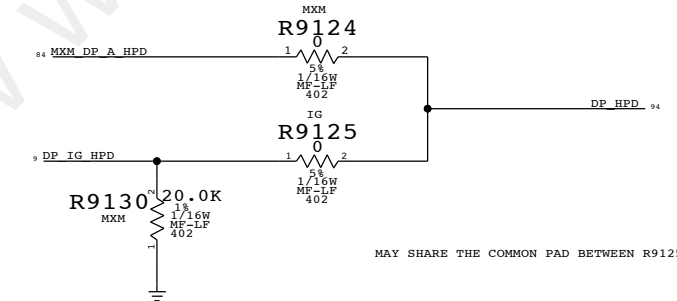
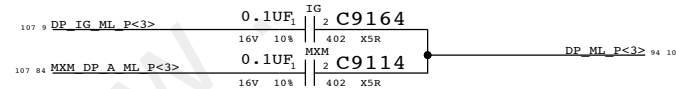
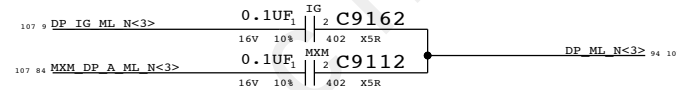
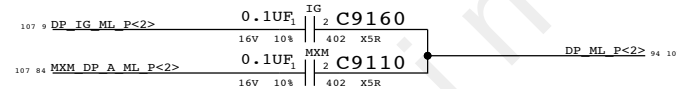
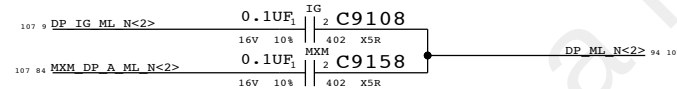
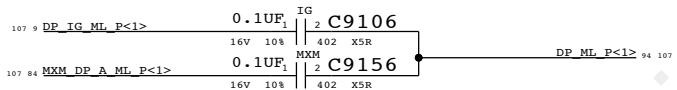
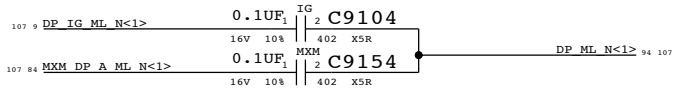
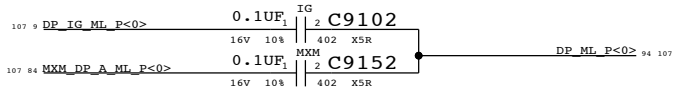
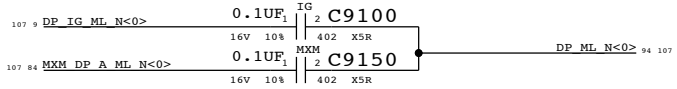
D

C

B

A

K50 NOTE: PLACE THESE CAPACITORS ATLEAST 1INCH AWAY FROM DP CONNECTOR
DCOX: PLACE AT MXM CONNECTOR IF THERE IS ROOM



SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
DP MUX SUPPORT			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
		REVISION	
		A.0.0	
		BRANCH	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
CURRENT DESIGN SHEET		PAGE	
		91 OF 110	
		SHEET	
		SHEET	OF <TOTAL>

<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>



8	7	6	5	4	3	2	1
D							
C							
B							
A							
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51

SYNC DATE=10/01/2008

PAGE TITLE

BLANK PAGE



Apple Inc.

DRAWING NUMBER

051-7845

SIZE

D

REVISION

A.0.0

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

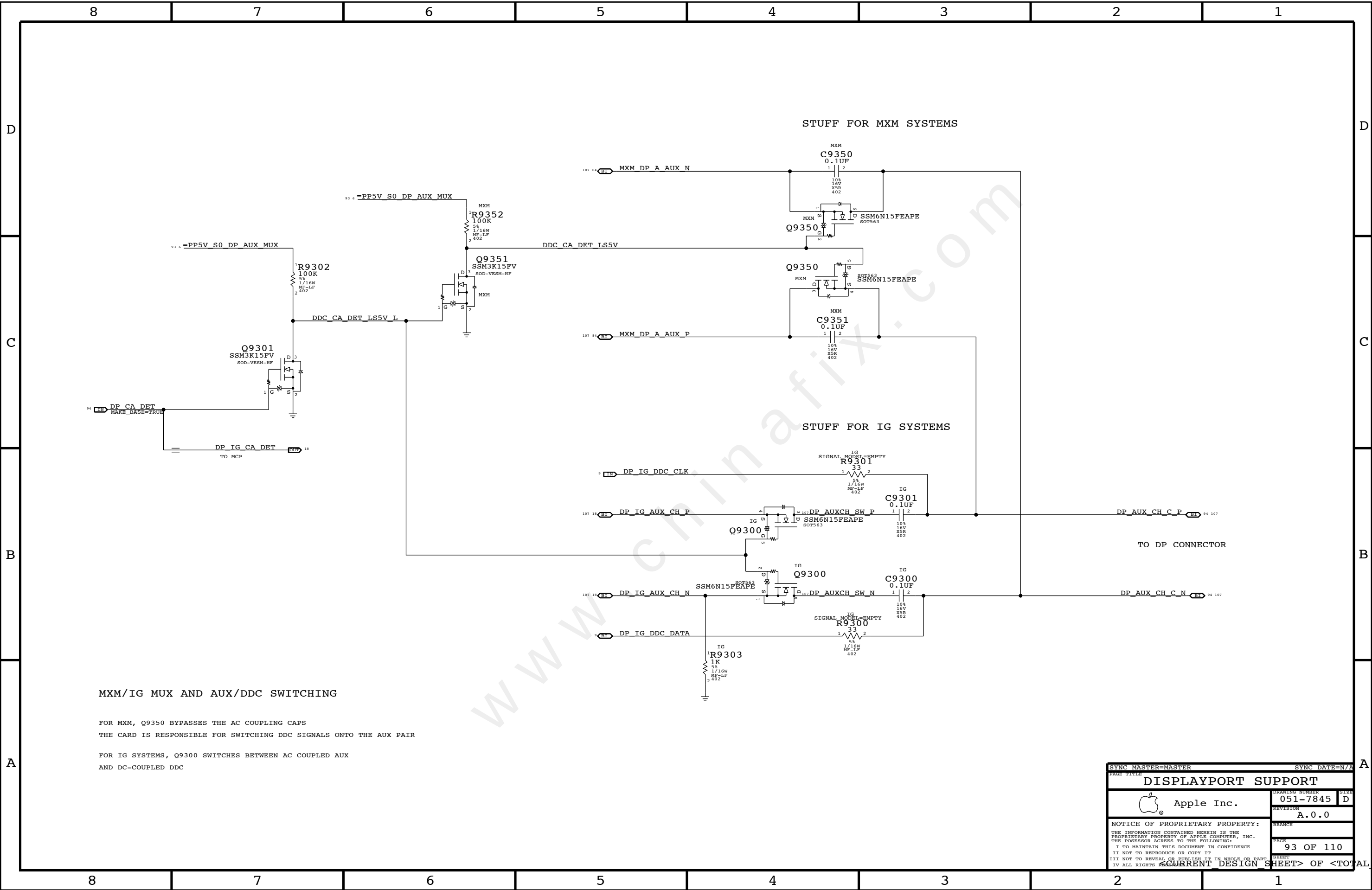
BONCH

PAGE

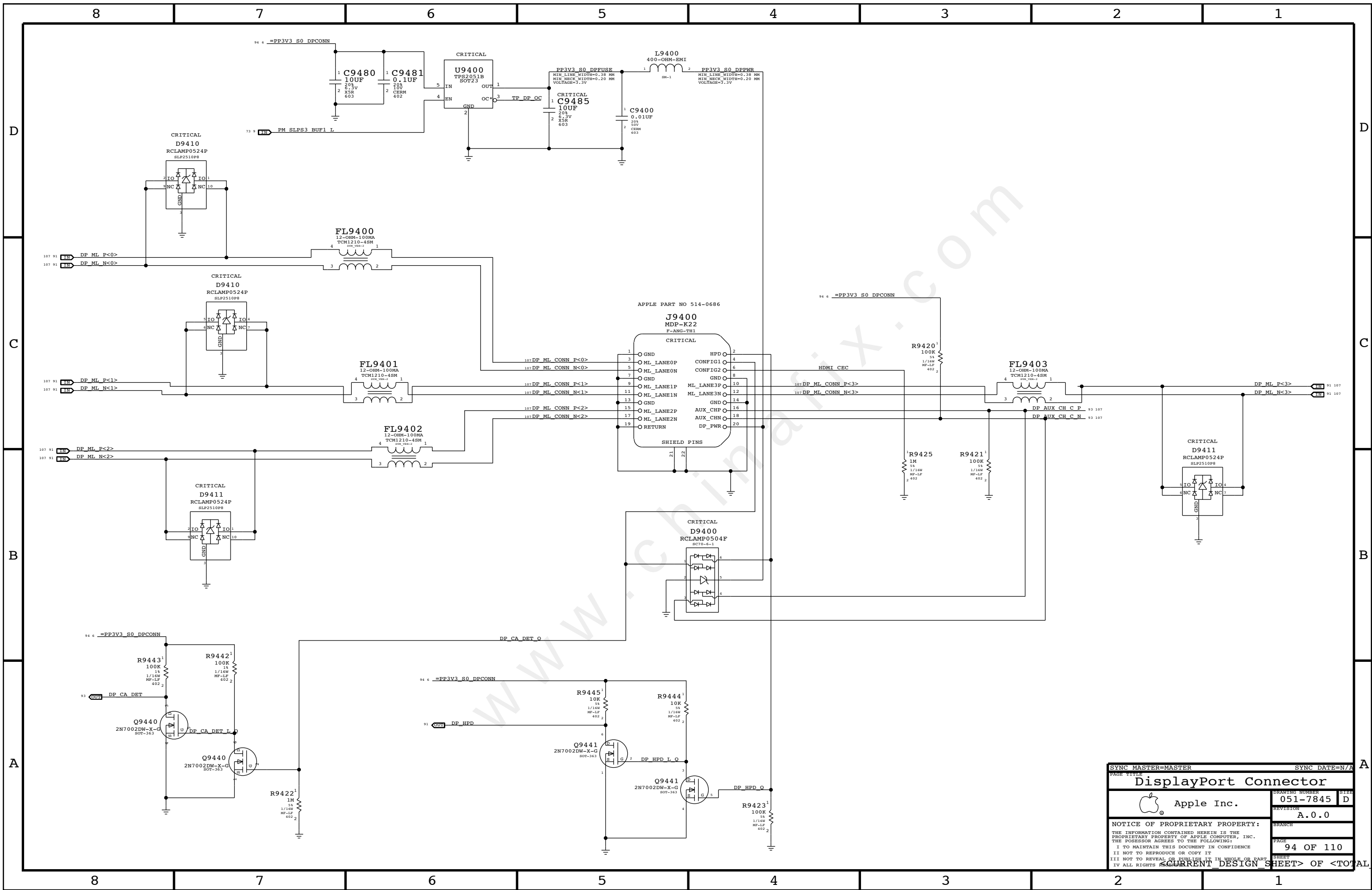
92 OF 110

SHEET

<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>



SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
DISPLAYPORT SUPPORT			
 Apple Inc.		DRAWING NUMBER	SHEET
		051-7845	D
		REVISION	
		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		93 OF 110	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>	
IV ALL RIGHTS RESERVED			



SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
DisplayPort Connector			
 Apple Inc.		DRAWING NUMBER	SHEET
		051-7845	D
		REVISION	
		A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		94	OF 110
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
CURRENT DESIGN SHEET			OF <TOTAL>



	8	7	6	5	4	3	2	1
D								
C								
B								
A								
	8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		95 OF 110	
		SHEET	

[illegible]



8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	97 OF 110
		SHEET	



	8	7	6	5	4	3	2	1
D								
C								
B								
A								
	8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	
		98 OF 110	
		SHEET	
<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>			



8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

www.chinafix.com

SYNC MASTER=K51		SYNC DATE=12/08/2008	
PAGE TITLE			
BLANK PAGE			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7845	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		A.0.0	
		BONCH	
		PAGE	99 OF 110
		SHEET	



8

7

6

5

4

3

2

1

FSB (Front-Side Bus) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FSB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
FSB_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
FSB_DSTB_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	*	=2x_DIELECTRIC	?
FSB_DSTB	*	=3x_DIELECTRIC	?
FSB_ADDR	*	=STANDARD	?
FSB_ADSTB	*	=2x_DIELECTRIC	?
FSB_1X	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_DSTB	TOP,BOTTOM	=5x_DIELECTRIC	?
FSB_ADDR	TOP,BOTTOM	=3x_DIELECTRIC	?
FSB_ADSTB	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_1X	TOP,BOTTOM	=3x_DIELECTRIC	?

All 2x/1x/Async FSB signals with impedance requirements are 50-ohm single-ended.
All 4x FSB signals with impedance requirements are 42-ohm single-ended.

FSB 4x signals / groups shown in signal table on right.
Signals within each 4x group should be matched within 5 ps of strobe.
DSTB# complementary pairs should be matched within 1 ps of each other, all DSTB#s matched to +/- 90 ps. (Tighter than MCP79)
Spacing is 2x dielectric between DATA#, DINV# signals, with 3x dielectric spacing to the DSTB#s.
DSTB# complementary pairs are spaced normally and are NOT routed as differential pairs.

FSB 2X signals / groups shown in signal table on right.
Signals within each 2x group should be matched within 20 ps. ADSTB#s should be matched +/- 300 ps.
Spacing is 1x dielectric between ADDR#, REQ# signals, with 2x dielectric spacing to ADSTB#s.

FSB 1X signals shown in signal table on right.
Signals within each 1x group should be matched to CPU clock, +/-1000 mils.
Design Guide recommends each strobe/signal group is routed on the same layer.
Intel Design Guide recommends FSB signals be routed only on internal layers.

NOTE: Intel Design Guide allows closer spacing if signal lengths can be shortened.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2
SOURCE: Santa Rosa Platform DG, Rev 1.5 (#22294), Sections 4.2 & 4.3

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	0.175 MM	0.175 MM

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_SMIL	*	0.2 MM	?
CPU_COMP	*	0.6 MM	?
CPU_GTLREF	*	0.6 MM	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	0.6 MM	?

SR DG recommends at least 25 mils, >50 mils preferred

MOST CPU SIGNALS WITH IMPEDANCE REQUIREMENTS ARE 50-OHM SINGLE-ENDED.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2
SOURCE: Santa Rosa Platform DG, Rev 0.9 (#20517), Sections 4.4 & 5.8.2.4

MCP FSB COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_FSB_COMP	*	0.2 MM	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.4

FSB Clock Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_FSB_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.5

CPU / FSB Net Properties

NET_TYPE

ELECTRICAL_CONSTRAINT_SET

PHYSICAL

SPACING

Group 0

FSB_42S

FSB_DATA

FSB D L<15..0>

10 14

Group 1

FSB_42S

FSB_DATA

FSB DINV L<0>

10 14

Group 2

FSB_42S

FSB_DATA

FSB DSTB L P<0>

10 14

Group 3

FSB_42S

FSB_DATA

FSB DSTB L N<0>

10 14

Group 0

FSB_42S

FSB_DATA

FSB D L<31..16>

10 14

Group 1

FSB_42S

FSB_DATA

FSB DINV L<1>

10 14

Group 2

FSB_42S

FSB_DATA

FSB DSTB L P<1>

10 14

Group 3

FSB_42S

FSB_DATA

FSB DSTB L N<1>

10 14

Group 0

FSB_42S

FSB_DATA

FSB D L<47..32>

10 14

Group 1

FSB_42S

FSB_DATA

FSB DINV L<2>

10 14

Group 2

FSB_42S

FSB_DATA

FSB DSTB L P<2>

10 14

Group 3

FSB_42S

FSB_DATA

FSB DSTB L N<2>

10 14

Group 0

FSB_42S

FSB_DATA

FSB D L<63..48>

10 14

Group 1

FSB_42S

FSB_DATA

FSB DINV L<3>

10 14

Group 2

FSB_42S

FSB_DATA

FSB DSTB L P<3>

10 14

Group 3

FSB_42S

FSB_DATA

FSB DSTB L N<3>

10 14

Group 0

FSB_50S

FSB_ADDR

FSB A L<16..3>

10 14

Group 1

FSB_50S

FSB_ADDR

FSB REQ L<4..0>

10 14

Group 2

FSB_50S

FSB_ADDR

FSB ADSTB L<0>

10 14

Group 3

FSB_50S

FSB_ADDR

FSB ADSTB L<1>

10 14

Group 0

FSB_50S

FSB_1X

FSB ADS L

10 14

Group 1

FSB_50S

FSB_1X

FSB BREQ0 L

10 14

Group 2

FSB_50S

FSB_1X

FSB BREQ1 L

10 14

Group 3

FSB_50S

FSB_1X

FSB BNR L

10 14

Group 4

FSB_50S

FSB_1X

FSB BPRI L

10 14

Group 5

FSB_50S

FSB_1X

FSB DBSY L

10 14

Group 6

FSB_50S

FSB_1X

FSB DEFER L

10 14

Group 7

FSB_50S

FSB_1X

FSB DRDY L

10 14

Group 8

FSB_50S

FSB_1X

FSB HIT L

10 14

Group 9

FSB_50S

FSB_1X

FSB HITM L

10 14

Group 10

FSB_50S

FSB_1X

FSB LOCK L

10 14

Group 11

FSB_50S

FSB_1X

FSB CPURST L

10 13 14

Group 12

FSB_50S

FSB_1X

FSB RS L<2..0>

10 14

Group 13

FSB_50S

FSB_1X

FSB TRDY L

10 14

Group 14

CPU_50S

CPU_AGTL

CPU A20M L

10 14

Group 15

CPU_50S

CPU_AGTL

CPU BSEL<2..0>

10 14

Group 16

CPU_50S

CPU_BN1L

CPU FERR L

10 14

Group 17

CPU_50S

CPU_AGTL

CPU IGNNE L

10 14

Group 18

CPU_50S

CPU_AGTL

CPU INIT L

10 14

Group 19

CPU_50S

CPU_AGTL

CPU INTR

10 14

Group 20

CPU_50S

CPU_AGTL

CPU NMI

10 14

Group 21

CPU_50S

CPU_AGTL

CPU PROCHOT L

11 14 50

Group 22

CPU_50S

CPU_AGTL

CPU PWRGD

11 13 14

Group 23

CPU_50S

CPU_AGTL

CPU SMI L

10 14

Group 24

CPU_50S

CPU_AGTL

CPU STRCLK L

10 14

Group 25

CPU_50S

CPU_BN1L

PM THRMTRIP L

11 14 50

Group 26

CPU_50S

CPU_AGTL

FSB CPUSLP L

11 14

Group 27

CPU_50S

CPU_AGTL

CPU DPSLP L

11 14

Group 28

CPU_50S

CPU_AGTL

CPU DPRSTP L

11 14

Group 29

MCP_50S

MCP_FSB_COMP

MCP BCLK VML COMP VDD

14

Group 30

MCP_50S

MCP_FSB_COMP

MCP BCLK VML COMP GND

14

Group 31

MCP_50S

MCP_FSB_COMP

MCP CPU COMP VCC

14

Group 32

MCP_50S

MCP_FSB_COMP

MCP CPU COMP GND

14

Group 33

CLK_FSB_100D

CLK_FSB

FSB CLK CPU P

10 14

Group 34

CLK_FSB_100D

CLK_FSB

FSB CLK CPU N

10 14

Group 35

CLK_FSB_100D

CLK_FSB

FSB CLK ITP P

13 14

Group 36

CLK_FSB_100D

CLK_FSB

FSB CLK ITP N

13 14

Group 37

CLK_FSB_100D

CLK_FSB

FSB CLK MCP P

14

Group 38

CLK_FSB_100D

CLK_FSB

FSB CLK MCP N

14

Group 39

CPU_50S

CPU_AGTL

CPU IERR L

10

Group 40

CPU_50S

CPU_GTLREF

CPU GTLREF0

10 11 29

Group 41

CPU_50S

CPU_GTLREF

CPU GTLREF1

10 11 29

Group 42

CPU_27P4S

CPU_COMP

CPU COMP<8>

11

Group 43

CPU_27P4S

CPU_COMP

CPU COMP<3>

11

Group 44

CPU_27P4S

CPU_COMP

CPU COMP<2>

11

Group 45

CPU_27P4S

CPU_COMP

CPU COMP<1>

11

Group 46

CPU_27P4S

CPU_COMP

CPU COMP<0>

11

Group 47

CPU_50S

CPU_ITP

CPU XDP TDI

11 13

Group 48

CPU_50S

CPU_ITP

CPU XDP TDO

11 13

Group 49

CPU_50S

CPU_ITP

CPU XDP TMS

11 13

Group 50

CPU_50S

CPU_ITP

CPU XDP TCK

11 13

Group 51

CPU_50S

CPU_ITP

CPU XDP TRST L

11 13

Group 52

CPU_50S

CPU_ITP

CPU XDP BPM L<5..0>

11 13

Group 53

CPU_50S

CPU_ITP

CPU XDP BPMB<3..0>

11 13

Group 54

CPU_50S

CPU_ITP

XDP CPURST L

13

Group 55

CPU_50S

CPU_BN1L

CPU VID<7..0>

12 71

Group 56

CPU_27P4S

CPU_VCCSENSE

CPU VCC PKG SENSE P

12 71

Group 57

CPU_27P4S

CPU_VCCSENSE

CPU VCC PKG SENSE N

12 71

Group 58

CPU_27P4S

CPU_VCCSENSE

VR CPU VSNS R P

71

Group 59

CPU_27P4S

CPU_VCCSENSE

VR CPU VSNS R_N

71

FSB 4x Signal Groups

FSB 2x Signals

FSB 1x Signals

8

7

6

5

4

3

2

1

A

B

C

D

SYNC MASTER=MASTER

SYNC DATE=N/A

CPU/FSB Constraints

Apple Inc.

051-7845

A.0.0

100 OF 110

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

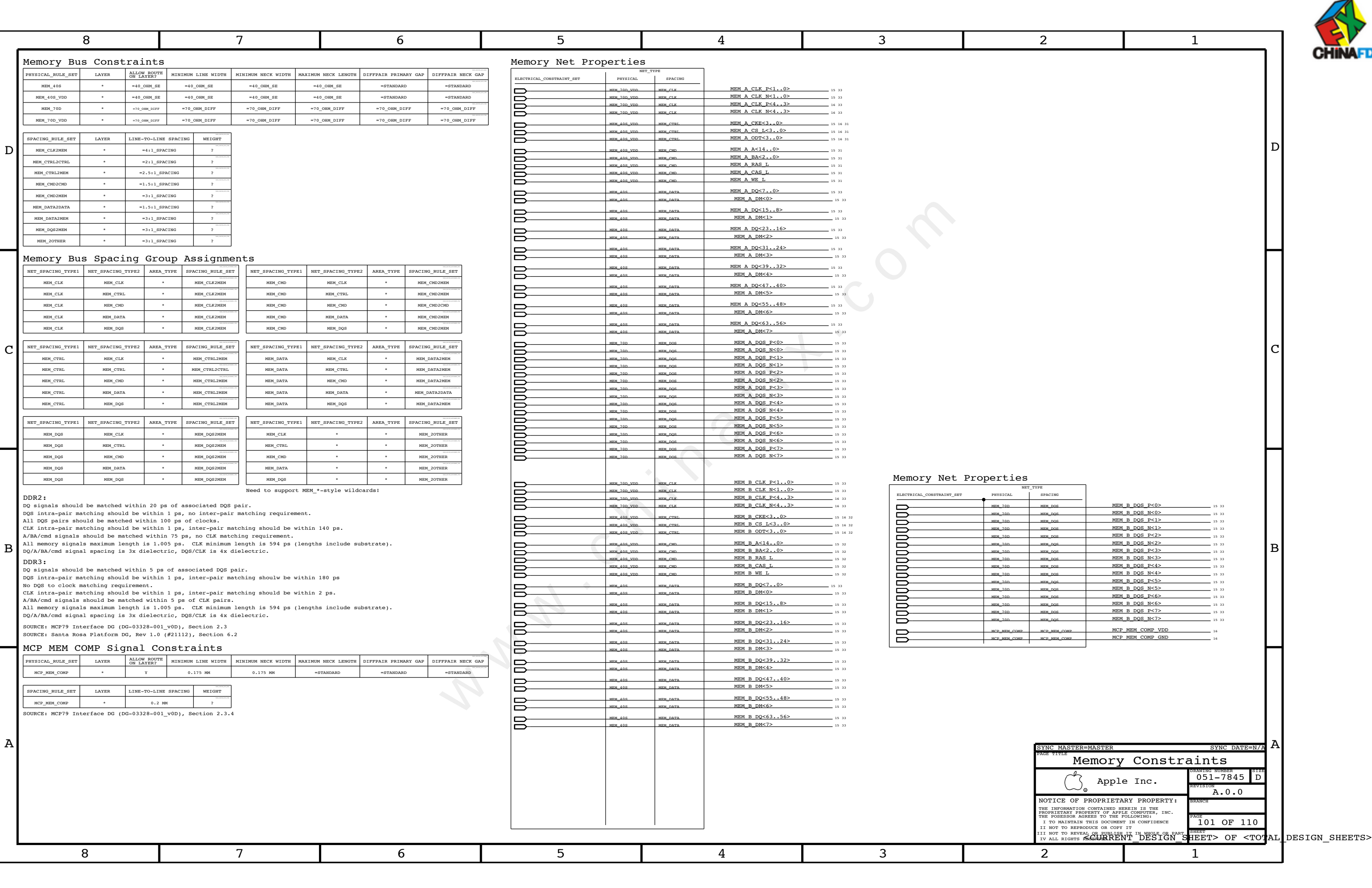
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

CURRENT DESIGN SHEET> OF <TOTAL DESIGN_SHEETS>



[illegible]



8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																																																										
PCI Bus Constraints																																																																																																																																																																																																																																																																																																	
<table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW ROUTE ON LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>PCI_55S</td><td>*</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr><tr><td>CLK_PCI_55S</td><td>*</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr></table>								PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD	CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																		
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																																																																																																																																																																																																																																																																																										
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																																										
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																																										
<table><tr><th>SPACING_RULE_SET</th><th>LAYER</th><th>LINE-TO-LINE SPACING</th><th>WEIGHT</th></tr><tr><td>PCI</td><td>*</td><td>=STANDARD</td><td>?</td></tr><tr><td>CLK_PCI</td><td>*</td><td>0.2 MM</td><td>?</td></tr></table>								SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	PCI	*	=STANDARD	?	CLK_PCI	*	0.2 MM	?																																																																																																																																																																																																																																																																														
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT																																																																																																																																																																																																																																																																																														
PCI	*	=STANDARD	?																																																																																																																																																																																																																																																																																														
CLK_PCI	*	0.2 MM	?																																																																																																																																																																																																																																																																																														
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.8.																																																																																																																																																																																																																																																																																																	
LPC Bus Constraints																																																																																																																																																																																																																																																																																																	
<table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW ROUTE ON LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>LPC_55S</td><td>*</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr><tr><td>CLK_LPC_55S</td><td>*</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr></table>								PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD	CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																		
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																																																																																																																																																																																																																																																																																										
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																																										
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																																										
<table><tr><th>SPACING_RULE_SET</th><th>LAYER</th><th>LINE-TO-LINE SPACING</th><th>WEIGHT</th></tr><tr><td>LPC</td><td>*</td><td>0.15 MM</td><td>?</td></tr><tr><td>CLK_LPC</td><td>*</td><td>0.2 MM</td><td>?</td></tr></table>								SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	LPC	*	0.15 MM	?	CLK_LPC	*	0.2 MM	?																																																																																																																																																																																																																																																																														
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT																																																																																																																																																																																																																																																																																														
LPC	*	0.15 MM	?																																																																																																																																																																																																																																																																																														
CLK_LPC	*	0.2 MM	?																																																																																																																																																																																																																																																																																														
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.9.1.																																																																																																																																																																																																																																																																																																	
USB 2.0 Interface Constraints																																																																																																																																																																																																																																																																																																	
<table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW ROUTE ON LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>MCP_USB_RB1AS</td><td>*</td><td>=STANDARD</td><td>0.2 MM</td><td>0.2 MM</td><td>=STANDARD</td><td>=STANDARD</td><td>=STANDARD</td></tr><tr><td>USB_90D</td><td>*</td><td>=90_OHM_DIFF</td><td>=90_OHM_DIFF</td><td>=90_OHM_DIFF</td><td>=90_OHM_DIFF</td><td>=90_OHM_DIFF</td><td>=90_OHM_DIFF</td></tr></table>								PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	MCP_USB_RB1AS	*	=STANDARD	0.2 MM	0.2 MM	=STANDARD	=STANDARD	=STANDARD	USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF																																																																																																																																																																																																																																																																		
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																																																																																																																																																																																																																																																																																										
MCP_USB_RB1AS	*	=STANDARD	0.2 MM	0.2 MM	=STANDARD	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																																										
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF																																																																																																																																																																																																																																																																																										
<table><tr><th>SPACING_RULE_SET</th><th>LAYER</th><th>LINE-TO-LINE SPACING</th><th>WEIGHT</th></tr><tr><td>USB</td><td>*</td><td>=2x_DIELECTRIC</td><td>?</td></tr></table>								SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	USB	*	=2x_DIELECTRIC	?																																																																																																																																																																																																																																																																																		
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT																																																																																																																																																																																																																																																																																														
USB	*	=2x_DIELECTRIC	?																																																																																																																																																																																																																																																																																														
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.10.1.																																																																																																																																																																																																																																																																																																	
SMBus Interface Constraints																																																																																																																																																																																																																																																																																																	
<table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW ROUTE ON LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>SMB_55S</td><td>*</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr></table>								PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																										
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																																																																																																																																																																																																																																																																																										
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																																										
<table><tr><th>SPACING_RULE_SET</th><th>LAYER</th><th>LINE-TO-LINE SPACING</th><th>WEIGHT</th></tr><tr><td>SMB</td><td>*</td><td>=2x_DIELECTRIC</td><td>?</td></tr></table>								SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SMB	*	=2x_DIELECTRIC	?																																																																																																																																																																																																																																																																																		
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT																																																																																																																																																																																																																																																																																														
SMB	*	=2x_DIELECTRIC	?																																																																																																																																																																																																																																																																																														
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.11.1.																																																																																																																																																																																																																																																																																																	
HD Audio Interface Constraints																																																																																																																																																																																																																																																																																																	
<table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW ROUTE ON LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>HDA_55S</td><td>*</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr></table>								PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																										
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																																																																																																																																																																																																																																																																																										
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																																										
<table><tr><th>SPACING_RULE_SET</th><th>LAYER</th><th>LINE-TO-LINE SPACING</th><th>WEIGHT</th></tr><tr><td>HDA</td><td>*</td><td>=2x_DIELECTRIC</td><td>?</td></tr><tr><td>MCP_HDA_COMP</td><td>*</td><td>0.2 MM</td><td>?</td></tr></table>								SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	HDA	*	=2x_DIELECTRIC	?	MCP_HDA_COMP	*	0.2 MM	?																																																																																																																																																																																																																																																																														
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT																																																																																																																																																																																																																																																																																														
HDA	*	=2x_DIELECTRIC	?																																																																																																																																																																																																																																																																																														
MCP_HDA_COMP	*	0.2 MM	?																																																																																																																																																																																																																																																																																														
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.12.1.																																																																																																																																																																																																																																																																																																	
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.13.																																																																																																																																																																																																																																																																																																	
SPI Interface Constraints																																																																																																																																																																																																																																																																																																	
<table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW ROUTE ON LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>SPI_55S</td><td>*</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr></table>								PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																										
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																																																																																																																																																																																																																																																																																										
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																																										
<table><tr><th>SPACING_RULE_SET</th><th>LAYER</th><th>LINE-TO-LINE SPACING</th><th>WEIGHT</th></tr><tr><td>SPI</td><td>*</td><td>0.2 MM</td><td>?</td></tr></table>								SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPI	*	0.2 MM	?																																																																																																																																																																																																																																																																																		
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT																																																																																																																																																																																																																																																																																														
SPI	*	0.2 MM	?																																																																																																																																																																																																																																																																																														
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.14.																																																																																																																																																																																																																																																																																																	
XTAL Constraints																																																																																																																																																																																																																																																																																																	
<table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW ROUTE ON LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>CLK_MCP_XTAL</td><td>*</td><td>=100_OHM_DIFF</td><td>=100_OHM_DIFF</td><td>=100_OHM_DIFF</td><td>=100_OHM_DIFF</td><td>=100_OHM_DIFF</td><td>=100_OHM_DIFF</td></tr></table>								PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	CLK_MCP_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF																																																																																																																																																																																																																																																																										
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																																																																																																																																																																																																																																																																																										
CLK_MCP_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF																																																																																																																																																																																																																																																																																										
<table><tr><th>SPACING_RULE_SET</th><th>LAYER</th><th>LINE-TO-LINE SPACING</th><th>WEIGHT</th></tr><tr><td>XTAL</td><td>*</td><td>=4x_DIELECTRIC</td><td>?</td></tr></table>								SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	XTAL	*	=4x_DIELECTRIC	?																																																																																																																																																																																																																																																																																		
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT																																																																																																																																																																																																																																																																																														
XTAL	*	=4x_DIELECTRIC	?																																																																																																																																																																																																																																																																																														
<table><tr><th colspan="3">ELECTRICAL_CONSTRAINT_SET</th><th colspan="2">NET_TYPE</th><th></th></tr><tr><th colspan="3"></th><th>PHYSICAL</th><th>SPACING</th><th></th></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>PCI_55S</td><td>PCI</td><td>PCI REQ0 L 19</td></tr><tr><td>PCI_55S</td><td>PCI</td><td>PCI REQ1 L 19</td></tr><tr><td>CLK_PCI_55S</td><td>CLK_PCI</td><td>PCI CLK33M MCP R 19</td></tr><tr><td>CLK_PCI_55S</td><td>CLK_PCI</td><td>PCI CLK33M MCP 19</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>LPC_55S</td><td>LPC</td><td>LPC AD<3..0> 19 49 51</td></tr><tr><td>LPC_55S</td><td>LPC</td><td>LPC AD R<3..0> 19</td></tr><tr><td>LPC_55S</td><td>LPC</td><td>LPC FRAME L 19 49 51</td></tr><tr><td>LPC_55S</td><td>LPC</td><td>LPC FRAME R L 19</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>LPC_55S</td><td>LPC</td><td>LPC RESET L 9 19</td></tr><tr><td>CLK_LPC_55S</td><td>CLK_LPC</td><td>LPC CLK33M SMC R 9 19</td></tr><tr><td>CLK_LPC_55S</td><td>CLK_LPC</td><td>LPC CLK33M SMC 9 49</td></tr><tr><td>CLK_LPC_55S</td><td>CLK_LPC</td><td>LPC CLK33M LPCPLUS 9 51</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>CLK_LPC_55S</td><td>CLK_LPC</td><td>PM CLK32K SUSCLK R 9 21</td></tr><tr><td>CLK_LPC_55S</td><td>CLK_LPC</td><td>PM CLK32K SUSCLK 9 49</td></tr><tr><td>MCP_USB_RB1AS</td><td></td><td>MCP USB_RB1AS_GND 20</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_EXT A_P 20 46</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>USB_90D</td><td>USB</td><td>USB_EXT A_N 20 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_PORT0_P 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_PORT0_N 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_EXT B_P 20 46</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>USB_90D</td><td>USB</td><td>USB_EXT B_N 20 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_PORT1_P 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_PORT1_N 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_EXTC_P 20 46</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>USB_90D</td><td>USB</td><td>USB_EXTC_N 20 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_PORT2_P 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_PORT2_N 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_EXT D_P 20 46</td></tr><tr><td rowspan="3"></td><td rowspan="3"></td><td rowspan="3"></td><td>USB_90D</td><td>USB</td><td>USB_EXT D_N 20 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_D_MUXED_P 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_D_MUXED_N 46</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>USB_90D</td><td>USB</td><td>USB_PORT3_P 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_PORT3_N 46</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_CAMERA_P 20 47</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_CAMERA_N 20 47</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>USB_90D</td><td>USB</td><td>USB_CAMERA_I_P 47 110</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_CAMERA_I_N 47 110</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_BT_P 20 47</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_BT_N 20 47</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>USB_90D</td><td>USB</td><td>USB_BT_L_P 47 110</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_BT_L_N 47 110</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_IR_P 20 47</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_IR_N 20 47</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>USB_90D</td><td>USB</td><td>USB_IR_L_P 47 110</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_IR_L_N 47 110</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_SDCARD_P 20 47</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_SDCARD_N 20 47</td></tr><tr><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td>USB_90D</td><td>USB</td><td>USB_SDCARD_I_P 47 110</td></tr><tr><td>USB_90D</td><td>USB</td><td>USB_SDCARD_I_N 47 110</td></tr><tr><td rowspan="4"></td><td rowspan="14"></td><td rowspan="14"></td><td>MCP_55S</td><td>SPI</td><td>SPI_CLK_R 21 51 61</td></tr><tr><td>MCP_55S</td><td>SPI</td><td>SPI_CLK 41</td></tr><tr><td>MCP_55S</td><td>SPI</td><td>SPI_MOSI_R 21 51 61</td></tr><tr><td>MCP_55S</td><td>SPI</td><td>SPI_MOSI 41</td></tr><tr><td>MCP_55S</td><td>SPI</td><td>SPI_MISO 21 51 61</td></tr><tr><td>MCP_55S</td><td>SPI</td><td>SPI_MISO_R 41</td></tr><tr><td>MCP_55S</td><td>SPI</td><td>SPI_CS0_R_L 21 51</td></tr><tr><td>MCP_55S</td><td>SPI</td><td>SPI_CS0_L 51</td></tr><tr><td>HDA_55S</td><td>HDA</td><td>HDA_BIT_CLK 21 62</td></tr><tr><td>MCP_HDA_COMP</td><td></td><td>MCP_HDA_PULLDN_COMP 21</td></tr><tr><td>HDA_55S</td><td>HDA</td><td>HDA_BIT_CLK_R 21</td></tr><tr><td>HDA_55S</td><td>HDA</td><td>HDA_RST_L 21 62</td></tr><tr><td>HDA_55S</td><td>HDA</td><td>HDA_RST_R_L 21</td></tr><tr><td>HDA_55S</td><td>HDA</td><td>HDA_SDOUT 21 62</td></tr><tr><td>HDA_55S</td><td>HDA</td><td>HDA_SDOUT_R 21</td></tr><tr><td>HDA_55S</td><td>HDA</td><td>HDA_SYNC 21 62</td></tr><tr><td>HDA_55S</td><td>HDA</td><td>HDA_SYNC_R 21</td></tr><tr><td>HDA_55S</td><td>HDA</td><td>HDA_SDINO 21 62</td></tr><tr><td>HDA_55S</td><td>HDA</td><td>AUD_SDI_R 62</td></tr><tr><td></td><td>HDA</td><td>AUD_SPDIF_IN 9 66</td></tr><tr><td></td><td>HDA</td><td>AUD_SPDIF_OUT 62 66</td></tr><tr><td></td><td>HDA</td><td>AUD_SPDIF_CHIP 62</td></tr><tr><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4"></td><td>CLK_MCP_XTAL</td><td>XTAL</td><td>MCP_CLK25M_XTALOUT 21 28</td></tr><tr><td>CLK_MCP_XTAL</td><td>XTAL</td><td>MCP_CLK25M_XTALIN 21 28</td></tr><tr><td>CLK_MCP_XTAL</td><td>XTAL</td><td>RTC_CLK32K_XTALOUT 21 28</td></tr><tr><td>CLK_MCP_XTAL</td><td>XTAL</td><td>RTC_CLK32K_XTALIN 21 28</td></tr></table>								ELECTRICAL_CONSTRAINT_SET			NET_TYPE						PHYSICAL	SPACING					PCI_55S	PCI	PCI REQ0 L 19	PCI_55S	PCI	PCI REQ1 L 19	CLK_PCI_55S	CLK_PCI	PCI CLK33M MCP R 19	CLK_PCI_55S	CLK_PCI	PCI CLK33M MCP 19				LPC_55S	LPC	LPC AD<3..0> 19 49 51	LPC_55S	LPC	LPC AD R<3..0> 19	LPC_55S	LPC	LPC FRAME L 19 49 51	LPC_55S	LPC	LPC FRAME R L 19				LPC_55S	LPC	LPC RESET L 9 19	CLK_LPC_55S	CLK_LPC	LPC CLK33M SMC R 9 19	CLK_LPC_55S	CLK_LPC	LPC CLK33M SMC 9 49	CLK_LPC_55S	CLK_LPC	LPC CLK33M LPCPLUS 9 51				CLK_LPC_55S	CLK_LPC	PM CLK32K SUSCLK R 9 21	CLK_LPC_55S	CLK_LPC	PM CLK32K SUSCLK 9 49	MCP_USB_RB1AS		MCP USB_RB1AS_GND 20	USB_90D	USB	USB_EXT A_P 20 46				USB_90D	USB	USB_EXT A_N 20 46	USB_90D	USB	USB_PORT0_P 46	USB_90D	USB	USB_PORT0_N 46	USB_90D	USB	USB_EXT B_P 20 46				USB_90D	USB	USB_EXT B_N 20 46	USB_90D	USB	USB_PORT1_P 46	USB_90D	USB	USB_PORT1_N 46	USB_90D	USB	USB_EXTC_P 20 46				USB_90D	USB	USB_EXTC_N 20 46	USB_90D	USB	USB_PORT2_P 46	USB_90D	USB	USB_PORT2_N 46	USB_90D	USB	USB_EXT D_P 20 46				USB_90D	USB	USB_EXT D_N 20 46	USB_90D	USB	USB_D_MUXED_P 46	USB_90D	USB	USB_D_MUXED_N 46				USB_90D	USB	USB_PORT3_P 46	USB_90D	USB	USB_PORT3_N 46	USB_90D	USB	USB_CAMERA_P 20 47	USB_90D	USB	USB_CAMERA_N 20 47				USB_90D	USB	USB_CAMERA_I_P 47 110	USB_90D	USB	USB_CAMERA_I_N 47 110	USB_90D	USB	USB_BT_P 20 47	USB_90D	USB	USB_BT_N 20 47				USB_90D	USB	USB_BT_L_P 47 110	USB_90D	USB	USB_BT_L_N 47 110	USB_90D	USB	USB_IR_P 20 47	USB_90D	USB	USB_IR_N 20 47				USB_90D	USB	USB_IR_L_P 47 110	USB_90D	USB	USB_IR_L_N 47 110	USB_90D	USB	USB_SDCARD_P 20 47	USB_90D	USB	USB_SDCARD_N 20 47				USB_90D	USB	USB_SDCARD_I_P 47 110	USB_90D	USB	USB_SDCARD_I_N 47 110				MCP_55S	SPI	SPI_CLK_R 21 51 61	MCP_55S	SPI	SPI_CLK 41	MCP_55S	SPI	SPI_MOSI_R 21 51 61	MCP_55S	SPI	SPI_MOSI 41	MCP_55S	SPI	SPI_MISO 21 51 61	MCP_55S	SPI	SPI_MISO_R 41	MCP_55S	SPI	SPI_CS0_R_L 21 51	MCP_55S	SPI	SPI_CS0_L 51	HDA_55S	HDA	HDA_BIT_CLK 21 62	MCP_HDA_COMP		MCP_HDA_PULLDN_COMP 21	HDA_55S	HDA	HDA_BIT_CLK_R 21	HDA_55S	HDA	HDA_RST_L 21 62	HDA_55S	HDA	HDA_RST_R_L 21	HDA_55S	HDA	HDA_SDOUT 21 62	HDA_55S	HDA	HDA_SDOUT_R 21	HDA_55S	HDA	HDA_SYNC 21 62	HDA_55S	HDA	HDA_SYNC_R 21	HDA_55S	HDA	HDA_SDINO 21 62	HDA_55S	HDA	AUD_SDI_R 62		HDA	AUD_SPDIF_IN 9 66		HDA	AUD_SPDIF_OUT 62 66		HDA	AUD_SPDIF_CHIP 62				CLK_MCP_XTAL	XTAL	MCP_CLK25M_XTALOUT 21 28	CLK_MCP_XTAL	XTAL	MCP_CLK25M_XTALIN 21 28	CLK_MCP_XTAL	XTAL	RTC_CLK32K_XTALOUT 21 28	CLK_MCP_XTAL	XTAL	RTC_CLK32K_XTALIN 21 28
ELECTRICAL_CONSTRAINT_SET			NET_TYPE																																																																																																																																																																																																																																																																																														
			PHYSICAL	SPACING																																																																																																																																																																																																																																																																																													
			PCI_55S	PCI	PCI REQ0 L 19																																																																																																																																																																																																																																																																																												
			PCI_55S	PCI	PCI REQ1 L 19																																																																																																																																																																																																																																																																																												
			CLK_PCI_55S	CLK_PCI	PCI CLK33M MCP R 19																																																																																																																																																																																																																																																																																												
			CLK_PCI_55S	CLK_PCI	PCI CLK33M MCP 19																																																																																																																																																																																																																																																																																												
			LPC_55S	LPC	LPC AD<3..0> 19 49 51																																																																																																																																																																																																																																																																																												
			LPC_55S	LPC	LPC AD R<3..0> 19																																																																																																																																																																																																																																																																																												
			LPC_55S	LPC	LPC FRAME L 19 49 51																																																																																																																																																																																																																																																																																												
			LPC_55S	LPC	LPC FRAME R L 19																																																																																																																																																																																																																																																																																												
			LPC_55S	LPC	LPC RESET L 9 19																																																																																																																																																																																																																																																																																												
			CLK_LPC_55S	CLK_LPC	LPC CLK33M SMC R 9 19																																																																																																																																																																																																																																																																																												
			CLK_LPC_55S	CLK_LPC	LPC CLK33M SMC 9 49																																																																																																																																																																																																																																																																																												
			CLK_LPC_55S	CLK_LPC	LPC CLK33M LPCPLUS 9 51																																																																																																																																																																																																																																																																																												
			CLK_LPC_55S	CLK_LPC	PM CLK32K SUSCLK R 9 21																																																																																																																																																																																																																																																																																												
			CLK_LPC_55S	CLK_LPC	PM CLK32K SUSCLK 9 49																																																																																																																																																																																																																																																																																												
			MCP_USB_RB1AS		MCP USB_RB1AS_GND 20																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_EXT A_P 20 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_EXT A_N 20 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_PORT0_P 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_PORT0_N 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_EXT B_P 20 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_EXT B_N 20 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_PORT1_P 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_PORT1_N 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_EXTC_P 20 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_EXTC_N 20 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_PORT2_P 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_PORT2_N 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_EXT D_P 20 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_EXT D_N 20 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_D_MUXED_P 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_D_MUXED_N 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_PORT3_P 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_PORT3_N 46																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_CAMERA_P 20 47																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_CAMERA_N 20 47																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_CAMERA_I_P 47 110																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_CAMERA_I_N 47 110																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_BT_P 20 47																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_BT_N 20 47																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_BT_L_P 47 110																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_BT_L_N 47 110																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_IR_P 20 47																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_IR_N 20 47																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_IR_L_P 47 110																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_IR_L_N 47 110																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_SDCARD_P 20 47																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_SDCARD_N 20 47																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_SDCARD_I_P 47 110																																																																																																																																																																																																																																																																																												
			USB_90D	USB	USB_SDCARD_I_N 47 110																																																																																																																																																																																																																																																																																												
			MCP_55S	SPI	SPI_CLK_R 21 51 61																																																																																																																																																																																																																																																																																												
			MCP_55S	SPI	SPI_CLK 41																																																																																																																																																																																																																																																																																												
			MCP_55S	SPI	SPI_MOSI_R 21 51 61																																																																																																																																																																																																																																																																																												
			MCP_55S	SPI	SPI_MOSI 41																																																																																																																																																																																																																																																																																												
MCP_55S			SPI	SPI_MISO 21 51 61																																																																																																																																																																																																																																																																																													
MCP_55S			SPI	SPI_MISO_R 41																																																																																																																																																																																																																																																																																													
MCP_55S			SPI	SPI_CS0_R_L 21 51																																																																																																																																																																																																																																																																																													
MCP_55S			SPI	SPI_CS0_L 51																																																																																																																																																																																																																																																																																													
HDA_55S			HDA	HDA_BIT_CLK 21 62																																																																																																																																																																																																																																																																																													
MCP_HDA_COMP				MCP_HDA_PULLDN_COMP 21																																																																																																																																																																																																																																																																																													
HDA_55S			HDA	HDA_BIT_CLK_R 21																																																																																																																																																																																																																																																																																													
HDA_55S			HDA	HDA_RST_L 21 62																																																																																																																																																																																																																																																																																													
HDA_55S			HDA	HDA_RST_R_L 21																																																																																																																																																																																																																																																																																													
HDA_55S			HDA	HDA_SDOUT 21 62																																																																																																																																																																																																																																																																																													
HDA_55S	HDA	HDA_SDOUT_R 21																																																																																																																																																																																																																																																																																															
HDA_55S	HDA	HDA_SYNC 21 62																																																																																																																																																																																																																																																																																															
HDA_55S	HDA	HDA_SYNC_R 21																																																																																																																																																																																																																																																																																															
HDA_55S	HDA	HDA_SDINO 21 62																																																																																																																																																																																																																																																																																															
HDA_55S	HDA	AUD_SDI_R 62																																																																																																																																																																																																																																																																																															
	HDA	AUD_SPDIF_IN 9 66																																																																																																																																																																																																																																																																																															
	HDA	AUD_SPDIF_OUT 62 66																																																																																																																																																																																																																																																																																															
	HDA	AUD_SPDIF_CHIP 62																																																																																																																																																																																																																																																																																															
			CLK_MCP_XTAL	XTAL	MCP_CLK25M_XTALOUT 21 28																																																																																																																																																																																																																																																																																												
			CLK_MCP_XTAL	XTAL	MCP_CLK25M_XTALIN 21 28																																																																																																																																																																																																																																																																																												
			CLK_MCP_XTAL	XTAL	RTC_CLK32K_XTALOUT 21 28																																																																																																																																																																																																																																																																																												
			CLK_MCP_XTAL	XTAL	RTC_CLK32K_XTALIN 21 28																																																																																																																																																																																																																																																																																												
<table><tr><td colspan="3">SYNC MASTER=MASTER</td><td colspan="5">SYNC DATE=N/A</td></tr><tr><td colspan="3">PAGE TITLE</td><td colspan="5">MCP Constraints 2</td></tr><tr><td colspan="3" rowspan="2"></td><td colspan="2">DRAWING NUMBER</td><td colspan="3">SIZE</td></tr><tr><td colspan="2">051-7845</td><td colspan="3">D</td></tr><tr><td colspan="3" rowspan="2"></td><td colspan="2">REVISION</td><td colspan="3">A.0.0</td></tr><tr><td colspan="2">BRANCH</td><td colspan="3"></td></tr><tr><td colspan="3" rowspan="2">NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED</td><td colspan="2">PAGE</td><td colspan="3">103 OF 110</td></tr><tr><td colspan="2">SHEET</td><td colspan="3" rowspan="2"></td></tr><tr><td colspan="3"></td><td colspan="5"><CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS></td></tr></table>								SYNC MASTER=MASTER			SYNC DATE=N/A					PAGE TITLE			MCP Constraints 2								DRAWING NUMBER		SIZE			051-7845		D						REVISION		A.0.0			BRANCH					NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			PAGE		103 OF 110			SHEET								<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>																																																																																																																																																																																																																															
SYNC MASTER=MASTER			SYNC DATE=N/A																																																																																																																																																																																																																																																																																														
PAGE TITLE			MCP Constraints 2																																																																																																																																																																																																																																																																																														
			DRAWING NUMBER		SIZE																																																																																																																																																																																																																																																																																												
			051-7845		D																																																																																																																																																																																																																																																																																												
			REVISION		A.0.0																																																																																																																																																																																																																																																																																												
			BRANCH																																																																																																																																																																																																																																																																																														
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			PAGE		103 OF 110																																																																																																																																																																																																																																																																																												
			SHEET																																																																																																																																																																																																																																																																																														
			<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>																																																																																																																																																																																																																																																																																														
8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																																																										

[illegible]



8

7

6

5

4

3

2

1

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FW_110D	*	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

FireWire Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	FW_110D	FW_TP	FW PORT0_TPA_P 42 43
	FW_110D	FW_TP	FW PORT0_TPA_N 42 43
	FW_110D	FW_TP	FW PORT0_TPB_P 42 43
	FW_110D	FW_TP	FW PORT0_TPB_N 42 43
PORT 1 & 2 NOT USED			
	FW_110D	FW_TP	FW P0_TPA_L_P 42
	FW_110D	FW_TP	FW P0_TPA_L_N 42
	FW_110D	FW_TP	FW P0_TPB_L_P 42
	FW_110D	FW_TP	FW P0_TPB_L_N 42

8

7

6

5

4

3

2

1

FireWire Constraints

SYNC MASTER=MASTER		SYNC DATE=N/A		
FireWire Constraints				
	Apple Inc.	DRAWING NUMBER	051-7845	
		REVISION	A.0.0	
NOTICE OF PROPRIETARY PROPERTY:		PAGE		
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		105 OF 110		
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET		
II NOT TO REPRODUCE OR COPY IT		<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>		
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART				
IV ALL RIGHTS RESERVED				

8

7

6

5

4

3

2

1

www.chinafix.com

[illegible]

8

7

6

5

4

3

2

1

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	0.08 MM	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
LVDS_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
MCP_DV_COMP	*	Y	0.5 MM	0.5 MM	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?
LVDS	TOP,BOTTOM	=4x_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Sections 2.5.3 & 2.5.4.

ELECTRICAL_CONSTRAINT_SET ASSIGNED IN CONT. MGR.	NET_TYPE			
	PHYSICAL	SPACING		
	DP_100D	DISPLAYPORT	MXM DP A ML P<3..0>	84 91
	DP_100D	DISPLAYPORT	DP IG ML P<3..0>	9 91
	DP_100D	DISPLAYPORT	MXM DP A ML N<3..0>	84 91
	DP_100D	DISPLAYPORT	DP IG ML N<3..0>	9 91
	DP_100D	DISPLAYPORT	DP ML P<3..0>	91 94
	DP_100D	DISPLAYPORT	DP ML N<3..0>	91 94
	DP_100D	DISPLAYPORT	DP ML CONN P<3..0>	94
	DP_100D	DISPLAYPORT	DP ML CONN N<3..0>	94
	DP_100D	DISPLAYPORT	DP IG_AUX_CH_P	18 93
	DP_100D	DISPLAYPORT	DP IG_AUX_CH_N	18 93
	DP_100D	DISPLAYPORT	DP_AUXCH_SW_P	93
	DP_100D	DISPLAYPORT	DP_AUXCH_SW_N	93
	DP_100D	DISPLAYPORT	DP_AUX_CH_C_P	93 94
	DP_100D	DISPLAYPORT	DP_AUX_CH_C_N	93 94
	DP_100D	DISPLAYPORT	MXM_DP_A_AUX_P	84 93
	DP_100D	DISPLAYPORT	MXM_DP_A_AUX_N	84 93
	LVDS_100D	LVDS	LVDS_IG_A_CLK_P	18 89
	LVDS_100D	LVDS	LVDS_IG_A_CLK_N	18 89
	LVDS_100D	LVDS	LVDS_IG_A_DATA_P<3..0>	18 89
	LVDS_100D	LVDS	LVDS_IG_A_DATA_N<3..0>	18 89
	LVDS_100D	LVDS	LVDS_IG_B_CLK_P	18 89
	LVDS_100D	LVDS	LVDS_IG_B_CLK_N	18 89
	LVDS_100D	LVDS	LVDS_IG_B_DATA_P<3..0>	18 89
	LVDS_100D	LVDS	LVDS_IG_B_DATA_N<3..0>	18 89
	LVDS_100D	LVDS	LVDS_A_DATA_P<3..0>	89
	LVDS_100D	LVDS	LVDS_A_DATA_N<3..0>	89
	LVDS_100D	LVDS	LVDS_B_DATA_P<3..0>	89
	LVDS_100D	LVDS	LVDS_B_DATA_N<3..0>	89
	LVDS_100D	LVDS	LCD_CONN_A_DATA_P<3..0>	89 90
	LVDS_100D	LVDS	LCD_CONN_A_DATA_N<3..0>	89 90
	LVDS_100D	LVDS	LCD_CONN_B_DATA_P<3..0>	89 90
	LVDS_100D	LVDS	LCD_CONN_B_DATA_N<3..0>	89 90
	LVDS_100D	LVDS	LVDS_A_CLK_P	89
	LVDS_100D	LVDS	LVDS_A_CLK_N	89
	LVDS_100D	LVDS	LVDS_B_CLK_P	89
	LVDS_100D	LVDS	LVDS_B_CLK_N	89
	LVDS_100D	LVDS	LCD_CONN_A_CLK_P	89 90
	LVDS_100D	LVDS	LCD_CONN_A_CLK_N	89 90
	LVDS_100D	LVDS	LCD_CONN_B_CLK_P	89 90
	LVDS_100D	LVDS	LCD_CONN_B_CLK_N	89 90
	MCP_DV_COMP		MCP_HDMI_RSET	18 26
	MCP_DV_COMP		MCP_HDMI_VPROBE	18 26

8

7

6

5

4

3

2

1

GRAPHICS CONSTRAINTS

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER
051-7845

REVISION
A.0.0

BRANCH

PAGE
107 OF 110

SHEET

SYNC MASTER=MASTER

SYNC DATE=N/A

<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>



8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																									
<table><tr><th>SPACING_RULE_SET</th><th>LAYER</th><th>LINE-TO-LINE SPACING</th><th>WEIGHT</th></tr><tr><td>GND</td><td>*</td><td>=STANDARD</td><td>?</td></tr><tr><td>PPDDR_MEM</td><td>*</td><td>=STANDARD</td><td>?</td></tr></table> <table><tr><th>SPACING_RULE_SET</th><th>LAYER</th><th>LINE-TO-LINE SPACING</th><th>WEIGHT</th></tr><tr><td>GND_P2MM</td><td>*</td><td>0.20 MM</td><td>1000</td></tr><tr><td>PWR_P2MM</td><td>*</td><td>0.20 MM</td><td>1000</td></tr></table> <table><tr><th>NET_SPACING_TYPE1</th><th>NET_SPACING_TYPE2</th><th>AREA_TYPE</th><th>SPACING_RULE_SET</th></tr><tr><td>MEM_CLK</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>MEM_CMD</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>MEM_CTRL</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>MEM_DATA</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>MEM_DQS</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>MEM_CLK</td><td>PPDDR_MEM</td><td>*</td><td>PWR_P2MM</td></tr><tr><td>MEM_CMD</td><td>PPDDR_MEM</td><td>*</td><td>PWR_P2MM</td></tr><tr><td>MEM_CTRL</td><td>PPDDR_MEM</td><td>*</td><td>PWR_P2MM</td></tr><tr><td>MEM_DATA</td><td>PPDDR_MEM</td><td>*</td><td>PWR_P2MM</td></tr><tr><td>MEM_DQS</td><td>PPDDR_MEM</td><td>*</td><td>PWR_P2MM</td></tr></table> <table><tr><th>NET_SPACING_TYPE1</th><th>NET_SPACING_TYPE2</th><th>AREA_TYPE</th><th>SPACING_RULE_SET</th></tr><tr><td>CLK_PCIE</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>PCIE</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>SATA</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>USB</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr></table> <table><tr><th>NET_SPACING_TYPE1</th><th>NET_SPACING_TYPE2</th><th>AREA_TYPE</th><th>SPACING_RULE_SET</th></tr><tr><td>CLK_FSB</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>CPU_COMP</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>CPU_GTLREF</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>CPU_VCCSENSE</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>FSB_DSTB</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr></table> <table><tr><th>NET_SPACING_TYPE1</th><th>NET_SPACING_TYPE2</th><th>AREA_TYPE</th><th>SPACING_RULE_SET</th></tr><tr><td>CLK_PCIE</td><td>PWR</td><td>*</td><td>PWR_P2MM</td></tr></table> <table><tr><th>NET_SPACING_TYPE1</th><th>NET_SPACING_TYPE2</th><th>AREA_TYPE</th><th>SPACING_RULE_SET</th></tr><tr><td>THERMAL</td><td>*</td><td>*</td><td>4:1_SPACING</td></tr><tr><td>SWITCHNODE</td><td>*</td><td>*</td><td>SWITCHNODE</td></tr><tr><td>THERMAL</td><td>PWR</td><td>*</td><td>PWR_P2MM</td></tr><tr><td>THERMAL</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr><tr><td>AUDIO</td><td>*</td><td>*</td><td>AUDIO</td></tr></table> <table><tr><th>NET_PHYSICAL_TYPE</th><th>AREA_TYPE</th><th>PHYSICAL_RULE_SET</th></tr><tr><td>THERM_DIFF</td><td>*</td><td>1:1_DIFFPAIR</td></tr><tr><td>SNS_DIFF</td><td>*</td><td>1:1_DIFFPAIR</td></tr></table> <table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW ROUTE ON LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>MEM_40S OVERRIDE</td><td>TOP</td><td>OVERRIDE</td><td>OVERRIDE</td><td>0.1 MM OVERRIDE</td><td>600 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr><tr><td>MEM_40S_VDD OVERRIDE</td><td>TOP</td><td>OVERRIDE</td><td>OVERRIDE</td><td>0.1 MM OVERRIDE</td><td>600 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr><tr><td>MEM_70D OVERRIDE</td><td>TOP</td><td>OVERRIDE</td><td>OVERRIDE</td><td>0.1 MM OVERRIDE</td><td>600 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr><tr><td>PCIE_90D OVERRIDE</td><td>TOP</td><td>OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td><td>500 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr><tr><td>USB_90D OVERRIDE</td><td>TOP</td><td>OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td><td>500 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr><tr><td>MCP_DV_COMP OVERRIDE</td><td>TOP</td><td>OVERRIDE</td><td>OVERRIDE</td><td>0.1 MM OVERRIDE</td><td>500 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr><tr><td>MCP_MEM_COMP OVERRIDE</td><td>TOP</td><td>OVERRIDE</td><td>OVERRIDE</td><td>0.1 MM OVERRIDE</td><td>500 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr><tr><td>MCP_MII_COMP OVERRIDE</td><td>TOP</td><td>OVERRIDE</td><td>OVERRIDE</td><td>0.1 MM OVERRIDE</td><td>500 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr><tr><td>MCP_USB_RBIAS OVERRIDE</td><td>TOP</td><td>OVERRIDE</td><td>OVERRIDE</td><td>0.1 MM OVERRIDE</td><td>500 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr><tr><td>MCP_DV_COMP OVERRIDE</td><td>*</td><td>OVERRIDE</td><td>OVERRIDE</td><td>0.25 MM OVERRIDE</td><td>250 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr><tr><td>CPU_27P4S OVERRIDE</td><td>BOTTOM</td><td>OVERRIDE</td><td>OVERRIDE</td><td>0.23 MM OVERRIDE</td><td>100 MIL OVERRIDE</td><td>OVERRIDE</td><td>OVERRIDE</td></tr></table>	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	GND	*	=STANDARD	?	PPDDR_MEM	*	=STANDARD	?	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	GND_P2MM	*	0.20 MM	1000	PWR_P2MM	*	0.20 MM	1000	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	MEM_CLK	GND	*	GND_P2MM	MEM_CMD	GND	*	GND_P2MM	MEM_CTRL	GND	*	GND_P2MM	MEM_DATA	GND	*	GND_P2MM	MEM_DQS	GND	*	GND_P2MM	MEM_CLK	PPDDR_MEM	*	PWR_P2MM	MEM_CMD	PPDDR_MEM	*	PWR_P2MM	MEM_CTRL	PPDDR_MEM	*	PWR_P2MM	MEM_DATA	PPDDR_MEM	*	PWR_P2MM	MEM_DQS	PPDDR_MEM	*	PWR_P2MM	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	CLK_PCIE	GND	*	GND_P2MM	PCIE	GND	*	GND_P2MM	SATA	GND	*	GND_P2MM	USB	GND	*	GND_P2MM	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	CLK_FSB	GND	*	GND_P2MM	CPU_COMP	GND	*	GND_P2MM	CPU_GTLREF	GND	*	GND_P2MM	CPU_VCCSENSE	GND	*	GND_P2MM	FSB_DSTB	GND	*	GND_P2MM	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	CLK_PCIE	PWR	*	PWR_P2MM	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	THERMAL	*	*	4:1_SPACING	SWITCHNODE	*	*	SWITCHNODE	THERMAL	PWR	*	PWR_P2MM	THERMAL	GND	*	GND_P2MM	AUDIO	*	*	AUDIO	NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET	THERM_DIFF	*	1:1_DIFFPAIR	SNS_DIFF	*	1:1_DIFFPAIR	PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	MEM_40S OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	600 MIL OVERRIDE	OVERRIDE	OVERRIDE	MEM_40S_VDD OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	600 MIL OVERRIDE	OVERRIDE	OVERRIDE	MEM_70D OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	600 MIL OVERRIDE	OVERRIDE	OVERRIDE	PCIE_90D OVERRIDE	TOP	OVERRIDE	OVERRIDE	OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE	USB_90D OVERRIDE	TOP	OVERRIDE	OVERRIDE	OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE	MCP_DV_COMP OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE	MCP_MEM_COMP OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE	MCP_MII_COMP OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE	MCP_USB_RBIAS OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE	MCP_DV_COMP OVERRIDE	*	OVERRIDE	OVERRIDE	0.25 MM OVERRIDE	250 MIL OVERRIDE	OVERRIDE	OVERRIDE	CPU_27P4S OVERRIDE	BOTTOM	OVERRIDE	OVERRIDE	0.23 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE	K50/K51 SPECIFIC NET PROPERTIES						
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT																																																																																																																																																																																																																																																													
GND	*	=STANDARD	?																																																																																																																																																																																																																																																													
PPDDR_MEM	*	=STANDARD	?																																																																																																																																																																																																																																																													
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT																																																																																																																																																																																																																																																													
GND_P2MM	*	0.20 MM	1000																																																																																																																																																																																																																																																													
PWR_P2MM	*	0.20 MM	1000																																																																																																																																																																																																																																																													
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET																																																																																																																																																																																																																																																													
MEM_CLK	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
MEM_CMD	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
MEM_CTRL	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
MEM_DATA	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
MEM_DQS	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
MEM_CLK	PPDDR_MEM	*	PWR_P2MM																																																																																																																																																																																																																																																													
MEM_CMD	PPDDR_MEM	*	PWR_P2MM																																																																																																																																																																																																																																																													
MEM_CTRL	PPDDR_MEM	*	PWR_P2MM																																																																																																																																																																																																																																																													
MEM_DATA	PPDDR_MEM	*	PWR_P2MM																																																																																																																																																																																																																																																													
MEM_DQS	PPDDR_MEM	*	PWR_P2MM																																																																																																																																																																																																																																																													
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET																																																																																																																																																																																																																																																													
CLK_PCIE	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
PCIE	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
SATA	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
USB	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET																																																																																																																																																																																																																																																													
CLK_FSB	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
CPU_COMP	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
CPU_GTLREF	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
CPU_VCCSENSE	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
FSB_DSTB	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET																																																																																																																																																																																																																																																													
CLK_PCIE	PWR	*	PWR_P2MM																																																																																																																																																																																																																																																													
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET																																																																																																																																																																																																																																																													
THERMAL	*	*	4:1_SPACING																																																																																																																																																																																																																																																													
SWITCHNODE	*	*	SWITCHNODE																																																																																																																																																																																																																																																													
THERMAL	PWR	*	PWR_P2MM																																																																																																																																																																																																																																																													
THERMAL	GND	*	GND_P2MM																																																																																																																																																																																																																																																													
AUDIO	*	*	AUDIO																																																																																																																																																																																																																																																													
NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET																																																																																																																																																																																																																																																														
THERM_DIFF	*	1:1_DIFFPAIR																																																																																																																																																																																																																																																														
SNS_DIFF	*	1:1_DIFFPAIR																																																																																																																																																																																																																																																														
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																																																																																																																																																																																																																																																									
MEM_40S OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	600 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
MEM_40S_VDD OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	600 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
MEM_70D OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	600 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
PCIE_90D OVERRIDE	TOP	OVERRIDE	OVERRIDE	OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
USB_90D OVERRIDE	TOP	OVERRIDE	OVERRIDE	OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
MCP_DV_COMP OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
MCP_MEM_COMP OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
MCP_MII_COMP OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
MCP_USB_RBIAS OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
MCP_DV_COMP OVERRIDE	*	OVERRIDE	OVERRIDE	0.25 MM OVERRIDE	250 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
CPU_27P4S OVERRIDE	BOTTOM	OVERRIDE	OVERRIDE	0.23 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE																																																																																																																																																																																																																																																									
				ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING																																																																																																																																																																																																																																																									
						PDDR_MEM	=PP1V5_S3 MEM_A 6 30 31																																																																																																																																																																																																																																																									
						PDDR_MEM	=PP1V5_S3 MEM_B 6 30 32																																																																																																																																																																																																																																																									
						SWITCHNODE	VR_CPU_SW1 72																																																																																																																																																																																																																																																									
						SWITCHNODE	VR_CPU_SW2 72																																																																																																																																																																																																																																																									
						SWITCHNODE	VR_CPU_SW3 72																																																																																																																																																																																																																																																									
						SWITCHNODE	1V8_SW 80																																																																																																																																																																																																																																																									
						SWITCHNODE	1V1S5_SW 79																																																																																																																																																																																																																																																									
						SWITCHNODE	PVTT50_PHASE 76																																																																																																																																																																																																																																																									
						SWITCHNODE	3V3S5_SW 76																																																																																																																																																																																																																																																									
						SWITCHNODE	5VS3_SW 73																																																																																																																																																																																																																																																									
						SWITCHNODE	MCPCORE0_PHASE 74																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_T_DP1_DN6 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_T_DN1_DP6 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_T_DP2_DN3 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_T_DN2_DP3 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	CPU_THERMD_P 11 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	CPU_THERMD_N 11 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_T_DP4_DN5 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_T_DN4_DP5 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	MCP_THMDIODE_P 21 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	MCP_THMDIODE_N 21 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	MXM_PWRSRC_SENSOR_P 53																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	MXM_PWRSRC_SENSOR_N 53																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL																																																																																																																																																																																																																																																										
					THERM_DIFF	THERMAL	SENSE_1V5_S0_P 54																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SENSE_1V5_S0_N 54																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_LCD_P 55 110																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_LCD_N 55 110																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_ODD_P 55 110																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_ODD_N 55 110																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_CPU_H_P 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_CPU_H_N 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_MCP_P 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_MCP_N 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL																																																																																																																																																																																																																																																										
					THERM_DIFF	THERMAL	SNS_AMB_P 55 110																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_AMB_N 55 110																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_MXM_P 55																																																																																																																																																																																																																																																									
					THERM_DIFF	THERMAL	SNS_MXM_N 55																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS1_P 71 72																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS1_N 71 72																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_P 71																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_N 71																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS2_P 71 72																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS2_N 71 72																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_P 71																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_N 71																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS3_P 71 72																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS3_N 71 72																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_P 71																																																																																																																																																																																																																																																									
					SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_N 71																																																																																																																																																																																																																																																									
				12430		THERMAL	SMC_CPU_ISENSE 49 53																																																																																																																																																																																																																																																									
				12431		THERMAL	VR_CPU_IOUT 53 71																																																																																																																																																																																																																																																									
				12432		THERMAL	VR_ISNS_CPU_P 53																																																																																																																																																																																																																																																									
				12433		THERMAL	VR_ISNS_CPU_N 53																																																																																																																																																																																																																																																									
				12434		THERMAL	SNS_PS_CPU_ISNS 53																																																																																																																																																																																																																																																									
				12435		THERMAL	SMC_CPU_VSENSE 49 53																																																																																																																																																																																																																																																									
				12436		THERMAL	CPU_VCC_SENSE 12 53																																																																																																																																																																																																																																																									
				12437		THERMAL																																																																																																																																																																																																																																																										
				12438		THERMAL	SMC_GPU_VSENSE 49 53																																																																																																																																																																																																																																																									
				12439		THERMAL	SMC_GPU_ISENSE 49 53																																																																																																																																																																																																																																																									
				12440		THERMAL																																																																																																																																																																																																																																																										
				12441		THERMAL	SMC_1V5_S0_ISENSE 50 54																																																																																																																																																																																																																																																									
				12220		THERMAL	SMC_1V5_S0_ISENSE_R 54																																																																																																																																																																																																																																																									
				12221		THERMAL	SMC_1V5_S0_VSENSE 50 54																																																																																																																																																																																																																																																									
				12222		THERMAL	SMC_MCP_CORE_ISENSE 50 54																																																																																																																																																																																																																																																									
				12223		THERMAL	SMC_MCP_CORE_VSENSE 50 54																																																																																																																																																																																																																																																									
				12430		THERMAL	MCPCORE0_IMON 54 74																																																																																																																																																																																																																																																									
				12442		THERMAL	CPU_PECI_L 11 55																																																																																																																																																																																																																																																									
				12225		THERMAL	SMB_PECI_L 11 55																																																																																																																																																																																																																																																									
				12226		THERMAL	CPU_PECI_MCP 14 55																																																																																																																																																																																																																																																									
				12227		THERMAL																																																																																																																																																																																																																																																										
				12228		THERMAL	HDD_OOB_TEMP_FILT 55																																																																																																																																																																																																																																																									
				12229		THERMAL	HDD_OOB_TEMP 55																																																																																																																																																																																																																																																									
				12230		THERMAL	HDD_OOB_TEMP_R 55																																																																																																																																																																																																																																																									
				12231		THERMAL	SMC_HDD_OOB_TEMP 55																																																																																																																																																																																																																																																									



K50/K51 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS		BOARD AREAS		BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, BOTTOM		NO_TYPE, BGA_P1MM		MM	15.5.1

PHYSICAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT?	*	Y	=50_OHM_SE	=50_OHM_SE	100 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27F4_OHM_SE	TOP,BOTTOM	Y	0.300 MM	0.085 MM	=STANDARD		
27F4_OHM_SE	*	Y	0.275 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.085 MM	=STANDARD		
40_OHM_SE	*	Y	0.15 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	TOP,BOTTOM	Y	0.151 MM	0.085 MM	=STANDARD		
42_OHM_SE	*	Y	0.136 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.1 MM	0.085 MM	15 MM		
50_OHM_SE	*	Y	0.1 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD		
55_OHM_SE	*	Y	0.076 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
70_OHM_DIFF	ISL3,ISL6	Y	0.155 MM	0.085 MM	=STANDARD	0.135 MM	0.1 MM
70_OHM_DIFF	TOP,BOTTOM	Y	0.165 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL6	Y	0.099 MM	0.085 MM	12 MM	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3,ISL6	Y	0.081 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.091 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	TOP,BOTTOM	Y	0.075 MM	0.085 MM	=STANDARD	0.320 MM	0.15 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
POWER_WIDTH	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER	*	POWER_WIDTH
VR_CTL_PHY	*	POWER_WIDTH

CONSTRAINTS ARE BASED ON MCP79 DESIGN GUIDE DG-03328-001 V06
PCI,LPC,SMB,HDA,SPI,RGMII,SMBUS ARE ROUTED AS 55 OHM SE SIGNALS

SPACING RULE SET

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT?	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SPACING_0.5MM	*	0.5 MM	?
CLK_SPACING_0.6MM	*	0.6 MM	?
GND_P2MM	*	0.2 MM	1000
PHR_P2MM	*	0.2 MM	1000
SWITCHNODE	*	0.6 MM	1000

CONSTRAINTS FOR BGA AREA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	0.2 MM	?
BGA_P3MM	*	0.3 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
MEM_CLK	*	BGA_P1MM	BGA_P2MM
CLK_FSB	*	BGA_P1MM	BGA_P2MM
CLK_PCIE	*	BGA_P1MM	BGA_P1MM
FSB_DSTB	FSB_DSTB	BGA_P1MM	BGA_P1MM
CLK_LPC	*	BGA_P1MM	BGA_P1MM
CLK_PCI	*	BGA_P1MM	BGA_P1MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MCP_FSB_COMP	*	BGA_P1MM	BGA_P2MM
MCP_MEM_COMP	*	BGA_P1MM	BGA_P2MM
MCP_PEX_COMP	*	BGA_P1MM	BGA_P2MM

SYNC MASTER=MASTER

SYNC DATE=N/A

K22/K23 RULE DEFINITIONS

Apple Inc.

051-7845

A.0.0

109 OF 110

SHEET

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>

[illegible]